

INTRODUCTION PAGE

Project Description:	The primary goal of this database is to compile a set of cost indicators and values (including capital cost, O&M rate, discount rate, etc.) that enable comparison across regions. These indicators can function as a reference point for the transmission project cost assessments, by providing a greater level of transparency and gauging the impact of various factors affecting the costs.	
Tab Description:		
- Database (Comprehensive):	This tab contains all the detailed information of a project, which is divided into three major categories: "Project Basic Information," "Transmission Information," and "Cost Information." The data are manually collected from related project documents, with the sources attached at the end.	
- Database (line numbers):	This tab contains only valid data that can be used for statistical analysis and comparison. Projects lacking specific data on capital cost, discount rate, O&M rate, etc. are excluded. There are repeated project IDs in this tab, as one project may have multiple transmission segments with varying costs. The user may use the filter function to generate some priliminary comparisons over region, voltage range, time, etc.	
- Raw Dataset	This tab includes 469 projects tagged under the broad category of “energy transmission and distribution”; however, not all of them involves the construction or rehabilitation of transmission lines/substations. To obtain valid data suitable for this project, the projects were filtered by if they contained the word “transmission” in the title, under the assumption that they are the most relevant ones.	
Sample size:	The total smaple size is 75. However, depending on the data availability, the sample size may vary per each analysis.	
Adjusting cost estimates:	The collected cost information is in the nominal value of the particular year that the project appraisal document is approved. The further analysis is adjusted to the constant dollar value of 2010, according to the Manufactured Unit Index (MUV), published on the Bank’s website. The source of this information is attached in this dataset.	
Limitations:	i. The dataset contains a high level of qualitative data, manually collected from project documents. Teasing out individual data requires some assumptions and judgments, which may introduce biases to the analysis. ii. There is a high degree of variation regarding the quality of the detailed information, for both the technical and cost categories. Further, not all the information is available online. The lack of standard reporting format may also introduce certain biases and variances to the analysis. iii. While existing cost frameworks suggest variations by terrain type, land structures, vegetation, etc., such information remains fragmented, and the availability varies significantly across projects, making it challenging to collect and	
Data source:	World Bank Project Library,	https://projects.worldbank.org/en/projects-operations/projects-home
	World Bank Commodity Price Data,	https://thedocs.worldbank.org/en/doc/5d903e848db1d1b83e0ec8f744e55570-0350012021/related/CMO-Historical-Data-Annual.xlsx
Data collection timespan:	January - June 2023: Main data collection period After June 2023: Data updates	
Last edited:	March 13, 2026	

EXPLANATION OF MAIN DATA FIELDS

Project Basic Information

Restructuring / Closing	This field records the restructuring efforts with major project component changes. For projects that are closed, this field documents the expenditure information at project closing.
Project type	Transmission projects have two major types: construction of new lines versus expansion or upgrade of the existing system. This field captures this distinction, which may have cost differences.
Routing (existing vs new)	Depending on the project type, the project may utilize existing or new line routing.
Implementation delays (Y/N)	This is a binary data field examining if the project has experienced delays that result in the extension of the project closing date. This field also records the causes for such delays.
WB financing	This field captures how much lending the World Bank has provided for the target project. One project is likely to have multiple financing sources.

Transmission Information

Ownership	In most of the cases, the transmission facility is owned nationally. In some cases, the regional electricity facility has the ownership over the facility.
Route length (km)	The length of the transmission lines.
Voltage	Transmission-level voltages are usually 110 kV and above.
Elevation (AMSL)	The elevation of the transmission lines. AMSL stands for "above mean sea level."
Number of transmission towers	This field captures the number of total transmission towers for the project, combining different types of towers.
Number of substations	The total number of substations either constructed and/or rehabilitated in the project.
Conductor type	Double circuit vs single circuit.
Materials	Materials for the conductors.
Estimated transmission losses	The estimated loss of power or voltage of a transmitted wave or current in passing along a transmission line or path or through a circuit device.

Cost Information

Capital cost for transmission component	Capital costs are one-time expenses paid for things used in the production of goods or service. This data field captures the capital expenditure for the transmission line construction or upgrades.
Discount rate applied	A discount rate is the rate of return used to discount future cash flows back to their present value.
Operations & Maintenance rate	The Operation and Maintenance (O&M) cost of a component is the cost associated with operating and maintaining that component. The total O&M cost of the system is the sum of the O&M costs of each system component.
ESMP cost	ESMP stands for the Environmental and Social Management Plan while the cost measures the expenses of implementing the plan. There are frameworks under similar names with slight changes.
Resettlement / RAP cost	RAP stands for the Resettlement Action Plan. The purpose of a resettlement action plan is to specify all resettlement arrangements and the measures for avoiding, minimising or compensating losses or other negative social impacts resulting from resettlement.
Capital cost per km (tab: line numbers)	The capital cost per km, referred as the unit investment cost (UIC), is calculated by dividing the total capital cost for transmission by the lengths of the transmission lines.

No.	Project ID	Project Name	Start date	Closing date	Region	Country
1	P174812	Transmission Investment Plan (PIT) to support Post-COVID-19 Green Economic Recovery in Peru	Sep-21	Nov-26	Latin America and Caribbean	Peru
2	P171683	Electricity Sector Transformation and Resilient Transmission	Jun-21	Jan-28	Europe and Central Asia	Uzbekistan

3	P167843	Additional Financing For Dasu Hydropower Stage I Project (Transmission Line)	Mar-20	31-May-24	South Asia	Pakistan
4	P167898	Additional Financing for the Central Asia South Asia Electricity Transmission and Trade Project	May-19	December 31, 2028	South Asia	Pakistan (mainly), Kyrgyz Republic, Tajikistan, Afghanistan

5	P163752	AFR RI-3A Tanzania-Zambia Transmission Interconnector	Jun-18	Jun-26	Eastern and Southern Africa	Tanzania, Zambia
6	P146330	NG-Electricity Transmission Project	Feb-18	Jun-25	Western and Central Africa	Nigeria

7	P154987	National Transmission Modernization I Project	Dec-17	Apr-25	South Asia	Pakistan
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8	P147921	OMVS - TRANSMISSION EXPANSION PROJECT	May-17	Dec-22	Western and Central Africa	Senegal
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9	P157055	Cote d'Ivoire - Electricity Transmission and Access Project	Mar-17	Sep-23	Western and Central Africa	Cote d'Ivoire
10	P152755	Electricity Transmission and Reform Project	Dec-16	Jun-25	Western and Central Africa	Cameroon

11	P156584	Modernization and Upgrade of Transmission Substations	Nov-16	Jun-24	Europe and Central Asia	Uzbekistan
12	P146199	Electricity Transmission Network Improvement Project	Mar-15	Dec-24	Europe and Central Asia	Armenia

13	P146788	Second Power Transmission Project	Dec-14	Dec-26	Europe and Central Asia	Ukraine
14	P131558	Transmission Efficiency Project (TEP)	Aug-14	Dec-21	East Asia and Pacific	Vietnam

15	P147348	Transmission Grid Strengthening Project	May-14	Mar-23	Europe and Central Asia	Georgia
16	P145054	Central Asia South Asia Electricity Transmission and Trade Project (CASA-1000)	Mar-14	Dec-28	South Asia	Kyrgyz Republic, Tajikistan, Pakistan, Afghanistan

17	P129920	Bangladesh: Rural Electricity Transmission and Distribution Project	Feb-14	Jun-21	South Asia	Bangladesh
18	P123994	Indonesia Second Power Transmission Development Project	Jul-13	Dec-19	East Asia and Pacific	Indonesia

19	P115767	Nepal-India Electricity Transmission and Trade Project	Jun-11	Oct-21	South Asia	Nepal
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20	P132631	Additional Financing for Nepal-India Electricity Transmission and Trade Project (Additional to Project No. P115767)	Jun-13	N/A	South Asia	Nepal
21	P133184	Lusaka Transmission and Distribution Rehabilitation Project	May-13	Feb-22	Eastern and Southern Africa	Zambia

22	P124351	Kafue-Muzuma-Victoria Falls Regional Transmission Line Reinforcement Project	May-12	Jun-18	Eastern and Southern Africa	Zambia
23	P094919	WAPP: The First Phase of the Inter-Zonal Transmission Hub Project of the WAPP (APL3) Program	Jun-11	Dec-18	Western and Central Africa	Ghana, Burkina Faso

24	P112893	Kabeli Transmission Project	May-11	Dec-16	South Asia	Nepal
25	P114875	Second Transmission and Distribution Project Additional Financing	Mar-11	N/A	East Asia and Pacific	Vietnam
26	P119939	Talimarjan Transmission Project	Mar-11	Jun-16	Europe and Central Asia	Uzbekistan

27	P116919	Alma Transmission Project	Nov-10	Apr-15	Europe and Central Asia	Kazakhstan
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28	P111598	Tanzania - Backbone Transmission Investment Project	Aug-10	Dec-16	Eastern and Southern Africa	Tanzania
29	P117323	Indonesia Power Transmission Development Project	Jul-10	Oct-19	East Asia and Pacific	Indonesia
30	P116784	Morupule B Generation and Transmission Project	Oct-09	Sep-15	Eastern and Southern Africa	Botswana

31	P112516	Botswana - Morupule B Generation and Transmission Project	Oct-09	Aug-14	Eastern and Southern Africa	Botswana
32	P114766	Kazakhstan Moinak Electricity Transmission Project	Sep-09	Apr-13	Europe and Central Asia	Kazakhstan
33	P095982	Electricity Distribution and Transmission Improvement Project	Jun-08	Feb-14	South Asia	Pakistan

34	P096207	Power Transmission Project in Support of the Energy Sector Reform & Development Program	Aug-07	Jun-16	Europe and Central Asia	Ukraine
35	P084404	Mozambique Transmission Upgrade Project	Jul-07	Aug-16	Eastern and Southern Africa	Mozambique; Malawi

36	P094917	WAPP APL 1 (2nd Phase - COASTAL TRANSMISSION BACKBONE)	Jun-06	Jun-16	Western and Central Africa	Ghana, Benin
37	P095155	North-South Electricity Transmission Project	Oct-05	Nov-11	Europe and Central Asia	Kazakhstan

38	P084871	Second Transmission and Distribution Project	Jul-05	Jun-14	East Asia and Pacific	Vietnam
39	P075994	WAPP APL 1 (1st Phase - COASTAL TRANSMISSION BACKBONE)	Jun-05	Jun-13	Western and Central Africa	Ghana, Benin
40	P083341	Power Transmission Project	May-05	Dec-11	Europe and Central Asia	Azerbaijan

41	P064844	Rural Electrification and Transmission Project	Dec-03	Jan-12	East Asia and Pacific	Cambodia
42	P072018	Transmission Development Project	Jul-01	Dec-08	Western and Central Africa	Nigeria
43	P110729	Central Asia South Asia Electricity Transmission and Trade Project (CASA 1000)	N/A	N/A		

44	P171248	Support for Preparation of the Rural Electrification, Sebzor HPP and Khorog-Qozideh Power Transmission Line Projects	Oct-19	N/A	Europe and Central Asia	Tajikistan
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45	P159974	Enhancement and Strengthening of Power Transmission Network in Eastern Region	Mar-18	Dec-25	South Asia	Bangladesh
46	P146830	OMVG Interconnection Project	Apr-15	Jun-25	Western and Central Africa	Guinea, The Gambia, Guinea Bissau, Senegal

47	P160427	Temane Regional Electricity Project	Jun-19	Dec-26	Eastern and Southern Africa	Mozambique
48	P166042	Guinea – Mali Interconnection Project	Jul-18	Dec-26	Western and Central Africa	Guinea, Mali
49	P168185	Cameroon - Chad Power Interconnection Project	Jun-20	Dec-27	Western and Central Africa	Cameroon, Chad

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[P074011](#)

Ethiopia/Nile Basin Initiative: Ethiopia-Sudan Interconnector

Dec-07

Jun-13

Eastern and Southern Africa

Ethiopia

51

[P168273](#)

Tunisia Energy Sector Improvement Project

Jun-19

Sep-26

Middle East and
North Africa

Tunisia

52	P104265	MA-Support to ONE (Office National de l'Electricité)	Jun-08	Dec-15	Middle East and North Africa	Morocco
53	P113266	WAPP-APL4 (Phase 1): Côte d'Ivoire Liberia Sierra Leone and Guinea Regional Interconnector- Additional Financing	May-12	Mar-23	Western and Central Africa	Côte d'Ivoire, Liberia, Sierra Leone, and Guinea

54	P160829	Moldova Power System Development Project	8-May-19	30-Jun-26	Europe and Central Asia	Moldova
55	P082337	ECSEE APL 3 - FYR Macedonia	10-Jan-06	31-Jul-16	Europe and Central Asia	Macedonia
56	P088867	Energy Community of South East Europe (APL) Program - Serbia and Montenegro Component - Serbia Project	30-Jun-05	30-Jun-12	Europe and Central Asia	Serbia

57	P069258	AFCC2/RI-3A-Southern Afr Power Mrkt APL 1 (FY04)	11-Nov-03	30-Sep-16	Eastern and Southern Africa	DRC, Zambia
58	P164354	Mozambique - Malawi Regional Interconnector Project	17-Sep-19	30-Jun-25	Eastern and Southern Africa	Mozambique; Malawi

59	P173763	Second Djibouti-Ethiopia Power System Interconnection Project	29-Mar-22	31-Dec-26	Eastern and Southern Africa	Djibouti; Ethiopia
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[P162933](#)

North Core/Dorsale Nord Regional Power
Interconnector Project

31-Oct-18

15-Mar-26

Western and
Central Africa

Nigeria; Niger,
Bénin, Burkina
Faso

61	P105329	GMS Power Trade (Cambodia) Project	5-Jun-07	31-Dec-11	East Asia and Pacific	Cambodia
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62	P133305	Uganda Grid Expansion and Reinforcement Project (GERP)	31-May-16	31-Dec-24	Eastern and Southern Africa	Uganda
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63	P174034	Niger Accelerating Electricity Access Project (Haské)	10-Dec-21	31-May-27	Western and Central Africa	Niger
64	P095965	Siddhirganj Power Project	30-Oct-08	31-Dec-19	South Asia	Bangladesh

65	P126579	AFR RI-The Eastern Electricity Highway Project under the First Phase of the Eastern Africa Power Integration Program	12-Jul-12	31-Jan-24	Eastern and Southern Africa	Ethiopia; Kenya
66	P169117	Energy Supply Reliability and Financial Recovery	23-May-19	31-Dec-25	Europe and Central Asia	Georgia
67	P101645	TZ-Energy Development & Access Expansion Project	13-Dec-07	29-Sep-17	Eastern and Southern Africa	Tanzania

68	P103037	Electricity Expansion	27-May-10	31-Dec-17	Eastern and Southern Africa	Kenya
69	P110177	Cameroon - Kribi Gas Power Project	10-Nov-11	18-May-13	Western and Central Africa	Cameroon
70	P110075	Increased Access to Modern Energy	23-Jun-09	31-Dec-18	Western and Central Africa	Benin

71	P119737	Electricity Sector Development Project	30-Jun-11	31-Jan-19	Eastern and Southern Africa	Uganda
72	P097201	AFCC2/RI-Regional and Domestic Power Markets Development Project (Southern Africa Power Market Project: APL-1b)	29-May-07	30-Jun-18	Eastern and Southern Africa	DRC
73	P166170	Network Reinforcement and Access Project	27-Jun-19	31-Oct-24	Eastern and Southern Africa	Eswatini

74	P169561	Zanzibar Energy Sector Transformation and Access Project	3-Jun-21	31-Aug-27		Tanzania
					Eastern and Southern Africa	
75	P074011	Ethiopia/Nile Basin Initiative: Ethiopia-Sudan Interconnector	20-Dec-07	30-Jun-13	Eastern and Southern Africa	Ethiopia; Sudan
1	i	Project Information Document	https://documents1.worldbank.org/curated/en/797641635946063728/pdf/Per			
2	i	Project Information Document	https://documents1.worldbank.org/curated/en/184411624932251299/pdf/Uzl			
	ii	Environmental and Social Management Plan	https://documents1.worldbank.org/curated/en/598991619774144557/pdf/Env			
3	i	Project Information Document	https://documents1.worldbank.org/curated/en/394751582532536169/pdf/Prc			
	ii	Pakistan - First Stage of Dasu Hydropower Pr	https://documents1.worldbank.org/curated/en/509011585965715775/pdf/Pal			

4	iii	Environmental Assessment	https://documents1.worldbank.org/curated/en/283621574657323856/pdf/Env
	i	Project Information Document	https://documents1.worldbank.org/curated/en/837661554802057794/pdf/Prc
	ii	Project Paper	https://documents1.worldbank.org/curated/en/958811559613716291/pdf/PaI
5	i	Project Information Document	https://documents1.worldbank.org/curated/en/325581529551846176/pdf/TAI
	ii	Integrated Safeguards Data Sheet	https://documents1.worldbank.org/curated/en/596391516729600327/pdf/Afr
6	i	Project Information Document	https://documents1.worldbank.org/curated/en/704121518922836847/pdf/NIC
	ii	Restructuring Paper (July 2022)	https://documents1.worldbank.org/curated/en/099940007222236468/pdf/P14
	iii	Resettlement Plan	https://documents1.worldbank.org/curated/en/752361490870871770/pdf/SFC
7	i	Project Information Document	https://documents1.worldbank.org/curated/en/345091513911668260/pdf/PaI
	ii	Resettlement Plan (Nowshera)	https://documents1.worldbank.org/curated/en/119281505932517759/pdf/SFC
	iii	Resettlement Plan (Islamabad West)	https://documents1.worldbank.org/curated/en/311621508185891559/pdf/NT
	iv	Environmental Assessment (Nowshera)	https://documents1.worldbank.org/curated/en/460251505932289763/pdf/SFC
	v	Environmental Assessment (Islamabad West)	https://documents1.worldbank.org/curated/en/923091508185670599/pdf/SFC
8	i	Project Information Document	https://documents1.worldbank.org/curated/en/923211494813685983/pdf/Ser
	ii	Restructuring Paper (Nov 2020)	https://documents1.worldbank.org/curated/en/869081605116876552/pdf/Dis
	iii	Restructuring Paper (May 2022)	https://documents1.worldbank.org/curated/en/974141488150742132/pdf/ApI
	iv	Environmental Assessment	https://documents1.worldbank.org/curated/en/311611485929124835/pdf/SFC
9	i	Project Information Document	https://documents1.worldbank.org/curated/en/450031491098454445/pdf/CO
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/977331591916714067/pdf/Dis
	iii	Resettlement Plan	https://documents1.worldbank.org/curated/en/236401485783294340/pdf/SFC
	iv	Environmental Assessment	https://documents1.worldbank.org/curated/en/974181485934712736/pdf/SFC
10	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/876841481338869545/pdf/148
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/112641617403135424/pdf/Dis
11	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/635091480129220617/pdf/Uzl
12	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/342701468211797826/pdf/PAI
	ii	Rehabilitation of Ashnak substation - Environ	https://documents1.worldbank.org/curated/en/705191468002993548/pdf/E47
	iii	Rehabilitation of the Substation of Yerevan T	https://documents1.worldbank.org/curated/en/253971468003028813/pdf/E47
	iv	Restructuring Paper	https://documents1.worldbank.org/curated/en/899481576784084105/pdf/Dis
13	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/700241468306919407/pdf/PAI
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/572701557146098572/pdf/Dis
	iii	Restructuring Paper 2	https://documents1.worldbank.org/curated/en/124631636071392066/pdf/Dis
	iv	Environmental Assessment	https://documents1.worldbank.org/curated/en/742461468112151851/pdf/E47

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15	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/257621468023033600/pdf/PA
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/429681469765722976/pdf/Re
	iii	Environmental and Social Impact Assessment	https://documents1.worldbank.org/curated/en/099055107142225359/pdf/P1
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17	i	Project Information Document	https://documents1.worldbank.org/curated/en/938841468200648366/pdf/PA
	ii	Restructuring Paper (2nd)	https://documents1.worldbank.org/curated/en/758751590576527809/pdf/Dis
	iii	Restructuring Paper (3rd)	https://documents1.worldbank.org/curated/en/427001624431799450/pdf/Dis
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19	i	Project Information Document	https://documents1.worldbank.org/curated/en/158731468114883648/pdf/59
	ii	Environmental Assessment	https://documents1.worldbank.org/curated/en/590431468300543210/pdf/E2
	iii	Environmental Assessment	https://documents1.worldbank.org/curated/en/326131468113052042/pdf/E2
20	i	Project Information Document	https://documents1.worldbank.org/curated/en/633041468291684272/pdf/77
	ii	Integrated Safeguards Data Sheet	https://documents1.worldbank.org/curated/en/174951468302961076/pdf/80
	iii	Resettlement Plan	https://documents1.worldbank.org/curated/en/676341468122984067/pdf/RP
	iv	Environmental Assessment	https://documents1.worldbank.org/curated/en/608631468059934264/pdf/IPP
21	i	Project Information Document	https://documents1.worldbank.org/curated/en/227411468178168566/pdf/77
	ii	Implementation Completion Report Review	https://documents1.worldbank.org/curated/en/099320012062215082/pdf/BO
	iii	Resettlement Plan	https://documents1.worldbank.org/curated/en/528621468334783372/pdf/RP
	iv	Environmental Assessment	https://documents1.worldbank.org/curated/en/340031468350181720/pdf/E4
22	i	Project Information Document	https://documents1.worldbank.org/curated/en/874411468194347530/pdf/61
	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/389581545939779039/pdf/im
	iii	Resettlement Plan	https://documents1.worldbank.org/curated/en/713491468010000275/pdf/RP
	iv	Environmental Assessment	https://documents1.worldbank.org/curated/en/382331467990376750/pdf/E2
23	i	Project Information Document	https://documents1.worldbank.org/curated/en/326761468192866533/pdf/60
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	iii	Environmental Assessment	https://documents1.worldbank.org/curated/en/616731468210283938/pdf/E2
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24	i	Project Information Document	https://documents1.worldbank.org/curated/en/713331468122072134/pdf/58
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	iii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/334301484080235651/pdf/ICR
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	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/890391484078955108/pdf/ICR
	iii	Environmental Assessment	https://documents1.worldbank.org/curated/en/282431468318347508/pdf/E1
35	i	Project Information Document	https://documents1.worldbank.org/curated/en/603251468191674194/pdf/39
	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/141431491945390535/pdf/M2
36	i	Project Information Document	https://documents1.worldbank.org/curated/en/521621468008716086/pdf/36
	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/410221520607911958/pdf/FIN
37	i	Project Information Document	https://documents1.worldbank.org/curated/en/493141468047412318/pdf/32
	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/667841468039265236/pdf/ICR

38	i	Project Information Document	https://documents1.worldbank.org/curated/en/523511468346744205/pdf/316
	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/125911468115479745/pdf/ICR
39	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/413281468329997758/pdf/327
40	i	Project Information Document	https://documents1.worldbank.org/curated/en/335251468002660209/pdf/314
	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/965571468219011886/pdf/ICR
41	i	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/431961468225302550/pdf/ICR
	ii	Resettlement Plan	https://documents1.worldbank.org/curated/en/259511468770393755/text/RP
42	i	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/711941468292477625/pdf/ICR
44	i	Project Information Document	https://documents1.worldbank.org/curated/en/641441566910988408/pdf/Pro
45	i	Project Information Document	https://documents1.worldbank.org/curated/en/550691522548022924/pdf/Ba
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/258731616559897385/pdf/Dis
	iii	Environmental Assessment	https://documents1.worldbank.org/curated/en/653041557737637101/pdf/Init
46	i	Project Information Document	https://documents1.worldbank.org/curated/en/442701468194079362/pdf/89
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/099545105252218581/pdf/P1
	iii	Environmental Assessment	https://documents1.worldbank.org/curated/en/099525004132241631/pdf/P1
47	i	Project Information Document	https://documents1.worldbank.org/curated/en/603941561341688507/pdf/Mc
	ii	Environmental Assessment	https://documents1.worldbank.org/curated/en/837131551069472216/pdf/M2
48	i	Project Information Document	https://documents1.worldbank.org/curated/en/814001531711841488/pdf/GU
	ii	Environmental Assessment	https://documents1.worldbank.org/curated/en/859871526297089294/pdf/Ra
	iii	Environmental Assessment	https://documents1.worldbank.org/curated/en/785601526377951265/pdf/Ra
49	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/482631592619043723/pdf/Car
50	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/158671468281678195/pdf/32
	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/453861468252883220/pdf/ICR
51	i	Project Information Document	https://documents1.worldbank.org/curated/en/296941561687292260/pdf/Tu
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/929641590683093143/pdf/Dis
	iii	Resettlement Plan	https://documents1.worldbank.org/curated/en/704611556012892066/pdf/Car
	iv	Environmental Assessment	https://documents1.worldbank.org/curated/en/441351624451146918/pdf/Etu
52	i	Project Information Document	https://documents1.worldbank.org/curated/en/192371468057551166/pdf/43
53	i	Project Information Document	https://documents1.worldbank.org/curated/en/116431468006606003/pdf/67
	ii	Additional Financing	https://documents1.worldbank.org/curated/en/369271509740134599/pdf/Sie
54	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/713911557626447656/pdf/Mc
	ii	Resettlement Plan	https://documents1.worldbank.org/curated/en/519961545988915384/pdf/Re

	iii	Environmental Assessment	https://documents1.worldbank.org/curated/en/254311545999599347/pdf/ESI
55	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/576611468086049488/pdf/34
	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/755341486394764965/pdf/ICR
56	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/617281468165582712/pdf/32
	ii	Project Paper	https://documents1.worldbank.org/curated/en/329381468304814564/pdf/66
	iii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/297501468305943797/pdf/No
57	i	Implementation Completion and Results Rep	https://documents.worldbank.org/en/publication/documents-reports/docume
	ii	Additional Financing	https://documents1.worldbank.org/curated/en/588201468008760871/pdf/69
	iii	Restructuring Paper	https://documents1.worldbank.org/curated/en/588201468008760871/pdf/69
		Resettlement Plan	https://documents.worldbank.org/en/publication/documents-reports/docume
58	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/914891569031288591/pdf/Mc
		Environmental Assessment	https://documents1.worldbank.org/curated/en/275841559105418449/pdf/En
		Environmental Assessment	https://documents1.worldbank.org/curated/en/856011559194741084/pdf/En
59	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/947121653579297553/pdf/Mi
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/099330002102224513/pdf/P1
60	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/432331541438685373/pdf/no
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/642971597249080139/pdf/Dis
	iii	Restructuring Paper	https://documents1.worldbank.org/curated/en/529791634396135053/pdf/Dis
61	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/503531468047754008/pdf/37
62	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/512271467996734100/pdf/PA
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/099810206302228044/pdf/P1
	iii	Environmental Assessment	https://documents1.worldbank.org/curated/en/553121468302089363/pdf/SFC
63	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/509521641922250371/pdf/Nig
64	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/574001467997861308/pdf/41
	ii	Resettlement Plan	https://documents1.worldbank.org/curated/en/973631468014065661/pdf/RP
65	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/620191468212087246/pdf/69
	ii	Resettlement Plan	https://documents1.worldbank.org/curated/en/325171468198877328/pdf/RP
	iii	environmental Assessment	https://documents1.worldbank.org/curated/en/193031468009293834/pdf/E2
67	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/401371468313491911/pdf/40
	ii	Restructuring Paper	https://documents1.worldbank.org/curated/en/678881468131968475/pdf/61
68	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/155911468273019836/pdf/54
	ii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/328821531497639930/pdf/ICR

69	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/855321468010903834/pdf/53
	ii	Resettlement Plan	https://documents1.worldbank.org/curated/en/172391468231309155/pdf/RP
	iii	Resettlement Plan	https://documents1.worldbank.org/curated/en/930551468010901719/pdf/IPP
	iv	Environmental Assessment	https://documents1.worldbank.org/curated/en/593281468233037841/pdf/E1
70	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/644301468200933603/pdf/47
	ii	Environmental Assessment	https://documents1.worldbank.org/curated/en/971571468200932813/pdf/E2
	iii	Implementation Completion and Results Rep	https://documents1.worldbank.org/curated/en/891701570046002579/pdf/Be
71	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/559731468350143749/pdf/59
72	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/482781468211481560/pdf/39
73	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/232091560442722757/pdf/Esv
	ii	Environmental Assessment	https://documents1.worldbank.org/curated/en/099300008152289240/pdf/P1
	iii	Resettlement Plan	https://documents1.worldbank.org/curated/en/862431556875155858/pdf/Re
74	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/569391623031256913/pdf/Tar
	ii	Environmental Assessment	https://documents1.worldbank.org/curated/en/099140009062257320/pdf/P1
75	i	Project Appraisal Document	https://documents1.worldbank.org/curated/en/372861468281708815/pdf/41

Project Basic Information

Location	Project description
National: 10 regions of Peru that include Ancash, Arequipa, Junín, La Libertad, Piura, Puno, San Martín, Tumbes, Tacna, Ucayali, where monetary poverty rates ranged from 18.6 to 42.6 percent in 2020.	Component 1: Strengthening and expansion of substations and transmission lines (US\$65 million IBRD, US\$24.59 million public distribution companies) a. expansion and strengthening resilience of existing power transformation substations; b. repowering of existing transmission lines within existing rights of way; c. construction of new substations; and d. installation of new transmission lines in urban, peri-urban, and rural areas Component 2: Technical Assistance for Regulatory Strengthening (US\$1 million IBRD, US\$0.5 million MINEM) Component 3: Project management and capacity building (US\$ 4 million IBRD)
Andijan, Bukhara, Fergana, Kashkadarya, Khorezm, Navoi, Samarkand, Surkhandarya, Syrdarya, Tashkent regions, and Tashkent City.	Component 1: Digitalization of the electricity transmission sector (US\$115 million IDA credit) Component 2: Power grid strengthening and renewable energy integration (US\$239 million IDA credit and US\$43 million GCF loan) Component 3: NEGU institutional development and project implementation support (US\$16 million IDA credit and US\$4 million GCF grant)
Substation is built around farm land.	Component 4: Electricity Market Development (US\$10 million IDA credit)

The line route (Dasu-Mansehra-Islamabad) is dominated by high mountains	Component D1 (US\$654 million): Constructing a double circuit 765 kV transmission line from Dasu to Islamabad and a substation at Mansehra, to evacuate power from the Dasu hydropower plant. <i>Note: Under the revised plan the cost of the TL is US\$797.1 million, including financing cost. Financing will be provided by the original IDA credit (US\$15 million), the IBRD loan (US\$700 million), and the NTDC financing (US\$82.1 million). (ii)</i> Component D2 (US\$25 million): Supporting project management, technical assistance, training strategic studies, additional systems studies, and analytical work, pilot projects, preparation of future investments; and capacity building for NTDC as agreed with the Bank. Component D3 (US\$36.3 million): Implementing the environmental and social measures required under Safeguard Instruments for the Project
Khyber Pakhtunkhwa Province (KP), Pakistan, with unsecure environment	Component A: Construction of High Voltage Transmission Infrastructure (This project provides additional funding to CASA-1000 project. The AF will finance the financing gaps for the committed contracts for the Nowshera HVDC converter station under the CASA-1000 while the IsDB will finance the HVDC transmission line from Afghanistan border to the Nowshera converter station.) Component B: Technical Assistance and Project Implementation Support. Component C: Community Support Programs (CSPs) (i)

Transmission lines from Iringa (Tanzania) to Kabwe (Zambia), with an additional spur in Tanzania between Tunduma and Sumbawanga Traverses a wildlife migration corridor	Component 1: Transmission Infrastructure Extension (Estimated cost: US\$465 million) Component 2: TANESCO Transmission System Readiness for Interconnection and Corporate Commercial Management Improvements (Estimated cost: IDA Credit - US\$99 million) Component 3: Project Implementation Support and Capacity Building (Estimated cost: IDA Credit -US\$21 million equivalent) Component 4: Technical Assistance to EAPP (IDA grant - US\$10 million equivalent)
National	Component 1: Transmission Network Strengthening and Improvement (US\$408 million) (a) the upgrading and rehabilitation of up to 48 existing substations (US\$244.5M) (b) replacement of conductor on up to 13 132 kV transmission lines and conversion of up to two 132 kV lines from single circuit to double circuit (the total length of the transmission lines is about 1,260 km) (US\$49.5M); (c) the upgrading and expansion of the network's SCADA and telecommunication systems and construction of four regional control centers (US\$65M); and (d) installation of static var compensator (SVC) at the remotely located Gombe substation (US\$45M) Component 2: Capacity Building and Technical Assistance (US\$32 million equivalent, excluding contingency)

National (in various urban, peri-urban and rural areas. A large number of subprojects are located in Punjab Province and in the Islamabad Capital Territory and the remaining subprojects are located in the provinces of Sindh, Khyber Pakhtunkhwa (KP) and Balochistan. The majority of the sub stations will be located near the load centers, which are urban and peri- urban areas)	Component A: Expansion and Upgradation of the Transmission Network (estimated cost about US\$455.89 million) (a) expanding, augmenting, and upgrading selected existing 500 kV and 220 kV power substations and associated lines; and (b) constructing new 765 kV, 500 kV, and 220 kV substations and transmission lines. Component B: Deployment of the Enterprise Resource Planning (ERP) System (estimated cost US\$29.46 million) Component C: Project Management, Technical Assistance, and Capacity Building (estimated cost US\$20 million) (i)
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Kayes (Mali) to Tambacounda (Senegal): mainly of scarce woody savannah; 4 protected forest areas in Senegal near Tambacounda could not be avoided. Forest losses will be compensated (200ha) The line passes for 17.9 km through the Goudiry Protected Forest, for 12.5 km through the Bala East, 12.8 km through the Bala West, and 5.9 km through the Botou Protected Forests. About 200 ha of forest will be lost.

Component 1: RIMA Reinforcement and Expansion (US\$94 million);
Subcomponent 1.1: Kayes-Tambacounda Transmission Line (US\$83.6 million)
Subcomponent 1.2: SCADA System (US\$10.4 million)
Component 2: Technical Assistance (US\$8 million)

urban, peri-urban and rural areas in the entire country	Component 1: Reinforcement of Transmission Systems (US\$115.6 million) (i) the construction of a new substation at Gagnoa and the related transmission and distribution lines (US\$14.1 million); (ii) the upgrade of three 90kV substations to 225kV at Youpougou1, Bia-Sud, and Treichville (US\$44.3 million); and (iii) the improvement of reliability of supply in 15 substations (US\$57.2 million). Component 2: Rehabilitation, Reinforcement, and Extension of Distribution Systems in Abidjan and 10 Regional Capital Cities (US\$ 95.4 million) Component 3: Rural Electrification and Support to Electricity for All Program (US\$ 100.6 million) Component 4: Strengthening Institutional Capacity of the Electricity Sector and Project Management (US\$ 13.4 million)
National	Component 1: Provision of support towards the operationalization of SONATREL's transmission activities (US\$30 million) Component 2: Strengthening of the National Transmission Grid (US\$335 million) Sub-component 2(a): Strengthening of Douala's Transmission System Sub-component 2(b): Strengthening of Yaoundé's Transmission System Sub-component 2(c): Strengthening and stabilization of the Three National Transmission Networks (South, East, and North) and their Interconnections Sub-component 2(d): Provision of "Smart Grid" Investment Package in the Transmission system Sub-component 2(e): Safeguard Activities Component 3: Project Management Support and Capacity Building (US\$10 million).

high priority substations located in Andijan, Bukhara, Jizzakh, Kashkadarya, Navoiy, Surkhandarya, Samarkand, Fergana, Namangan Regions and Tashkent City and Region	Component 1. Electricity Transmission System Upgrade (rehabilitation, upgrade, and expansion of existing substations of 220 kV and higher voltage) (US\$191.3 million) Component 2: Institutional Development and Technical Assistance (US\$5 million)
Yerevan, Aragatsotni Marz	Component 1: Strengthening of the power transmission network (US\$60.04 million) Sub-component 1.1: Rehabilitation of Ashnak substation of HVEN (US\$20.53 million) Sub-component 1.2: Rehabilitation of YTPC's substation (US\$35.88 million) Sub-component 1.3: Purchase of Machinery and Equipment for Maintenance of the Power Transmission Network (US\$3.25 million) Sub-component 1.4: Project implementation support to HVEN substation (US\$187,500) Sub-component 1.5: Project implementation support to YTPC substation (US\$187,500) Component 2: Improvement of the power system management (US\$3.13 million) (back-up dispatch center) Component 3: Preparation of a new electricity generation project (US\$1.69 million)

National	Component 1: Rehabilitation of Transmission Substations (US\$268 million) (Novokyivska, October, Kremenchug, Zhytomyrska, Cherkassy, Sumy) Component 2: Electricity Market Enhancement (US\$107.925 million) Component 3: Improving of Institutional Capacity of MoECI for Reform Implementation (US\$2.50 million)
Greater Hanoi Area, the Greater Ho Chi Minh City Area/the Mekong Delta, and Central Region	Component 1: Enhancing Transmission Infrastructure (USD 625 million) Component 2: Developing Smart Grid Network: (USD 80 million) Component 3: Capacity Building (USD 25 million)

National	Component 1: Transmission System Strengthening (US\$52.35 million) Component 2: Wholesale Power Exchange Platform (US\$6.00 million) Component 3: Electricity Sector Strategic Environmental and Social Assessment (US\$1.00 million) Component 4: Project Management and Transmission Systems Studies (US\$0.50 million)
Cross nation	Component A. Construction of High Voltage Transmission Infrastructure (US\$945 million) Sub-component A1 – High Voltage Direct Current (HVDC) Transmission Line (US\$295 million) Sub-Component A2 – HVDC Converter Stations (US\$385 million) Sub-component A3 – High Voltage AC Transmission (HVAC) Interconnection between the Kyrgyz Republic and Tajikistan (US\$200 million) Sub-component A4 – Tajikistan grid reinforcement (US\$65 million) Component B. Technical Assistance and Project Implementation Support (US\$30 million) Component C. Community Support Programs (US\$70 million)

National	Component A: Rural Grid Augmentation and Rehabilitation (US\$671 million) - (Distribution - total 6,249 km) Component B: Transmission enhancement (US\$157 million) Component C: Institutional strengthening (US\$9 million)
Java-Bali, East Indonesia and West Indonesia	- Extension and rehabilitation of selected existing 150/20 kV substations and 70/20 kV substations, including adding one or more new transformers and associated equipment; and/or replacing existing transformers with new transformers and associated equipment with higher capacity and - Construction of selected new 150/20 kV substations, including installation of transformers and associated equipment. (i) Original: 325 million

Cross nation	Original: Total: \$202.3 million - Component A1: Muzaffarpur-Sursand Transmission Line - 90km 400kv double circuit (\$29.7 million) - Component A2: Dhalkebar-Bhittamod Transmission Line - 40km 400kv double circuit (\$17.9 million) - Component B1: H-D-D Transmission Line & Substations (including conductors) - 285 km 220kv double circuit (\$108.6 million) - Component B2: Measures for synchronization of the Nepal grid with the Indian grid (\$10.0 million) - Part C: Technical Advisory Services (\$11.0 million) Contingencies (\$15 million) + Interest During Construction (\$10 million)
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Cross nation	This project provides additional funding to a larger project connecting India and Nepal. Additional: Total: \$39 million 1. Conductor for H-D-D 400 kV transmission Line (\$23 million). 2. Hetauda-Bharatpur and Bharatpur-Bardaghat (70 km) 220 kV Lines and Associated Substations (\$10.47 million). 3. System Integrator for IFMIS of NEA (\$3.0 million) 4. Contingency (\$2.53 million)
Lusaka	Component 1: Rehabilitation of the 132kV and 88kV Transmission Network in Lusaka Area (US\$106 million) (3 new + 2 substation upgrades + replacement of 11kV switchgear in various locations) Component 2: Rehabilitation of the 33kV and 11kV Distribution Network in Lusaka Area (US\$94 million) Component 3: Technical Assistance and Project Supervision (US\$10 million)

Southern Region of Zambia, between Kafue West and Victoria Falls in Livingstone	Component 1: Upgrading of the existing 220kV transmission line (US\$24 million) Component 2: Reinforcement and Construction of Substations (US\$71 million) Component 3: Project Management, Environmental and Social Mitigation, and Training (US\$5 million)
Bolgatanga in the northern part of Ghana to the capital of Burkina Faso	Component 1: 225 kV Transmission Line between Bolgatanga, Ghana, and Ouagadougou, Burkina Faso ((US\$84.2 million) • Construction of 210 km of 225 kV transmission line; • Construction of a new 225/161 kV substation in Bolgatanga, Ghana; • Extension of the 225/90 kV Zagatouli substation and construction of a new 90/33 kV substation in Burkina Faso; • Installation of telecommunications systems and SCADA; • Implementation of Environmental Management and Resettlement Action Plan and strengthening the management of the Kaboré Tambi National Park Component 2: Reinforcement of the transmission grid in Ghana (US\$6 million) Component 3: Electrification of rural localities along the right of way in Burkina (US\$9 million) Component 4: Supervision/Owner's Engineer (US\$4.2 million) Component 5: Capacity building and institutional support to GRIDCO and SONABEL for project implementation (US\$7.3 million)

Eastern Nepal	Component 1. Kabeli Corridor 132 kV Transmission Line Component (IDA funding \$32.24) Component 2. Community-based Rural Electrification-Grid Extension Component (IDA funding \$4.36 million) Component 3. Rural Enhanced Energy Services Component (IDA funding \$1.40 million). Total: 47.68 million
	This project provides additional funding of US\$180.0 million to the project P084871. Additional financing has been reflected.
The new 500 kV substation and the interconnection of Talimarjan generation facilities with Karakul and Gujjar are located on the boundaries of the Talimarjan thermal power plant (TPP). The new power transmission line will start from Talimarjan TPP to South # East of Uzbekistan, passing the territory of Samarkand (87 km) and Kashkadarya (131 km) counties (oblast)	Component 1: Strengthening of Power Transmission Network (US\$151.04 million), including construction of (a) about 220 km single circuit 500 kV transmission line from Talimarjan TPP to Sogdiana substation; (b) 500/220 kV open switch-yard at Talimarjan TPP; (c) a bay extension at Sogdiana substation; and (d) a 500 kV connection line from the 500/220 kV open switch-yard at Talimarjan thermal power plant to the Karakul-Guzar transmission line. Component 2: Institutional Strengthening of UE through technical assistance (US\$ 10 million)

National	Part 1 - Construction of Transmission System Lines (\$105.41 million) (a) a 500 kV overhead transmission line (OHTL) from Alma substation to Almaty substation; (b) a 220 kV OHTL from Alma substation to Kensai substation; (c) incoming cross connection lines of the 220 kV OHTL from Almaty CHP-3 to Shelek substation for connection with Alma substation; (d) the incoming cross connection lines of the 220 kV OHTL from Almaty CHP-3 to Robot substation for connection with Alma substation; and (e) a 500 kV OHTL from Alma substation to YuKGRES substation. Part 2 - Construction, Extension and Modernization of Substations (\$80.07 million) (a) construction of a new 500 kV Alma substation and extension and modernization of the 500 kV Almaty substation; and (b) extension and modernization of YuKGRES substation. Part 3 –Consulting and Technical Services (\$3.2 million) Total: \$200.9 million
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Iringa and Shinyanga (Central of Tanzania)	Component 1: Construction of Backbone Transmission Line between Iringa and Shinyanga (US\$ 434.77 million) Component 1(a). Construction of OHL section (US\$398.71 million) Component 1(b): Upgrade of 220kV substations at Iringa, Dodoma, Singida and Shinyanga (US\$ 36.06 million) Component 2: Technical Assistance to TANESCO (US\$10 million)
National	1. Expansion and Construction of Transmission Substations in the Java-Bali System 1.1. Expansion of 500/150 kV substations in Java (US\$48.6 million) 1.2. 150 kV Millennium substation in West Java (US\$9.9 million) 1.3. Expansion of 150 kV substations in Central Java (US\$28.8 million) 2. Upgrading of and Expansion of Transmission Substations 2.1. Upgrading of 275 kV substations in South-Central Sumatra (US\$ 101.2 million) 2.2. Expansion of 150/20 kV substations in South-Central in South-Central Sumatra System (US\$25.8 million) Project total: \$254.1 million

Eastern Botswana	Component A(1)-Morupule B Power Station (estimated cost US\$1,211.3 million) Component A(2)-transmission system (estimated cost US\$275.2 million) Component A(3)-water supply system (estimated cost US\$53.0 million) Component B-alternative energy development: This component (to be implemented by MMEWR and estimated cost US\$6.8 million) Component C-institution and capacity building: This component (estimated cost US\$13.9 million)
National	A1 - Construction of OHTL from MHPP to Shelek SS. This involves the construction of about 97 km long 220 kV single-circuit OHTL. The design includes ACSR, horizontal phase arrangement, anchor angle steel towers, ground wire lightning protection, and high frequency carrier communication. A2 - Construction of OHTL from MHPP to Robot SS. 225 km; 220 kV B1 - Construction of 220/110 kV OSY at MHPP. B2 - Modernization of 220 kV Robot SS B3 - Modernization of 220 kV Shelek SS Component C - Consulting and Technical Services
National	A. Strengthening distribution networks (US\$253.4 million) B. Transmission network (US\$24.4 million): construction of a new 2x160 MVA, 220-kV substation at Kassowal, with associated 220-kV and 132-kV lines. Construction of the 132-kV lines from the Kassowal substation will be the responsibility of MEPCO, on whose service territory the substation is located. C. Technical Assistance and Capacity Building (US\$16.6 million) D. Energy Efficiency (US\$15.0 million)

National	Component A: Rehabilitation of transmission substations (US\$ 157.8 million) 10 substations Component B: Strengthening of transmission network (US\$ 47.0 million). This component includes expansion of the existing 330 kV Bar substation and construction of the 77 km long 330 kV transmission line Component C: Stabilization of the Crimea Electric Power Grid (US\$ 23.2 million). This component includes upgrading of the existing 220 kV Simferopol - Sebastopol transmission line and associated substations to the 330 kV voltage level, and upgrading of substation control systems at 330 kV and 220 kV substations in the Crimea electric power grid. Component D: UE Institutional development (US\$ 8.7 million) Component E: Implementation of the Grid Code (US\$ 1.3 million)
Cross nation	(A) Mozambique - Malawi Interconnection; (\$90.6 million : US\$47.1 million for Malawi, US\$43.5 million for Mozambique): Malawi 75km + Mozambique 135 km (B) Capacity Building and Technical Support for Upgrade and Expansion to Support Power Trading; (US\$4.6 million; US\$2.9 million for Malawi, US\$1.7 million for Mozambique) (C) Improved Infrastructure to Support Power Trading.(US\$14.5 million; US\$9.9 million for Malawi, US\$4.6 million for Mozambique)

Cross nation	Component 1 : Transmission Infrastructure Development \$34.55 million 1.1 Sub-station Expansion Works at Volta and Akosombo 1.2 Installation of Third Bulk Supply Point for the Accra/Tema Load Centers 1.3 Upgrade of Switchgear at the Kpong Generation Station 1.4 Engineering Consulting Services and Logistics Support Component 2: Upgrade of Transmission Control and Communications System - \$15.29 million Component 3: Upgrade of Strategic Power Generation Stations - \$6.4 million Component 4: Implementation of “WAPP Action Plans” - \$7.47 million
North-south	Component A. Construction of 500 kV Overhead Single Circuit Transmission Line (estimated cost: US\$113.8 million) Component B. Extension/Modernization of 1,150/500 kV Ekibastuz Air-Insulated Substation (estimated cost: US\$13.2 million) Component C. Extension/Modernization of 500 kV Agadyr Air-Insulated Substation (estimated cost: US\$32.7 million) Component D. Technical Assistance (US\$0.9 million)

Southern part	Component 1: Transmission and Distribution Systems Expansion and Reinforcement (\$297.75 million) 1.1 500 and 220kV Transmission System Expansion and Reinforcement, total cost \$237.37 million 1.2 220/110kV Gas-insulated switchgear (GIS) substations at\$48.64 million 1.3 Regional Distribution Systems Expansion and Reinforcement totaling \$11.75 million Component 2: Electricih, Market and System Operations Modernization (\$22.80 million) Component 3: Market Transition Support (\$1.65 million)
Cross nation	Component 1: Infrastructure Development A.1 330 kV Transmission Line from Aboadze to Volta A.2 330 kV substations at Aboadze and Volta A.3 Reinforcement and extension of the SCADA and communication system A.4 Environmental Impact and Mitigation Measures A.5 Upgrade of the Volta System Control Center (Tema) A.6 Upgrade of Switchgear at the Akosombo Generation Station of VRA A.7 Upgrade of Switchgear at Volta Sub-station of VRA Component 2: WAPP APL 1 Project Implementation Component 3: Technical Assistance for “WAPP Action Plan
Baku and other places	Component A: Power System Management Component B: Transmission Network Rehabilitation Component C: Management Assistance (US\$3.6 million) Component D: Project Implementation (US\$0.8 million) Total: \$55.41 million

National	Component A. The Transmission Line (TL) (Total cost US\$90.59 million) (a) the construction of a 109 km double circuit 230 kV line from the border with Vietnam to Phnom Penh and two associated substations; (b) reinforcement of the 115 kV grid around Phnom Penh involving about 20 km of 115 kV lines and modifications to three 115 kV substations and 22 kV extension; (c) the establishment of a National Control Center (NCC); and (d) building EDC's capacity Component B. Rural Electrification (Total cost US\$14.74 million) Component C. Pilot Rural Electrification Fund (Total cost 28.23 million) Component D. Institutional Development and Sector Reform (Total cost US\$9.33 million)
National	(a) Investment Component for enhancing transmission and system capacity (about US\$86 million); (b) Technical Assistance (TA) Component for establishing and developing TransysCo (about US\$12 million); (c) Social and Environmental Component for developing environmental institutions (about US\$2 million).
	Project dropped

	Component 1: Preparation of environmental and social documents required for Rural Electrification, Sebzor HPP, and Khorog-Qozideh power transmission line projects consistent with requirements of financiers. Component 2: Update of the feasibility study for Sebzor HPP to address specific technical issues of the project design and review of certain technical assumptions. Component 3: Completion of feasibility study for Khorog-Qozideh (close to Afghan border) power transmission line. Component 4: Disaster risk screening and mitigation assessment for PEC energy assets. Component 5: Preparation of bidding documents for construction of Khorog-Qozideh power transmission line. Component 6: Capacity building for PEC in construction, operation, and maintenance of renewable energy based micro-grids.
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National	Component 1: Enhancement and Strengthening of Power Network (US\$686.6 million) 1.1 New Substations. Thirteen new gas insulated substations (GISs) (one upgrade to 400/230/33 kV substation from 230/33 kV substation, one new 400/230/132 kV substation, two new 230/132 kV substations, and nine new 132/33kV substations) 1.2 Transmission Lines: One 230 kV high-capacity four circuit backbone transmission line (with twin Finch conductor per phase) through greater the Comilla region and four short distance 132 kV double circuit lines to connect the new 400/230 kV, 230/132 kV, and 132/33 kV substations to the existing ones in Comilla and Noakhali areas; 1.3 Replacement of substation and reconductoring: Replacement of the Halishahar 132/33 kV air insulated substation (AIS) by an advanced GIS and reconductoring of existing Sikalbaha- Cox's Bazar 132 kV line and Korerhat-Feni 132 kV line with higher capacity conductor. Component 2: Institutional Development and Implementation Support (US\$31.46 million)
Cross nation	Component 1: Extension of WAPP transmission network – OMVG Interconnection (Component cost US\$586 million) Sub-Component 1.1: Transmission lines (sub-component cost: US\$397 million) Sub-Component 1.2: Substations (sub-component cost: US\$189 million) Component 2: Technical Assistance to OMVG (Component cost US\$112 million) Sub-Component 2.1: Implementation Support (sub-component cost: US\$92 million) Sub-Component 2.2: Operations and Maintenance support to OMVG Transmission Company (sub-component cost: US\$20 million)

National	Component 1: Temane Transmission Project (Estimated cost: US\$533 million equivalent) Component 2: CTT Power Generation Plant (Estimated cost: US\$750 million) Component 3: Implementation Support, Technical Assistance and Capacity Building (Estimated cost US\$23 million)
Cross nation	Component 1: Power Transmission Infrastructure (US\$343.8 million equivalent) Sub-component 1-A: Construction of the Transmission Interconnector (US\$317.3 million equivalent) Sub-component 1-B: Implementation of the Environmental and Social Management Plans and Resettlement Action Plans (US\$26.6 million equivalent) Component 2: Implementation Support and Capacity Building (US\$37.0 million)
Cross nation	Component 1: Cameroon RIS-RIN Interconnection (US\$375 million equivalent) Sub-component 1.A: Construction of RIS-RIN Interconnection (US\$266 million) Sub-component 1.B: Project implementation support to SONATREL (US\$24 million) Sub-component 1.C: Implementation of the ESIA(s), ESMP(s) and RAP(s) (US\$83.5 million) Sub-component 1.D: Technical assistance for the negotiation of contracts for power trade and for commercialization of optical fiber (US\$1.5 million) Component 2: Cameroon-Chad Interconnection (US\$328 million equivalent) Component 3: Electricity access in Chad (US\$41 million equivalent)

Cross nation

Component 1: Construction of transmission interconnection between the towns of Bahir Dar and Metema (at the Ethiopia/Sudan border). (US\$39.55 million)

1.1 Transmission lines construction and expansion - US\$27.40 million

1.2 Extension and rehabilitation of substations - \$11.15 million

1.3 telecommunications system - \$1 million

Component 2: Institutional Strengthening and Capacity Building for Regional Development (US\$3.55 million)

National	Component 1: Strengthening the electricity transmission network (US\$131 million): Construction of approximately 384 km of high-voltage transmission lines, Construction, extension and/or reinforcement of the following substations: (i) construction of a 400/225 kV substation in Kondar to connect the 400 kV line with the existing 225 kV central corridor; and (ii) the extension of the existing substations at Thyna and Tataouine Component 2: Commercial performance improvement (US\$20 million) (i)
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National	Component 1. Transmission and distribution network (loan amount \$123 million) a) Construction of 392km of 400kV lines and 76 km of 225kV lines b) A 400/225 kV substation in Chamaia (2~450MVA) c) Expansion of substations at Mediouna (400/225 kV) and Tensift I1 (225/60kV). d) distribution network Component 2: Compact Fluorescent Lamps (CFL)-(loan amount: \$ 8.4 million) Component 3: Trading Desk (loan amount: \$ 7 million) Component 4: Wind data base (loan amount: \$ 1 million) Component 5: Technical Assistance Components (\$2.4 million)
Cross nation	Component 1.A: Power interconnection between Côte d'Ivoire, Liberia, Sierra Leone, and Guinea (US\$321.9 million) Sub-Component 1.A-1: Transmission and Distribution Interconnection (US\$193 million excluding contingencies, taxes and duties) Sub-Component 1.A-2: Substations Sub-Component 1.A-3: Supervisory Control and Data Acquisition (SCADA) Sub-Component 1.A-4: Installation and Supply of Compensation and Frequency Regulation Equipment Sub-Component 1.A-5: Implementation of the Environmental and Social Management Plans (ESMPs) and Resettlement Action Plansz Component 1.B: Institutional framework and project oversight (US\$45.4 million) Component 2.A: Supply Alternatives Studies and Project Preparation Support (US\$10 million) Component 2.B: Technical Assistance and Integration of WAPP network (US\$21.5 million)

North-south

Component 1: Construction of Vulcanesti-Chisinau overhead line (US\$46.5 million).
Component 2: Extension of two existing substations (US\$13.8 mln)
Component 3: Strengthening of power dispatch and metering system (US\$7.2 million)
Component 4: Technical Assistance and Project Management (US\$2.5 million)

Cross nation

Component 1: Expansion of Skopje 5 Substation (\$3.5 million)
Component 2: Rehabilitation and construction of interconnection and overhead lines (\$19.76 million)
Component 3: Upgrading of the existing EMS System and Planning Software (\$0.55 million)
Component 4: Upgrading and rehabilitation of 110kV substations (\$14.60 million)
Component 5: Support for institutional development (\$1.42 million)
Component 6: Office Equipment and Training for PIU (\$0.024 million)

Component 1: 110 kV Substations and Related Activities
Component 2: 110 kV Interconnecting Transmission Lines and Related Activities

Component 1: Support to the Southern African Power Pool (SAPP) Coordination Center (appraisal: US\$3.36 million; actual: US\$3.36 million).

Component 2: An Increase of the Capacity of the Transmission Connection between the Democratic Republic of Congo (DRC) and the SAPP through Zambia (appraisal: US\$186.11 million; actual: US\$593.80 million)

Component 3: An Increase of the Capacity of the Transmission Connection between the Zambian Border with the DRC and Luano Substation near Kitwe in Zambia (appraisal: US\$9.70 million; actual: US\$20 million).

Component 4: A Feasibility and Preliminary Design Study of an Interconnector between Zambia and Tanzania that would connect the SAPP to the market that would be formed by interconnecting Tanzania, Uganda and Kenya (appraisal: US\$1.02 million; actual: US\$1.02 million).

Component 1: Mozambique-Malawi Power Transmission Infrastructure (US\$127 million equivalent, of which IDA grant is US\$41 million, and IDA credit is US\$13 million)

Component 2: Technical Assistance and Capacity Building (US\$3 million equivalent, of which IDA credit of US\$2 million and IDA grant of US\$1 million)

Cross nation

Cross nation

Cross nation

Component 1: Transmission line from Galafi to Nagad substation and Nagad substation extension (estimated cost: US\$48.76 million, IDA credit: US\$41 million; AfDB: US\$7.76 million)

Subcomponent 1.1: Transmission line from Galafi to Nagad substation (cost: US\$41 million, IDA credit: US\$41 million; AfDB: US\$0 million).

Subcomponent 1.2: Nagad substation extension (estimated cost: US\$7.76 million, IDA credit: US\$0 million; AfDB: US\$7.76 million - parallel financing from the AfDB)

Component 2: Project Management, Owner's Engineer, Incremental Operating Costs and Technical Assistance (estimated cost: US\$9.18 million, IDA credit: US\$5.0 million; AfDB: US\$4.18 million)

Component 3: Contingency Emergency Response Component (estimated cost: US\$0 million, IDA credit: US\$0 million)

Cross nation

Component 1: Power Interconnection between Nigeria, Niger, Benin and Burkina Faso (US\$500.0 million equivalent, of which IDA Credit US\$370.3 million; AfDB US\$95.6 million; AFD US\$34.1 million).

Sub-component 1.A: Construction of Transmission Lines (US\$322.0 million equivalent, of which IDA US\$278.8 million; the AfDB US\$43.1 million)

Sub-component 1.B: Construction and Extension of Sub-stations (US\$168.7 million equivalent, of which IDA US\$87.2 million; AfDB US\$49.9 million; AFD US\$31.5 million)

Sub-component 1.C: Expansion and/or Rehabilitation of Dispatching Centers (US\$9.3 million equivalent, of which IDA US\$4.2 million; AfDB US\$2.5 million; AFD US\$2.5 million)

Component 2: Rural Electrification in Burkina Faso linked to the North Core Regional Interconnector (US\$69.5 million equivalent, of which IDA US\$53.4 million; EU US\$16.1 million)

Component 3: Institutional Framework and Project Oversight (US\$60.8 million equivalent, of which IDA US\$35.1 million; AfDB US\$20.8 million; Government of Nigeria US\$4.9 million)

Component 4: Strengthening Institutional Capacity for Regional Electricity Trade (US\$10.0 million equivalent, of which IDA US\$6.7 million; AfDB US\$3.3 million)

Component 1: Development of cross-border 115 kV Transmission Line linking Lao PDR (Ban Hat) and Cambodia (Stung Treng) to facilitate import of lower cost electricity from Lao PDR to displace higher cost diesel-fueled generation sources in Cambodia.

Component 2: Development of cross-border 115 kV Transmission Line linking Vietnam (Tan Bien) and Cambodia (Kampong Cham) for bilateral power exchanges, beginning with imports of lower cost electricity from Vietnam to displace diesel-fueled generation in Cambodia, and leading over the long term to the transfer of surplus hydropower from Cambodia to Vietnam.

Component 3: Development of the Xeset 1 to Saravan segment of the proposed 115 kV Transmission Link that would eventually interconnect electricity grids from (western) Thailand through (southern) Laos to (southern) Cambodia.

Component 4: Development (feasibility, design, building, commissioning) of a modern Load Dispatch Center (LDC) for Lao PDR to facilitate regional power trade

Component A: Construction of Transmission Infrastructure (Estimated Cost US\$106 million, of which IDA US\$80 million Equivalent and GoU US\$26 million)

Subcomponent A-1: Transmission Line (LGNA)

Sub-component A-2: Substation Works

Subcomponent A-3: Engineering Construction and Supervision Consultant

Component B: Project Implementation and Operational Support to UETCL (Estimated Cost US\$11.8 million, of which IDA US\$11 million Equivalent and GoU US\$0.8 million)

Component C: Sectoral Strengthening Support (Estimated Cost US\$4 million, of which IDA US\$3.5 million Equivalent and GoU US\$0.5 million)

Component 1. Reinforcement and expansion of transmission and distribution networks to increase access to grid electricity services (US\$195 million equivalent, of which IDA US\$195 million)

Subcomponent 1.1. Construction of the transmission interconnector between Western grid and the CenterEastern grid (US\$90 million equivalent, of which IDA US\$90 million)

Subcomponent 1.2. Increasing access to grid electricity through densification, reinforcement of existing systems and construction of new distribution networks (US\$85 million equivalent, of which IDA US\$85 million)

Subcomponent 1.3. Construction of service connections and support to productive uses of energy for new electricity consumers (US\$20 million equivalent, of which IDA US\$20 million)

Component 2. Reinforcing the ecosystem for the development of solar PV powered mini grids in rural areas (US\$40 million equivalent, of which IDA US\$40 million)

Component 3. Increasing access to solar off-grid electricity for public institutions, households and productive uses of energy and clean and efficient cooking solutions (US\$50.0 million equivalent, of which IDA US\$42.5 million and ESMAP MDTF US\$7.5 million)

Component 4. Utility operational performance improvement, institutional strengthening, technical assistance and implementation support (US\$32.5 million equivalent, of which IDA US\$32.5 million)

Component (a) - 300 MW gas turbine power station (US\$ 277.9 million equivalent)

Component (b) – Transmission evacuation component (US\$ 53.9 million equivalent)

Component (c) – Gas supply pipeline (US\$ 104.3 million equivalent)
Technical assistance - 33.9 million

Cross nation

Component A. Construction of a High Voltage Direct Current (HVDC)
Transmission Interconnection between Ethiopia and Kenya
Component B. Project Management and Capacity Building

Component 1: Strengthening of the power transmission network
(€57.80 million, including €56.05 million IBRD loan and €1.75 million
Government co-financing)
Sub-component 1.1: Construction of Jvari-Tskaltubo 500 kV power
transmission line and 500/220 kV Tskaltubo substation (€56.05
million, including €54.3 million IBRD loan and €1.75 million
Government co-financing for land acquisition and resettlement)
Sub-component 1.2: Project implementation support (estimated cost
of €1.75 million)
Component 2: Support to financial recovery of GSE and preparatory
work to access the capital markets (€5.79 million to be financed by
IBRD).

Component A: Ongrid Investments (US\$85.8 million)
Component B: Offgrid Investment Framework and Pilot Investments
(US\$16 million IDA, US\$6.5 million GEF)
Component C: Sectoral Technical Assistance (US\$3.2 million)

Component A: Geothermal Generation (US\$1,035 million: IDA, US\$120 million; JICA, US\$323 million; AFD, US\$210 million; EIB, US\$168 million; KfW, US\$84 million; KenGen, US\$ 130 million).

Component B: Transmission (US\$72.5 million of which IDA US\$64.5 million; and KETRACO US\$8 million)

Component C: Distribution (US\$272 million: IDA, US\$134 million; AFD, US\$10 million; KPLC, US\$30 million, GPOBA, US\$5 million, REA US\$2 million; users US\$91 million)

Component D: Sector Institutional Development and Operational Support (US\$11.5 million).

(i) the development, construction, and operation of a new 216 (nameplate) MW natural gas-fired power plant (with LFO as alternative fuel) including auxiliary equipment and

(ii) the development and construction of a new 100-km 225-kV double-circuit transmission line between the Kribi power plant and the existing Mangombe 225/90-kV substation at Edéa in Littoral Province, including substations, transformers, and auxiliary equipment.

Component A: Electrical Network Upgrading (US\$98.32 million, of which US\$43.32 million from IDA).

A1: Construction of 161 kV Interconnection between Onigbolo and Parakou (IDA-, EIB-, and CEB-financed)

A2: Construction of 161 kV Interconnection between SakCtC and Porto-Novo (KfW and CEB-financed)

A3: Associated Supervision Consulting Services (IDA-financed)

A4: Rehabilitation and Reinforcement of the SBEE network (IDA and SBEE-financed)

A5: Improving Lighting and Appliance Efficiency (GEF-financed).

Component B: Electrification and Modern Energy Services (US\$65.50 million, of which US\$12.00 million from IDA)

Component C: Sustainable Energy Services (US\$6.68 million, all of which IDA-financed)

Component A - Construction of a 220 kV Kawanda-Masaka
Transmission Line: 128,281,202
Component B – Technical Assistance to UETCL: US\$7.6 million.
Component C – Community Support Projects and Technical
Assistance to MEMD: US\$11.7 million

Component 1 : Generation (US\$ 226.7 million)
Component 2: Transmission (US\$ 93.8 million)
Component 3: Distribution (US\$ 88.5 million):
Component 4: Capacity Building and Governance (US\$ 41.2 million):
Component 5: Project Execution (US\$ 48.8 million):

Component 1: Reinforcement of the Transmission and Distribution
Grid - Euro 27.7 million (US\$31 million equivalent) of which IBRD Euro
25 million
Component 2: Electricity Access Expansion - Euro 10.7 million (US\$12
million equivalent) of which IBRD Euro 8.9 million
Component 3: Analytical Support and Capacity Building - Euro 1.8
million (US\$2 million equivalent) of which IBRD Euro 1.8 million
Component 4: Contingent Emergency Response Component (EUR 0
million: to be capitalized in the event of an emergency)

Island

Component 1. Renewable Energy and Storage Infrastructure Development (US\$36.5 million: US\$16.5 million IDA equivalent, US\$18 million CTF loan, US\$2 million CTF grant)
Component 2. Grid modernization and access scale-up (US\$86 million IDA equivalent)
Sub-component 2.1. 132kV Backbone Transmission Infrastructure (US\$44 million IDA equivalent).
Component 3. Sector Institutional Strengthening and Project Implementation Support (US\$14 million: US\$13 million IDA equivalent and US\$1 million CTF)

Cross nation

Component 1: Construction of transmission interconnection between the towns of Bahir Dar and Metema (at the Ethiopia/Sudan border).
Component 2: Institutional Strengthening and Capacity Building for Regional Development (US\$3.55 million)

[Peru-Transmission-Investment-Plan-PIT-to-Support-Post-COVID-19-Green-Economic-Recovery-in-Peru.pdf](#)
[Pakistan-Electricity-Sector-Transformation-and-Resilient-Transmission-Project.pdf](#)
[Environmental-and-Social-Management-Plan-ESMP-Electricity-Sector-Transformation-and-Resilient-Transmission-Project-Information-Document-Integrated-Safeguards-Data-Sheet-Additional-Financing-For-Dasu-Hydropower-Project-First-Stage-of-Dasu-Hydropower-Project-Additional-Financing.pdf](#)

Restructuring / Closing	Project type (if new construction)	Routing (Existing vs new)
N/A	Upgrade	Existing
N/A	Upgrade	New

N/A

Construction

New

Additional financing to P145054

Construction
(See parent project)

(See parent project)

N/A	Construction	New
Component 1: Transmission Network Strengthening and Improvement (US\$ 363.4 million) (a) The upgrading and rehabilitation of up to 56 existing substations (\$300 million) (b) replacement of conductor on existing double circuits of 132 kV lines (\$5.4 million) (c) Supply and installation of a functional SCADA EMS and Telecommunication System on the Nigerian Transmission Network (US\$ 58 million) Component 2: Capacity Building and Technical Assistance (ii) Restructured mainly due to substantial social impact and high ROW compensation	Upgrade (replace conductors)	Existing

N/A	Construction and upgrade	New
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Component 1: RIMA Reinforcement and Expansion Subcomponent 1.1: Kayes-Tambacounda Transmission Line (US\$ 62 million) Subcomponent 1.2: Reinforcement of RIMA's OPGW (\$4.7 million) Subcomponent 1.3: Supply and installation of three 20 MVA 225/90/30 kV transformers at the Bakel, Matam and Dagana substations (\$7.5 million) Subcomponent 1.4: Acquisition of a 20 MVA 225/90/30 kV mobile substation (\$2.4 million) Subcomponent 1.5: Acquisition of a washing truck for High Voltage (HV) lines (\$1.6 million) Component 2: Technical Assistance (\$18.8 million) Restructured due to delays in procurement and implementation. However, the price for TL was significantly lower in bidding. (ii) (iii) [added 2/26/2025]: The project was restructured thrice mainly because of a lower contract price for the HV transmission line and related substations. The November 2020 and May 22 restructurings were triggered by savings that resulted from a lower contract price than anticipated for the construction of the HV KayesTambacounda transmission line and related substations. (high competition in equipment market in Senegal)	Construction	New
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Restructuring due to implementation savings: Component 1: US\$89.93 million (i) US\$14.4 million; (ii) US\$35.7 million; and (iii) US\$39.8 million. Component 2: US\$ 73.1 million Component 3: US\$ 149.2 million Component 4: US\$ 14.3 million (ii)	Upgrade and Construction (mainly the substations)	N/A
Component 1: US\$30 million Component 2: US\$287.66 million - relocation of US\$53 million contingency fees to RAP as the GOC is unable to provide \$50 million for the RAP compensation. Sub-component 2(f): Acquisition of utility vehicles for SONATREL's maintenance fleet (restructuring). Component 3: US\$10 million (ii)	Upgrade	N/A

Changes in financing components; extension of project duration [added 2/2025] This delay was primarily caused by the COVID-19 pandemic and the unbundling of Uzbekistan’s power sector, which took place just before the pandemic (refer to the section of Key Factors during Implementation). While these changes caused temporary disruptions to Project implementation, the unbundling of the power sector overall contributed positively to Project efficiency.	Upgrade (mainly substations)	N/A
Component 1: US\$49.71 million due to savings from strong competition in bidding [As of Feb 2025] - Completion reports not published yet.	Upgrade (mainly substations)	N/A

First restructuring in May 2019: Savings from implementation in the amount of US\$168 million: (i) high level of competition among twelve to fifteen bidders per tender package; (ii) actual costs turned out to be much lower than the cost estimates developed during project preparation; (iii) reduction of the cost of the Gas Insulated Switchgear substations due to technology advances; and (iv) devaluation of the local currency (UAH) during the implementation period. Utilize US\$108 million for financing of rehabilitation of 15 additional substations under Component 1 of the Project. (ii) (The total amount under component 1 should be \$208 million) Second restructuring in November 2021: Withdrawal of 9 substations due to high tendering price (iii)	Upgrade (mainly substations)	N/A
The investment cost was estimated using a rough conservative approach at US\$500 million at appraisal including US\$50 million for contingencies. The actual cost was US\$340 million. The reasons for this significantly lower investment cost were as follows: (a) Decrease in steel prices from US\$1,000 per metric ton to US\$400 that resulted in a saving of US\$30 million; (b) competitive bidding that is estimated to have resulted in a cost saving of US\$70 million; and (c) effective contract management practices that are estimated to have lowered the construction cost by US\$10 million. In total, the actual project cost was US\$160 million lower than the cost estimated at appraisal including the allocation for contingencies that was not used. The project cost was overestimated at appraisal. (ii)	Construction	New

Bidding for the contract resulted in a total contract price of US\$36.65 million resulting in US\$9 million saving. The saving will be used for financing of the rehabilitation and upgrade of the three lines (ii)

Construction

New

Restructuring in 2018/19 does not change project component.

Construction

New

Project went through 3 rounds of restructuring. The first two rounds extended the project closing date due to delays in implementation; the thrid round proposed changes in project costs, thanks to efficient procurement:

Component A: \$478.97 million
Component B: \$98.17 million
Component C: \$7.23 million

(iii)

Construction and upgrade

Existing + new

Extension and rehabilitation of transmission substations and construction of new transmission substations under Group 1 and 2.

Restructured due to lower-than-expected electricity demand growth
(ii)

Total: 159.03 million after two restructuring

Construction and upgrade
(mainly substations)

N/A

See project P132631.

Construction

New

Changes in component B - Total project cost (incl. additional financing) is \$237.9 million	Construction	New
2018: Funds relocation with savings on Component 1from revised procurement packaging and closer implementation support from the World Bank. Changes in component 2: Component 1: 106 million Component 2: 130 million Component 3: 10 million (ii) Closing actual spending: Component 1: 94.84 million Component 2: 43.06 million Component 3: 15.28 million (ii)	Construction and upgrade (mainly substations)	N/A

Actual cost:
Component 1: US\$24.15 million
Component 2: US\$43.49 million
Component 3: US\$3.22 million

Savings associated with the competitive tendering, process and exchange rate changes of the US\$ against the Special Drawing Rights during implementation.

Upgrade

N/A

Actual cost:
Component 1: 84.2 million
Component 2: 6 million
Component 3: 6 million
Component 4: 7.2 million
Component 5: 6.3 million (ii)

Construction

New

Actual cost: Component 1: 15.97 million Component 2: 7.44 million Component 3: 2.2 million Total: 29.61 million (ii)	Construction	New
1.(e) rehabilitation activities of the transmission network in South-Western region of Uzbekistan, i.e. Samarkand, Bukhara, Navoyi, and Kashkadarya provinces’. Savings of \$48 million due to substantially lower equipment prices offered under the international competitive bidding process than estimated at appraisal. (ii) Actual spending (Bank financing): Component 1: US\$93.22 million Component 2: US\$3.25 million (iii)	Construction	New

Actual costs:

Part 1: 97.1 million

Part 2: 81 million

Part 3: 1.5 million

Total: 186.9 million (ii)

Construction and upgrade

New

Actual cost: Component 1: US\$222.06 million (a): \$178.52 million (b): \$43.52 million (including \$12 million for EMSP) Component 2: US\$6.20 million (ii) Low prices offered by the bidders under international competitive bidding for the construction of the transmission line. The decrease in the price of electrical steel from US\$2.25 per pound in 2008 to about US\$1.00 per pound in 2011 was the main reason for lower than expected offers, in addition to the competitive bidding process. (ii)	Construction	New
Actual cost: 1: US\$108.8 million 2: US\$79.3 million 3: Preparation of hydro power project: 2.25 million Total: \$190.44 million	Construction and upgrade (mainly substations)	N/A
Note this project only has an information document without other ones. The project may have been closed without operation.		

Actual cost: Component A(1)-US\$1,065 million Component A(2)-98.2 million - TL: \$49.8 million; transformers: \$11.2 million; substation: \$24.8 million; Reactive compensation equipment: \$5.1 million; taxes: 7.3 million Component A(3)-\$25.4 million Component B-\$6.8 million Component C-\$13.9 million (ii)	Construction	New
Actual cost: Component A : US\$44.0 million Component B : US\$18.8 million Component C : US\$2.5 million	Construction (upgrade of substations)	New
Actual cost: A. \$133.45 million B. \$20.12 million C. \$5.46 million D. 0 Total: \$159.67 million (with front end fees)	Construction (mainly substation)	N/A

Actual cost: Component A: US\$187.8 million Component B: US\$31.5 million Component C: US\$5.9 million Component D: 0 Savings due to contract cancellation in Crimea and Eastern Ukraine	Construction and upgrade	New
Restructuring / Closing: Component A: Consulting Engineering Services (US\$0.8 million) Component B: Capacity Building and Technical Support to EdM (US\$1.6 million) Component C: Improved Transmission Infrastructure (US\$8.2 million) Component D: Tete Substation Upgrades and Mobile Substation (US\$ 3.5 million) (ii)	Upgrade (mainly substations)	N/A

Actual cost
Component 1: \$35.04 million
Component 2: \$22.2 million
Component 3: \$10.41 million
Component 4: \$2.23 million

Upgrade (mainly substations) N/A

Acutal cost:

Component A: US\$84.54 million
Component B: US\$10.46 million
Component C: US\$32.65 million
Component D: US\$0.59 million
Component E Modernization of Transmission SSs (component added in 2009): US\$13.99 million

savings in taxes and efficiently administered procurement.

Construction New

Actual cost: See additional information for transmission costs (ii)	Upgrade	Existing
1. 330 kV Aboadze-Volta line \$53.21 million -> \$71.95 million 2. Volta substation and Akosombo switchyard upgrade \$14 million -> 13.55 million 3. Project implementation support \$7.50 million -> \$8.38 million 4. Technical assistance \$1.50 million -> \$0.82 million	Construction and upgrade	New
Actual cost: Component A: US\$35.7 million Component B: US\$14.7 million Component C: US\$2.2 million Component D: US\$1.5 million Total: \$54.1 million	Upgrade	Existing

Actual total cost: 131.6 million; however, the completion report does not match..	Construction and upgrade	New
Actual cost: (a) US\$90.46 million (b) US\$12.53 million (c) US\$0.25 million (d) US\$19.11 million	Upgrade	N/A

Only at concept stage		
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The project will have significant savings from the IDA SUW credit amount. The prices received for all the contracts so far have been low compared to the estimated price due to the fall in prices of aluminum, copper and steel in the global market in the last two years. Based on the joint assessment of project savings and financing requirements for the remaining IDA SUW financed components of the project during MTR, the total amount of IDA SUW credit available for cancellation is US\$50 million (ii)

Construction and upgrade

Existing + new

all of the required safeguards instruments and supervision arrangements needed for critical environmental and social impacts have been duly implemented

Construction

New

N/A	Construction	New
N/A	Construction	New
Managerial matters	Construction	New

Actual cost:
1.1: 27.45 million
1.2 11.62 million
Component 2: 3.2 million (ii)

Construction and upgrade

Existing + new

Restructuring:
1. Add a new substation, which was under the responsibility of private investment;
2. Drop the 100km line connecting the solar PV power as this falls under the private sector's responsibility (15.5 million) (ii)

Construction and upgrade

Existing + new

Additional cost of EURO 30 million	Construction and upgrade	New
[2/2025]: The revised PDO clarified that most electricity cost savings would occur in Liberia and Sierra Leone, where cheaper CLSG Interconnector imports would replace oil-based thermal power. The 2017 Additional Financing addressed a gap caused by exchange rate losses (39%), contingencies (34%), cost increases (12%), shield wire technology (10%), and post-Ebola impacts (5%). Project delays have led to overlapping debt servicing and early operations. By 2017, construction costs rose to \$489M, with completion delayed to 2018–2020. Total cost: \$249.38M for 1,299 km.	Construction	New

N/A

Construction

Existing

Construction and upgrade

Y: Additional financing; management

existing

Existing + new (mainly new ones)

eight restructurings

Y: DRC’s post-conflict situation and findings of the technical complexity of the project that increased the cost of rehabilitating the DC transmission line and its substations.

New construction

Existing + new

Including: P126421; P105654; P082825

y

New construction

Existing + new

N/A

New construction

Y

New construction

Existing

N/A

Construction

Existing

New construction

No completion report available yet as of 2/9/2025

Y

New construction

Y: Implementation delay

Closing for Component A at \$261.9M; savings due to competitive procurement

Project total \$823.6M due to canceling of component

New construction

Restructuring: change of due date

Additional financing: Transmission and Distribution contract price increases (ii)

Actual cost:
Geothermal Generation: 171.02 million
Transmission :59 million
Distribution: 160.50 million
Sector Institutional Development and Operational Support: 10.48 million

Construction

N/A

Construction

Actual at Project Closing (US\$, Millions): 56.32;

The project costs were less than originally projected, mainly because of the project restructurings that eliminated components that were not performing (see Table 1 for a complete accounting of project costs). (iii)

Construction

The GoU had formally requested the World Bank to drop some activities from the project and cancel SDR 25.37 million (about US\$41.07 million). These were funds that were savings realized from the bidding process of construction activities under Component A.

Construction

Total final cost: 81.75 million

- 1. Generation: (estimated cost at appraisal: US\$226.7 million, revised to US\$460.0 million. Actual cost: US\$454.7 million)
 - 2. Transmission (estimated cost at appraisal: US\$93.8 million; revised marginally to US\$92 million. Actual cost: US\$24.2 million)
 - 3. Distribution: (estimated cost at appraisal: US\$88.5 million; revised to US\$189.9 million. Actual cost: US\$118.2 million)
 - 4. Capacity Building and Governance: (estimated cost at appraisal US\$41.2 million; revised cost: US\$81.3 million. Actual cost: US\$38.7 million)
 - 5. Project Execution: (estimated cost at appraisal: US\$48.8 million; revised to US\$59.8 million. Actual cost: US\$61.1 million)
- Construction and upgrade

N/A

Construction

N/A

Construction and upgrade

N/A

Construction and upgrade

[mission-P171683.pdf](#)
[wer-Phase-I-Project-Transmission-Line-P167843.pdf](#)

Implementation Delays (Y/N)	Total project cost	WB Financing	Cost by component (WB component)
N/A	95	70	- Transmission component: 65 - Technical assistance: 1 - Management and capacity: 4
N/A	427	380	- Tech upgrade: \$115 million; - Transmission component: \$239 million (+\$43 million GCF loan); - Capacity building: \$16 million (+ US\$4 million GCF grant); - Market Development: \$10 million

Y: Land acquisition responsibility; land valuation; terrain difficulties; tribal nature; inadequate tender design	700	700	- Transmission component: 654 - Management: 25 - SES: 36.3
Y: inadequate market responses on the procurement package	(See parent project)	65 Relate to the parent project	- Transmission component: 40 - Management: 7 - Contingency: 18

N	605	465	- Transmission component: 335 - Tech upgrade: 99 - Management: 31
Y: Travel restrictions, document approval	490	486	- Transmission component: 404 - Capacity building and technical assistance: 32 - Contingency: 50

N	536.33	425	- Transmission component: 344.56 - Technical upgrade: 29.46 - Management: 20 - Other/contingency: 30
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Y:
COVID, procurement, political
situation (in Mali),
management, implementing
the RAP and clearing ROW

103

97

- Transmission component: 88
- Technical assistance: 9

Y: Procurement, COVID, environment concerns	325	325	- Transmission component: 115.6 - Distribution: 196 - Capacity: 13.4
Y: procurement, preparation and finalization of Resettlement Action Plans (RAPs), publication of expropriation and compensation decrees.	375	325	- Transmission component: 285 - capacity: 40

N	196.3	150	- Transmission component: 145 - capacity : 5
Y: Procurement, commissioning	64.86	52	- Transmission component: 48.02 - management: 2.5 - Other:1.35

y: Procurement, management, COVID	378	330	- Transmission component: 225 - Market: 51 - Cpacity: 2.5 - Contingency and other: 51
Y: resettlement matters	731.25	500	- Transmission component: 429.45 - technical upgrade: 55 - capacity: 14.3 - Other: 1.25

Y: resettlement matters	61.88	60	- Transmission component: 404 - Capacity building and technical assistance: 32 - Contingency: 50 - Other: 0.15
Y: Procurement	1170	526.5	N/A

Y:
Land acquisition

837

600

- Distribution: 485
- Transmission: 106
- capacity: 9

Y:
procurement, management

159

137.6

- Transmission component:
137.6

Y:
procurement, management,
land acquisition and
resettlement, earthquake

202.3

119

N/A

Y: procurement, management, land acquisition and resettlement, earthquake	39	39	- Transmission component: 34 - management: 3 - other: 2.5
Y: Procurement, land acquisition, technical issues, COVID	210	105	- Transmission component: 95 - technical assistance: 10

N/A	100	90	- Transmission component: 53.25 - management: 4.5 - contingency: 2.25
Y: Procurement	111	41.9	- Transmission component: 25.4 - Distribution: 9 - capacity: 10.5

Y: Procurement, management, resettlement	47.69	38	N/A
Y: Management	171.04	110	- Transmission component: 90 - capacity: 10 - Contingency: 10

No	200.9	78	N/A
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Y: Procurement, government management.	468.45	150	- Transmission component: 134.5 - technical assistance: 10 - other: 5.5
Y: Management, contract	254.1	225	- Transmission component: 225

Y: Procurement, studies	1662	136.4	
Y: Procurement, construction	69	48	N/A
Y: procurement, management	309.9	256.7	- Distribution: 206.1 - transmission: 19.5 - capacity: 30.6

Y: Construction - substation	238	200	N/A
Y: Procurement, restructuring process	13.8	13.8	- Transmission: 11.7 - capacity: 2.4

Y : Procurement, disbursement, management	75	60	N/A
Y: Sustation	160.84	113.14	N/A

Y: Inflation, management	324.2	200	- Transmission component: 176.55 - Market support: 23.5
Y: Procurement, rerouting, management matters, financing	83	40	N/A
Y: Management	55.4	48	

Y: Procurement, management, land acquisition	67.92	40	- Transmission component: 16.97 - Distribution: 12.81 - fund: 5.1 - capacity: 5.12
Y: government	113.35	100	

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N/A	718.06	450.64	- Transmission component: 426.25 - Capacity: 24.39
Y: Financing, RAP, management	711	200	-Transmission component: 163 - technical support: 37

N/A	1426	444	- Transmission component: 291 - generation: 120 (guarantee) - capacity: 9
N/A	380.8	84	- Transmission component: 71.8 - capacity: 12.2
N/A	744	385	- Transmission component: 385

Y:
contractual, construction
procedures, comissioning,
political - custom

43.1

41.05

- Transmission component:
39.55
- capacity: 1.5

N/A	151	151	- Transmission component: 131 - capacity: 20
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Y: Contracting, management	150	150	
Y: Covid, construction, procurement,	427.5	176	- Transmission component: 84.8 - Management support: 45 - Technical assistance: 31.5 - Other: 14.63

N/A

75

70 N/A

Y: additional financing

39.71

- Transmission: 8.91
- Technical: 0.47
- Implementation support:
25 10.72

Y: supply (ii)

Y: DRC’s post-conflict situation and findings of the technical complexity of the project that increased the cost of rehabilitating the DC transmission line and its substations.

- implementation and slow procurement and disbursements
- delays in the construction and rehabilitation
- delays in payment of the government counterpart funds

(iii)

Y: Covid-19; Resettlement Action Plans (RAPs) for the both countries; institutional; recruitment; procurement

130

57 - transmission: 54
- technical: 3

N/A

66.94

46 - Transmission: 41
- Project management: 5

Y: COVID on the hiring process
of the Auditor.
late securing of the financing
(AfDB and KfW), delays in
clearance of line routing, and
disputes between the
contractor and the
Government of Benin.; rewrite
the bidding documents,
procurement; internal
administrative procedures

640.2

465.5

N/A

22.197 million

18.5 million

- Transmission system: 15
- Implementation: 1.75
- capacity: 1.2

Y: COVID-19; (i) Right of Way (RoW) acquisition due to the long procedural steps involving other sector agencies especially the Ministry of Lands; (ii) long lead times associated with the Implementing Agency (UETCL)/Government clearances; (iii) change in the technical design for the River Nile tower crossing; (iv) initial low mobilization by contractors and (v) global shortage of semiconductor chips. I	127.3 million (i)	100 million (i)
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317.5 million (i)

310.0 million (i)

Y:in procurement and lengthy
soil consolidation issues;
management

350 million (i)

470.0 million (i)

Y: Construction; right of way contested; COVID, procurement; bankruptcy	1,262.5 million (i)	683.9 million (i)	589 million for transmission; 3 million for management & capacity; 90 million for other (i)
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Y: rerouting;	63.60 millon (i)	62.00 milllon (i)	
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Y: procurement	US\$ 147.96 million	105 million (i)	- Component A: 85.8 - Offgrid electrification: 16 - Technical: 3.2
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Y:(due to right-of-way,
compensation, and contractor
issues which caused the
Component to be classified as
Moderately Unsatisfactory on
the Implementation Status and
Results Report of June 2015

N/A for the transmission component, which was completed on schedule.	US\$ 350.00 million	US\$ 82.00 million
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Y: a slow procurement process, low technical capacity in implementing bodies, and difficulties with some of the contractors (iii)	US\$ 163.19 million	US\$ 101.59 million
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Y: land acquisition challenges;
procurement, host government 153.2 million
communication

120 million

- transmission: 79.2;
- Technical: 19.5
- other: 21.4

Y: - difficulties in coordination
between key stakeholders
and weaknesses in 499 million (i)
implementing arrangements;
procurement

296.7 million (i)

- Generation: 198.3
- distributionL 26.2
- capacity: 34.7
- implementation: 37.5

N/A 40.2 million (i)

35.7 million (i)

- transmission: 25
- distribution: 8.9
- support: 1.8

N/A	142 million (i)	117 million (i)	- Renewable Energy and Storage Infrastructure Development: 16.5 - transmission: 86 - Implementation Support: 14.5
N/A for the transmission component, which was completed on schedule.	43.10 million(i)	41.05 million (i)	- Transmission: 39.55 - implementation support: 1.5

Ownership	Route length (km)	Voltage	Elevation (AMSL)
National Transmission Utility	20 km (i)	220/138 kv (i)	Not available
National Transmission Utility	290 km (i)	110/220/500 kv (i)	Not available

National Transmission Utility	255 km (i)	765 kv (i)	1000-3200 m (i)
National Transmission Utility	100 km (i)	500 kv (i)	Not available

National Transmission Utility	620 km (i)	400 kv (i)	592 - 1834m (environmental assessment)
National Transmission Utility	Replacement of conductor on existing double circuits of 132kV transmission line of approximately 137 km (ii) [1260 km (replacement of conductors on 13 segments of the 132 kV transmission system with a total length of about 960 km and the reconstruction of two segments of 132 kV lines of about 300 km) (i)]	330/132/33 kv (i)	15 - 600 m (envrionmental assessment)

National Transmission Utility	131 km (i)	765/500/220 kv (i)	100-3200 m (iii)
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Intergovernmental Organization	288 km (i)	225 kv (i)	Not available
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National Transmission Utility	6 (1 km for 225 kv and 5 km for 90kv) (i)	225/90 (i)	Not available
National Transmission Utility	650 (i);	225/90 (i)	Not available

National Transmission Utility	N/A	220/500 (i)	Not available
National Transmission Utility	N/A	220/110/10 kV (35 kv)	865 to 1430m (iv)

National Transmission Utility	N/A	330/110	Not available
National Transmission Utility	1040.2 km (i); Review: Rehabilitation: 260.60 km Construction: 980.76 km (ii)	220/500 (i)	Not available

National Transmission Utility	Construction: 142 km	220 kv (i)	500 m to 2150 m (iii)
National Transmission Utility	750 km: A1 - 120 km in Tajikistan, 560 km in Afghanistan, 70 km in Pakistan 475 km: A2 - 450 km in Kyrgyz Republic and 25 km in Tajikistan	500 kv (i)	500 m to 2000 m (environmental assessment)

National Transmission Utility	389 km (280 km rehabilitation; 109 km construction) (ii) (Construction: 4 circuit-km circuit-km 230 kv line + 170 circuit-km 132kv Line; Rehabilitation: 458 circuit-km 132kv Line; Second circuit: 51 circuit-km 132 kv line) (i)	230/132 kv (i)	Not available
National Transmission Utility	up to 5km connection line (i)	150/70 kv (i)	Not available

National Transmission Utility	415 km	400 kv (i)	Not available
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National Transmission Utility	N/A	400/220 kv (i)	Not available
National Transmission Utility	Rehabilitation: 75 km Construction: 5 km (i)	132/88 kv (i)	1200 - 1300 m

National Transmission Utility	348 km rehabilitaiton (i)	330/220kV (i)	1200 m on average (1130 - 1400 m) (iv)
National Transmission Utility	210 km (i)	225 kv (i)	Not available

National Transmission Utility	90 km (i)	132 kv (i)	Not available
National Transmission Utility			
National Transmission Utility	216 km (iii)	500 kv (i)	450 – 1,150 m below sea level (v)

National Transmission Utility

485.4 km (i)

500/220 kv (i)

Not available

National Transmission Utility	683.2 km (i)	400/220 kv (i)	750 - 1650 m
National Transmission Utility	N/A	500/275/150 kv (i)	Not available

National Transmission Utility	317 km (i) \$2.27 million /km	400/220 kv (i)	1000 m (i)
National Transmission Utility	322 km (i)	220 kv (i)	Not available
Private companies for distribution; National transmission utility for transmission	277.50 km (i)	220/132 kv (i)	Not available

National Transmission Utility	72 km (ii)	750/330/220 kv (i)	245 m to 369 m (iii)
National Transmission Utility	N/A	220 kv (i)	Not available

Intergovernmental
Organization

219 km (ii)

330/161 kv (ii)

Not available

National Transmission Utility

462.57 km (ii)

500 kv (i)

Not available

National Transmission Utility	621.23 km(ii)	500/220/110 kv (i)	0 - 50 m
National Transmission Utility	215 km (i)	330 kv (i)	Not available
National Transmission Utility	850 km (ii) (replacement)	330/220/110 kv (i)	Not available

National Transmission Utility	109 km (i) + 20 km upgrades	230/115 kV (i)	Not available
National Transmission Utility	N/A	330/132 kV (i)	Not available

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National Transmission Utility	1066.1 km (i)	400/230/132 kv (i)	Not available
National Transmission Utility	1677 km (i)	225/30 kv	Not available

National Transmission Utility	563 km (i)	400 kv (i)	Not available
National Transmission Utility	714 km (i)	225 kv (i)	Not available
National Transmission Utility	1556 km (i)	225 kv (i)	Not available

National Transmission Utility

296.5 km for single circuit;
36.5 km double circuit (i) -
construction;
reinforcement expansion
of the existing 260 km single
circuit 230 kV line

230/220 kV (i)

Not available

National Transmission Utility	384 km (i)	400/225 kV (i)	Not available
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National Transmission Utility	392km of 400kV lines; 76 km of 225kV lines (i)	400/225 kV (i)	Not available
Regional Transmission Company (RTC)	1,349 km (i)	225 kv (i)	Not available

National Transmission Utility 158 km(i) 400 kv(i)

National Transmission Utility

19 km (i) 400 kv (i)

National Transmission Utility 34.9 km (i) 110 kV (i)

National Transmission Utility from Inga to Kasumbalesa: 286
km (i) 220 kv (i)
from DRC to Zambia: 40 km (i)

National Transmission Utility Mozambique: 142 km
Malawi: 76 km 400 kv (i)

National Transmission Utility 192 km (i) 230 kV (i) Not available

National Transmission Utility	875 km of 330 kV line (62 km of 330 kV lines in Nigeria, 425 km in Niger, 381 km in Burkina Faso, and 12 km in Benin)	330/225 kv (i)
	24 km of 225 kV	

National Transmission Utility	See the line numbers for detailed costs for each segment	115 kv	5-44 m
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National Transmission Utility

314 km (i)

132 kv (i)

600-1400 m (iii)

National Transmission Utility 270 km (i) 132 kv (i)

National Transmission Utility 11 km (i) 230 kV (i)

National Transmission Utility	1,045 km (433 km of the line will be in Ethiopia and 612 km in Kenya.) (i)	500 kv (i)	1000 - 2000 m (iii)
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	77 km (i)	500 kv (i)
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National Transmission Utility	51.7 km (i)	132 kv (i)
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National Transmission Utility	Kindaruma-Mwingi-Garissa line: 230 km; Eldoret-Kitale line: 60 km Kisii-Awendo line: 44 km	132 kv (i)	70 - 1700 m
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National Transmission Utility	100 km	225 kv (i)	below 100 m above sea level (iv)
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National Transmission Utility	280 km (i)	161 kv (i)	
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National Transmission Utility	137 km (i)	220 kv (i)	
National Transmission Utility	260 km (i)	400/220 kv (i)	
National Transmission Utility	87 km (i)	132 kv (i)	60-1,860 masl

National Transmission Utility 100 km 15- 120 m (ii)

		132 kv (i)
National Transmission Utility	Construction and expansion of 296.5 km of transmission lines in Ethiopia, including expansion of Ethiopia’s existing 260 km single circuit line linking Bahir Dar to Shehedi and a new 36.5 km double circuit 230/220 kV transmission line linking Shehedi to Metema at the border with Sudan	230/220 kV (i)

Transmission Information

Number of transmission towers	Number of substations	Conductor type	Materials
Not available	21 (i)	Not available	Not available
Not available	22 (modernize) + 1 new 500 kV 'Koltsevaya' substation with associated transmission lines(i)	Double Circuit/Single Circuit (i)	Not available

674 (i)

1 (i)

HVAC double circuit (ii)

Not available

Not available

1 converter station (i)

HVDC (i)

Not available

1690 (environmental assessment)	4 (400/220/33 kV substation in Kisada; a 400/220/33 kV Mbeya substation in Iganjo; a 400/220/33 kV Tunduma substation in Chiwanda; and a 400/220/66/33 kV Sumbawanga substation in Sumbawanga and 400 kV switchgear in existing Iringa substation in Iringa) (ii)	Double Circuit/one Single Circuit (i)	ACSR Class AA (environmental assessment)
N/A (using existing towers)	48 (i) + 8 (ii)	Double Circuit (i)	Aluminium conductor composite core conductors (i)

For a section of 9.5 km, 29
towers will need to be built (iii)

35 (i)

Double Circuit (i)

Not available

717 (iv)	2 (1 new; 1 existing) (i)	Double Circuit (i)	fiber optic cable (24 pairs); aluminum alloy (AAAC), 228 mm ² , with two conductors per phase (i)
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4 for the 1 km 225 kv (iv)	19 (1 new, 3 upgrades, 15 improvement) (i)	Not available	Not available
Not available	52 (i)	Double Circuit/Single Circuit (i)	Not available

N/A

22 (i)

N/A

N/A

N/A

3 (i)

N/A

N/A

N/A	12 (iii)	N/A	N/A
See additional information	15 (five 500 kV substations (Hoa Binh, Ha Tinh, Da Nang, Pleiku, and Phu Lam) and 10 220 kV substations (Ha Dong, Ninh, Thai Nguyen, Thai Vinh, Krongbuk, Quy Nhon, Bao Loc, Cai Lay, My Tho, and Tra Noc))	Double Circuit/Single Circuit (i)	See additional information

398 (New steel towers will be placed at intervals ranging from 600 to 800 meters depending on topography; towers will be closer when there is little or no relief and farther apart in hilly or mountainous terrain) (iii)	N/A	Double Circuit (i)	ACSR; optical-fiber cable (OPGW 70 ACS – of an aluminum-coated steel type); AAAC conductors (iii)
3000 (i)	3 HVDC converter stations (I - A2)	Double Circuit (i)	

Not available	6 new; 10 expansion (for transmission section) (i)	Double Circuit (i)	Not available
N/A	78 (37 under group 1; 41 under group 2) (ii)	N/A	N/A

824 (i) (ii)

3 (i) (Construction)

Double Circuit (i)

aluminum steel-reinforced (i)

H-D-D + DB: 324.2 km - 585 towers (iii)	H-D-D + DB: 2 new + 2 upgrades (iii)	Double Circuit (i)	Aluminum Conductor Steel Reinforced 525 mm2 "Moose" (iii)
Mostly existing	3 new + 3 upgrades (ii)	Double Circuit/Single Circuit (i)	ACSR

Mostly existing

2 upgrades + 1 new (i)

Double Circuit (i)

twin Bison ACSR conductor

106 towers shall be
required for the approximately
39.3 km span (iii)
500 for 170.8km (iv)

2 new + 1 extension (i)

Double circuit (ii)

fiber optic and of stainless
steel loose tubes (i);
AAAC/TW (ii)

300 (i)
258 (ii)

4 (i) (Construction)

Double Circuit (i)

ACSR (ii)

562 (ii)

1 (ii)

Single Circuit (i)

Not available

Not available

1 new + 2 extensions (i)

Single Circuit (i)

AS400/51;
AS500/26
AS300/39
Ground wire;
OPGW, optic fibres

1660 (iii)	4 (i) (Upgrades)	Double Circuit (i)	Fiber Optic Communication Line (FOCL)
N/A	80 (ii)	N/A	

Not available	1 (i)	Not available	Not available
Not available	1 new + 2 upgrades (i)	Single circuit (i)	ACSR
120 km - 200 towers (iii)	1 new (i)	Double Circuit (i)	Not available

267 (iii)

13 upgrades (i)

Single Circuit (iii)

AA/ACS 0/57 (iii)

N/A

2 upgrades + 1 mobile
substation (ii)

Double Circuit (i)

OPGW (i)

N/A	2 upgrades (ii)	Single Circuit (ii)	N/A
Not available	2 upgrades (i) + 67 upgrades (Component E)(ii)	Single Circuit (i)	Fiber Optic Communication Line (i)

Not available	16 (additional information)	Double Circuit/Single Circuit (i)	Not available
430 (environmental studies)	2 new (i)	Double Circuit (i)	ACSR (i)
N/A	10 (ii)	Fiber Optic Communication Lines (ii)	Not available

370 for 230 kv (iii)	2 + 3 (i)	Double Circuit (i)	Not available
N/A	N/A	N/A	N/A

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884 (iii)	13 (i) (one upgrade to 400/230/33 kV substation from 230/33 kV substation, one new 400/230/132 kV substation, two new 230/132 kV substations, and nine new 132/33kV substations)	Double Circuit (i)	ACSR (iii)
3328 (iii)	15 (i)	Single circuit / double circuit (i)	AAAC for single circuit; ACSR/CGFO for double circuit (i)

1683 (ii)	3 new + 1 upgrade (i)	Single Circuit (i)	ACSR (i)
2082 (ii, iii)	5 new + 2 upgrades (i)	Double Circuit (i); one earthwire, and one Optical Fiber Ground Wire (OPGW)	Almelec 570 (i)
Not available	8 new + 4 extension (i)	Double Circuit/Single Circuit (i)	two-bundle aluminum conductor (Aster 570 mm ²) (i)

603 (i)

3 upgrades (i)

Double Circuit/Single Circuit (i)

AAAC 450; AAAC 181 (i)

197 km - 492 towers;
85 km - 167 towers (iii)

1 new + 2 upgrades (i)

Double Circuit/Single Circuit (i)

ACSR + AAAC (i)

Not available

6 (i)

Double Circuit/Single Circuit (i)

570 mm aluminum conductors
(i)

Not available

construction of 11 225kV
substations (i)

Double Circuit (i)

OPGW (i)

511 (ii)	Single Circuit (i)	three bundles of 300 mm2 aluminum steel reinforced conductor (i)
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1.4 km - 5 towers 9.1 km - 38 towers 5.3 km - existing 9.5 km - 40 towers 9.6 km - 56 towers Land - mainly agriculture	5 (new) (i)	Double Circuit (i) Single / double
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Double circuit (i)

485 in total (ii)

Mozambique: 2 (1 new + 1 upgrade)
Malawi: 1 upgrade

Single / double

ACSR triple bundle tern conductor (3x Tern)

548	1 upgrade (7.76 million)	Double circuit (i)	2*180 mm2 AAAC
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450 m per tower

For 330 kv: extend 2; construct
3

Double Circuit/Single Circuit (i)

two-bundle aluminum
conductor

Average span between towers:
80-100 m

6 upgrades

Double Circuit (i)

240 sq mm ACSR conductor
and Optical Fiber
Earth Wire

3 new + 1 extension (i)

Double circuit (i)

AAAC (ii)

3 new + 2 extension (i) double circuit (i)

41 (ii) 2 (i) double circuit (i)

2613 (i)

2 (i)

205	Double circuit (i)	400 mm2 ACSR conductor
Extension of two existing Substations; Supply and erection of 3 new 132/33 kV substations at Factory Zone II, Mbagala, Kurasini - 17 million (i)	double circuit (i)	

spacing between the towers is approx 300m.

Kindaruma-Mwingi-Garissa line: 2
Eldoret-Kitale line: 1
Kisii-Awendo line: 2

Double Circuit/Single Circuit (i) ACSR

The nominal spacing of the towers will be 350 m (iv)

1 new (iv)

double circuit (i)

3 x 366 mm2 (ASTER)

1 new + 3 upgrades (i)

Double circuit (i)

ACSR de 88,60 mm2 de section (ii)

1 new + 2 upgrades (i)

Double circuit (i)

240 mm2 twin AAAC conductor

Double circuit (i)

the transmission towers will likely be installed at approximately 300 meters intervals depending on the terrain

Double circuit (i)

3 new (i)

Double circuit (i)

AAAC 405 mm² (ii)

Double circuit (i)

Estimated transmission losses	Capital cost for transmission component	Discount rate applied	Operations & Maintenance rate
Not available	\$93.6 million (i)	8% (i)	3.59% (i)
Due to the enhanced voltage operation and elimination of overloading, construction of the new 500 kV Koltsevaya substation will decrease annual technical losses by 33.6 GWh (0.76 percent of total losses). (i)	US\$342 million; US\$288 million (excluding taxes) (i)	9.8% (i)	2% (i)

	US\$715.3 million; US\$81.75 million for front-end fee, commitment fee, and interest during construction (IDC). (ii)	Not available	Not available
477 GWh saved (ii)	US\$39.62 million (Goods, works, and services) (ii)	10% (ii)	See P16

3%	US\$443 million for transmission lines and substations (i) (project total capital cost is \$477 million) (total cost for TL: \$255 million; total cost for substations: \$180 million) (i)	10% (i)	2% (i)
N/A	Upgrading and rehabilitation of substations: \$245 million (incl. \$49 million for taxes); Reconducting of transmission lines: \$50 million (incl. \$10 million for taxes) (i) <i>Note: restructured July 2022</i>	7% (i)	2% (calculated based on operating expenses as of 31st December 2021, ii)

414 GWh saved (i)	\$430.69 million for goods, works, and services US\$350 million for the transmission line (i)	10% (i)	Not available
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208 GWh saved (i)	\$99 million (i)	6% (i); changed to 10% in 2020 (ii)	2% (i)
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Around 23 GWh saved from the transmission part (i)	\$108.4 million; 1.21 million for the transmission lines + 12.45 million for the associated S/S + 98.5 million for other S/S upgrades (i)	6 (i); changed to 10 in 2020 (ii)	5% (i)
From 6 to 4 percent	\$231 million (i)	6% (i)	5 % (i)

673 GWh saved	136.82 million (i)	10% (i)	3% (calculated based on capital cost)
Not available	Not available	8% (iv)	Not available

Not available	Not available	10% (i)	Not available
	The investment cost was estimated using a rough conservative approach at US\$500 million at appraisal including US\$50 million for contingencies. The actual cost was US\$340 million.	10% (i)	2% (ii)

	\$45.65 million for construction (i)	12% (i)	2% (i)
1.50%	\$801 million for transmission infrastructure (Kyrgyz Republic: \$160 million, Tajikistan: \$240 million, Pakistan: \$159 million, Afghanistan: \$242 million) (i)	10% (i)	Project Management Support (US\$4 million) B4 2% for HVAC DS&I and local grid reinforcement in each country; 3.9% for HVDC DS&I (i) <i>DS&I - Design, Supply and Installation</i>

2.70%	US\$127 million (i) Note - cost estimates for the turnkey projects: \$97.06 million: (\$35.69 million for upgrades and construction of transmission lines; \$18.46 million for upgrades of existing substations & construction of Sherpur (B) and Kurigram 132/33kV S/S; \$20.5 million for the new Keraniganj 230/132/33kV GIS S/S; \$22.66 million for the new Nawabganj, Sreenagar & Narail 132/33kV AIS S/S) (i)	12% (i)	4% (\$5.16 million for transmission part) (i)
N/A	\$318 million (infrastructure cost, out of \$346.43 million) (i)	Not available	\$19.3 million (1.5%; out of total \$124.6 million) of project management (ii) (Outside Java and Bali, the gap between the demand for electricity and the electrification levels remained large as PLN's operating costs are higher in lower-density rural areas and outer islands)

reduction of transmission losses by 2 percent (i)	\$28.86 million for A1 \$17.08 million for A2 \$126.76 million for B1 (i)	12% (i)	1.5% (i)
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Reduction in load loss: 250 MW (original project)			
Reduction of transmission losses by about 15 MW (additional financing)	Not available	12% (i)	1.5% (i)
245 GWh reduced (5%)	US\$240 million (project total) (i)	10% (i)	2% (i)

reduction in transmission losses estimated at about 19 GWh (i) The upgraded 330kV transmission line would have the capacity to transfer a power of 450MW with about 5.5% of losses Technical loss: 1–4.1% (ii)	US\$64.6 million as of completion (ii) Transmission: \$21.35 million Substation: \$43.07 million <i>Appraisal: US\$ 77.1 million Transmission lines: \$17.25 million; 2 substation upgrade: \$28.35 million + 1 substation construction: \$24.75 million Project management: \$7.1 million (i)</i>	10% (i)	2% (i)
Not available	- \$34.2 million for HV lines; - \$16.5 million Construction of Bolgatanga Substation in Ghana (225/161 kV) - \$19.7 million Construction of substations in Burkina Faso (90/33 kV) (i)	10% (i)	2.5% (i)

	\$12.65 million for the TLs \$13.28 million for 4 substations (i)	10% (i)	1.5% (i)
		10% (i)	
81 GWh saved (i)	Not available	10% (i)	1.5% (i)

Transmission lines: \$97.1 million;
Substations: \$81 million (i) (not sepcified for CAPEX)

8% (i)

Not available

5% reduction of transmission losses by 3 percent (i)	OHL section: US\$398.71 million Substations: US\$ 36.06 million (i) Actual: OHL: \$178.52 million Substations: \$31.54 million (ii)	10% (i)	1% (i)
	\$214.3 million (i) \$188.1 million (ii)	10% (ii)	3-5% (i)
			1.5% (i)

Not available	\$1,118 million (including generation section) (i)	12% (i)	O&M fixed costs: US\$25/kW O&M variable costs: US\$3/MWh (i)
Assumed at 6.5% (ii)	Not available	8% (i)	1.5% (i)
Not available	Not available	12% (i)	2 - 4% (ii)

transmission losses reduced by 14 MW (ii)	130.2 million - Component 1; Component 2: 34.9 million (i)	10% (i)	Not available
N/A	\$11.7 million (ii)	6% (ii)	2.5% (i)

Not available	\$20.49 million (ii) for transmission line construction and materials; \$16.56 million for the 2 substation upgrades (ii)	10% (i)	Not available
6.5% transmission losses have dropped from over 8.8 percent to 7.6 percent (ii)	See project description	10% (i)	1.1% (ii)

Not available	1.1 \$177.46 million 1.2 \$40.9 million 1.3 \$8.6 million (i) See additional info.	10% (ii)	2% (ii)
Not available	\$47.41 million for transmission lines; \$14 million for substation upgrades (i)	10% (i)	Not available
Not available	\$14.7 million for the substations (ii)	10% (i)	1.5% (i)

3%	See project description	12% (i)	4% (i)
Reduced transmission losses (from 10.7% to 4.0% at 330 kV and from 12.0% to 8.0% at 132 kV) (i)	40.91 million (i)	12% (i)	5% (i)

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2.70%	Not available	12% (i)	Not available
3% (i)	586 million (as excluded O&M and other)	10% (i)	2% (i)

	\$423.4 million (excluding VAT) (i)	10% (i)	3.5% (i)
5%	\$343.9 million (i) benchmarking and CBT for 225 kV transmission lines (US\$0.248 million/km) and 225/33 kV substations (US\$9.4 million) in West Africa	10% (i)	3% (i)
3% 4% in Chad - transmission loss reduction (i)	\$178 million for 517 km in Cameroon; \$87.9 million for 4 substations and equipment in Cameroon; \$245 million for transmission liens and substations cross-nation (i)	6% (i)	2% (i)

5% (i)

\$24 million for transmission
section (may incl. O&M);
\$9.65 million for substation (i)

10% (i)

Not available

Not available	1. New 100km 225 kV line to connect the Borj Bourguiba PV plant and Tataouine substation \$15.5 million 2. New 85 km 225 kV line (411mm²) to connect the Skhira site to existing Thyna substation \$18.6 3. New 192km 400 kV double line (bundle 2 x 570 mm²) to connect the Skhira site to new Kondar substation \$53.8 million 4. Line in-Line Out on the existing 225 kV line Bouficha-Sousse to connect on new substation Kondar 6.5km \$1.8 million 5. New 400/225 kV substation - \$27.0 million 6. Extension of the substation by adding Auto-Transformers 225/150 kV - \$6 million (i)	10% (i)	2% (i)
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3% (i)	\$92.83 million for transmission lines \$51.91 million for substations	10% (i)	Not available
	\$193 million for transmission lines; US\$99.8 million (AF) \$111 million for substations (i) benchmark on the market price (US\$0.203 million per km) (ii)	10% (i)	1% for transmission lines; 3% for substations; 10% for SCADA (i)

46.5 million (i)	6% (i)
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- 5.42 (This line will cost about Euro 4.5 Million to construct), (i) '- The Stip - Serbian: EUR 12.4 million (82 km/440 kv)	10% (i)
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\$2.5 million

limited to installation and civil work	12% (i)	2% (i)
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9% (i)	from Inga to Kasumbalesa: \$79.2 million (ii, based on the contract estimation) DRC to Zambia: \$18 million (i)	2% (i)
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12% (i)

Mozambique: 48.898; Malawi: 23.056	6% (i)	2% (i)
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\$41 million

See the line numbers for
detailed costs for each
segment

6% (i)

1% (i)

See the line numbers for
detailed costs for each
segment

10% (i)

2.5% (i)

3.3% (i)

\$ 58 million (i)

10% (i)

2% (i)

18.017 million (i)

10% (i)

12% (i)

2.5% (i)

4.53 million (i)

5% (i) 308.1 million (i) 10% (i) 2% (i)

60 million (i)

5.826 million (i) 10% (i) 2.5% (i)

4% (i)	64.5 million (including the substations) (i) 54 million (results report) (ii)	12% (i)	3.5% (i)
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	US\$24 million (i)	10% (i)	N/A
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	35.28 million (i)	10% (i)	N/A
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4% (i) 35,690,857 (i) 12% (i) 2.5% (i)

82.6 million (i) 12% (i) 2.5% (i)

6% (i) 1% (i)

16,000,000 USD (ii)

6.4% (i)

2% (i)

US\$27.40 million (i)

10% (i)

Cost Information			
Community development cost / ESMP	ESMP (Y/N)	RAP (Y/N)	Resettlement (RAP) costs
Not available			Not available
Not available			Not available

US\$6.0 million (ii)			US\$30.99 million; Total cost of ESMP implementation is USD 41.89 million including the RAPs budget, which is USD 30.99 million. (iii)
0; CSP: US\$ 15 million for the 145054 parent project (ii)			Not available

Not available	Y	Y	\$10 million for compensation (i) \$14.52 million for RAP (Resettlement plans)
Not available	Y	Y	RAP total: 4,000,000 (iii) Around \$90,000 - \$110,000 for environmental mitigation costs (environmental assessments)

a. Nowshera: \$0.8 million (v) b. Islamabad West: \$1.35 million (iv)	Y	Y	a. Nowshera: 2% (11.26/51.29 million) (iii) b. Islamabad West: 1.9% (15.05/79.36 million) (ii)
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Environment: US\$6,166,500
(iii)

Y

N

\$1,290/km for Mali (95,983m
in Mali);
\$1,172/km for Senegal
(192,221m in Senegal)
(included in ESMP) (iv)

\$1.3 million for environmental protection (iv)	Y	Y	\$3.6 million (iii);
Not available	Y	Y	\$50 million (i); \$62.31 million (ii)

Not available	Y	N	N/A
Not available	Y	N	N/A Environmental Management Plan \$203,500 for each substation (ii, iii)

Environmental Impact Assessment: \$421,952 for the six substations of component 1; \$30,870 for component 2 (iv)	Y	N	N/A
See additional information	Y	Y	US\$19 million for group 1 subprojects: For subproject in Group 1, there are totally 1785 affected households, including 11 households to be relocated and 259 households having severe impact on total land holdings. The estimated budget for resettlement related activities is 401,853,266,352 VND (equivalent to US\$ 19 million). (i)

Not available	Y	Y	\$14,693,538.40
Community support program: \$70 million (i - Component C) Environmental and social costs: \$20 million	Y	Y	\$4 million (Sub-component B2 – Environment and Social Management Support - incl. RAP)

225,000 (iii)	Y	N	Not available
Not available	Y	Y	Not available

US\$15 million for
environmental and social
costs (i);
ESMP - \$8.37 million (H-D-D)
(ii)
ESMP - \$783,995 (A2, D-B)(iii)

Y

Y

Not available

HDD+DB: Total Social Impact Assessment (SIA) and RAP is Rs 541.60 million. (US\$5.80 million) i) compensation for land Rs 142.42 million (US\$1.52 million) ii) ROW land use restriction (836.92 ha) Rs 299.7 million (US\$ 3.21 million) iii) community support Rs 20 million (US\$ 0.21 million) iv) setting up function public information centre and local consultative forums Rs 10 million (US\$ 0.11 million) v) rural electrification Rs 10 million (US\$ 0.11 million) vi) institutional strengthening (NEAIESSD) Rs 7.5 million (US\$0.08 million) (iv)	Y	Y	HDD+DB: US\$1.73 million (NRs. 161.45 million) (iii) - cost of mitigation and enhancement measures by RAP;
Environmental Mitigation and Monitoring Budget: US \$1,002,800 (iv)	Y	Y	US\$794,000 (Cost of compensation) (iii)

Environmental management: \$500,000 (i)	Y	Y	Implementation: USD 2,263,770 (iii)
\$5.4 million for environmental and social costs (i)	Y	Y	USD 645,095 (v) + 1,024,650 (vi)

ESMP: \$0.9 million (i)	Y	Y	\$0.2 million (i)
	Y	Y	USD 75,048.25 (i)

See land acquisition	Y	Y	See land acquisition
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US\$18.18 million on environment and social mitigation measures (i) Actual: \$12 million (ii)	Y	Y	US\$8 million (i)
Not available	Y	Y	Not available

Not available	Y	Y	Not available
Not available	Y	Y	Not available
ESMP: 5.488 million; Land acquisition: 5.3344 million (iii)	Y	Y	Not available

Not available	Y	Y	Not available
Not available	Y	Y	757,728.00 (iii)

Not available	Y	N	\$12.71 million for RAP + EMP (ii)
	Y	N	N/A

See additional information	Y	Y	See additional information
US2.41 million for Environmental Impact and Mitigation Measure for the transmission line (i)	Y	Y	Included in left (i)
Not available	Y	N	N/A

Not available	Y	Y	\$1,936,604.40 (iii)
\$0.25 million (i)	Y	N	N/A

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ESMP: US\$ 86028.25 (III)	Y	Y	US\$41 million (i)
Not available	Y	Y	\$3.4 million (i)

\$22 million (i) - environmental	Y	Y	\$17.1 million (i) - including land
Not available	Y	Y	\$26.6 million (for ESMP + RAP) (i)
Not available	Y	Y	\$80 million (Cameroon) (i)

Not available	Y	Y	US\$1.30 million (EMP + RAP) (i)
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50.6 k (iv)	Y	Y	2.6 million (iii)
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Not available	Y	Y	Not available
Community: \$861,128 (i)	Y	Y	\$10.14 milioni for both ESMP + RAP (i): ESMP: 2,623,281 (i) RAP: 7,519,689 (i)

Y

Y

Unit cost available

Y

Y

N/A

Y

Y

Y: Y Y K442 million (zambia) - 2003

Y:
Mozambique: 5,562,970 (ii) Y Mozambique: 4,016,939 (ii)
Malawi: 3,979,781 (iii) Malawi: 2,353,376 (iii)

Au moins 1.200.000 \$ (ii)	Y	Y	n/a
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16.2 million in total for both
ESMP + RAP

Y

Y

Y

Y

US \$ 538,382.13

\$1,615,000 Y

Y

\$11 million

US\$1 million for both ESMP +
RAP

Y

N/A

Y

Y

74511 USD (ii)

US\$30 million). This subcomponent will implement the Environmental and Social Management Plans as defined in the ESIs, Resettlement Policy Framework, and the RAPs.

Y

Y

US\$ 14,067,442.76 (ii)

Y

Y

Y

Y

9 million for total

10 million (includng RAP)	Y	Y	7 million for transmission section (i)
Community development: \$583,000 (iii)	Y	Y	\$5,901.997.70 + 25% Contingency (\$1,180,399.54) (ii)
3.88 million for both ESMP + RAP (i)	Y	Y	

		Y	12,950,207 (i)
	Y	Y	
USD465,000.00	Y	Y	\$5 million (iii)

The cost of implementing the Management Plan (ESMP) is estimated to be about USD 64,610, USD 74,355 for Monitoring and USD 550,000 for Capacity building activities and will be paid by project developer (ZECO).(ii)	Y	N/A
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EMP and RAP (US\$1.30 million).	Y	Y
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No	Project ID	Region	Country	Project Name	Start Date
1	P088867	Europe and	Serbia	Energy Commu	30-Jun-05
2	P105329	East Asia and	Cambodia	GMS Power Trad	5-Jun-07
3	P105329	East Asia and	Cambodia	GMS Power Trad	5-Jun-07
4	P105329	East Asia and	Laos	GMS Power Trad	5-Jun-07
5	P105329	East Asia and	Laos	GMS Power Trad	5-Jun-07
6	P112893	South Asia	Nepal	Kabeli Transmiss	May-11
7	P133305	Eastern and S	Uganda	Uganda Grid Exp	31-May-16
8	P174034	Western and	Niger	Niger Acceleratir	10-Dec-21
9	P101645	Eastern and S	Tanzania	TZ-Energy Develo	13-Dec-07
10	P103037	Eastern and S	Kenya	Electricity Expan	27-May-10
11	P169561	Eastern and S	Tanzania	Zanzibar Energy	3-Jun-21
12	P110075	Western and	Benin	Increased Access	23-Jun-09
13	P131558	East Asia and	Vietnam	Transmission Eff	August-14
14	P131558	East Asia and	Vietnam	Transmission Eff	August-14
15	P131558	East Asia and	Vietnam	Transmission Eff	August-14
16	P131558	East Asia and	Vietnam	Transmission Eff	August-14
17	P147348	Europe and C	Georgia	Transmission Gri	May-14
18	P084871	East Asia ar	Vietnam	Second Transn	July-05
19	P069258	Eastern and S	DRC, Zambia	AFCC2/RI-3A-Sou	11-Nov-03
20	P069258	Eastern and S	DRC, Zambia	AFCC2/RI-3A-Sou	11-Nov-03
21	P119737	Eastern and S	Uganda	Electricity Sector	30-Jun-11
22	P147921	Western and	Senegal	OMVS - TRANSM	May-17
36	P110177	Western and	Cameroon	Cameroon - Kribi	10-Nov-11
25	P094919	Western and	Ghana, Burkina Faso	WAPP: The First	June-11
23	P152755	Western and	Cameroon	Electricity Transr	December-16
24	P152755	Western and	Cameroon	Electricity Transr	December-16
27	P166042	Western and	Guinea, Mali	Guinea – Mali In	July-18
28	P166042	Western and	Guinea, Mali	Guinea – Mali In	July-18
26	P146830	Western and	Guinea, The Gambia, G	OMVG Interconr	April-15
30	P168273	Middle East a	Tunisia	Tunisia Energy S	June-19
31	P168273	Middle East a	Tunisia	Tunisia Energy S	June-19
32	P168273	Middle East a	Tunisia	Tunisia Energy S	June-19
29	P168185	Western and	Cameroon, Chad	Cameroon - Cha	June-20
33	P113266	Western and	Côte d'Ivoire, Liberia, Si	WAPP-APL4 (Pha	May-12
34	P162933	Western and	Burkina Faso	Regional Power:	31-Oct-18
35	P162933	Western and	Burkina Faso	Regional Power:	31-Oct-18
37	P173763	Eastern and S	Djibouti; Ethiopia	Second Djibouti-	29-Mar-22
38	P095965	South Asia	Bangladesh	Siddhirganj Powe	30-Oct-08
39	P074011	Eastern and S	Ethiopia; Sudan	Ethiopia/Nile Ba	20-Dec-07

40	P114766	Europe and Kazakhstan	Kazakhstan Mo	September-09
41	P096207	Europe and Ukraine	Power Transmi	August-07
42	P075994	Western and Ghana, Benin	WAPP APL 1 (1st	June-05
43	P162933	Western and Nigeria	Regional Power:	31-Oct-18
44	P162933	Western and Niger	Regional Power:	31-Oct-18
45	P162933	Western and Niger	Regional Power:	31-Oct-18
46	P162933	Western and Niger	Regional Power:	31-Oct-18
47	P162933	Western and Niger	Regional Power:	31-Oct-18
48	P162933	Western and Togo/Benin	Regional Power:	31-Oct-18
49	P162933	Western and Burkina Faso	Regional Power:	31-Oct-18
50	P112516	Eastern and S Botswana	Botswana - Mor	October-09
51	P163752	Eastern and S Tanzania, Zambia	AFR RI-3A Tanza	June-18
52	P152755	Western and Cameroon	Electricity Transr	December-16
53	P152755	Western and Cameroon	Electricity Transr	December-16
54	P115767	South Asia Nepal	Nepal-India Elect	June-11
55	P115767	South Asia Nepal	Nepal-India Elect	June-11
56	P115767	South Asia Nepal	Nepal-India Elect	June-11
57	P111598	Eastern and S Tanzania	Tanzania - Backb	August-10
58	P160427	Eastern and S Mozambique	Temane Regiona	June-19
59	P168273	Middle East a Tunisia	Tunisia Energy S	June-19
60	P160829	Europe and C Moldova	Moldova Power	8-May-19
61	P082337	Europe and C Macedonia	ECSEE APL 3 - FY	10-Jan-06
62	P164354	Eastern and S Mozambique; Malawi	Mozambique - M	17-Sep-19
63	P164354	Eastern and S Mozambique; Malawi	Mozambique - M	17-Sep-19
64	P097201	Eastern and S DR Congo	AFCC2/RI-Region	29-May-07
65	P167898	South Asia Pakistan (mainly), Kyrgyz	Additional Financ	May-19
66	P131558	East Asia and Vietnam	Transmission Effi	August-14
67	P145054	South Asia Kyrgyz Republic, Tajikist	Central Asia Sou	March-14
68	P145054	South Asia Kyrgyz Republic, Tajikist	Central Asia Sou	March-14
69	P095155	Europe and Kazakhstan	North-South Elk	October-05
70	P084871	East Asia ar Vietnam	Second Transn	July-05
71	P126579	Eastern and S Ethiopia; Kenya	AFR RI-The Easte	12-Jul-12
72	P169117	Europe and C Georgia	Energy Supply R	23-May-19
73	P167843	South Asia Pakistan	Additional Financ	March-20
74	P166170	Eastern and S Eswatini	Network Reinfor	27-Jun-19
75	P095982	South Asia Pakistan	Electricity Distrib	June-08
76	P174812	Latin America Peru	Transmission Inv	September-21
77	P084404	Eastern and S Mozambique; Malawi	Mozambique Tra	July-07
78	P157055	Western and Cote d'Ivoire	Cote d'Ivoire - El	March-17
79	P129920	South Asia Bangladesh	Bangladesh: Rur	February-14
80	P064844	East Asia and Cambodia	Rural Electrificat	December-03
81	P094917	Western and Ghana, Benin	WAPP APL 1 (2nd	June-06
82	P119939	Europe and C Uzbekistan	Talimarjan Trans	March-11

Start year	MUV Index	End Date	Inter-connection	Time Duration	Route length (km)
2005	87.70	30-Jun-12	N	7.01	34.9
2007	95.43	31-Dec-11	N	4.58	64
2007	95.43	31-Dec-11	N	4.58	56
2007	95.43	31-Dec-11	N	4.58	26
2007	95.43	31-Dec-11	N	4.58	25
2011	111.04	December-16	N	5.59	90
2016	94.03	30-Apr-24	N	7.92	314
2021	109.40	31-May-27	N	5.47	270
2007	95.43	29-Sep-17	N	9.80	51.7
2010	100.00	31-Dec-17	N	7.60	334
2021	109.40	31-Aug-27	N	6.25	100
2009	96.46	31-Dec-18	N	9.53	280
2014	108.25	December-21	N	7.34	50.6
2014	108.25	December-21	N	7.34	71.83
2014	108.25	December-21	N	7.34	31.1
2014	108.25	December-21	N	7.34	139.89
2014	108.25	March-23	N	8.84	142
2005	87.70	June-14	N	8.92	78.3
2003	79.62	30-Sep-16	N	12.90	286
2003	79.62	30-Sep-16	Y	12.90	40
2011	111.04	31-Jan-19	N	7.59	137
2017	97.29	December-22	N	5.59	288
2011	111.04	18-May-13	N	1.52	100
2011	111.04	December-18	Y	7.51	210
2016	94.03	June-25	N	8.58	12.5
2016	94.03	June-25	N	8.58	4
2018	101.86	December-26	Y	8.51	714
2018	101.86	December-26	Y	8.51	53.7
2015	97.87	June-25	Y	10.25	1677
2019	99.48	September-26	N	7.34	100
2019	99.48	September-26	N	7.34	85
2019	99.48	September-26	N	7.34	6.5
2020	97.90	December-27	Y	7.50	517
2012	110.23	March-23	Y	10.84	1,349
2018	101.86	15-Dec-24	N	6.13	9
2018	101.86	15-Dec-24	N	6.13	15
2022	110.40	31-Dec-26	Y	4.76	192
2008	102.83	31-Dec-19	N	11.18	11
2007	95.43	30-Jun-13	N	5.53	

2009	96.46	April-13	N	3.58	322
2007	95.43	June-16	N	8.84	77
2005	87.70	June-13	Y	8.01	215
2018	101.86	15-Dec-24	N	6.13	62
2018	101.86	15-Dec-24	N	6.13	28
2018	101.86	15-Dec-24	N	6.13	108
2018	101.86	15-Dec-24	N	6.13	180
2018	101.86	15-Dec-24	N	6.13	104
2018	101.86	15-Dec-24	Y	6.13	12
2018	101.86	15-Dec-24	N	6.13	381
2009	96.46	August-14	N	4.84	317
2018	101.86	June-26	Y	8.08	620
2016	94.03	June-25	N	8.58	167
2016	94.03	June-25	N	8.58	25
2011	111.04	October-21	N	10.34	90
2011	111.04	October-21	N	10.34	40
2011	111.04	October-21	N	10.34	285
2010	100.00	December-16	N	6.34	667
2019	99.48	December-26	N	7.59	563
2019	99.48	September-26	N	7.34	192
2019	99.48	31-Aug-24	N	5.32	158
2006	89.93	31-Jul-16	N	10.56	19
2019	99.48	30-Jun-24	N	4.79	142
2019	99.48	30-Jun-24	N	4.79	76
2007	95.43	30-Jun-18	N	11.10	
2019	99.48		Y		100
2014	108.25	December-21	N	7.34	39.97
2014	108.25	December-28	Y	14.85	750
2014	108.25	December-28	Y	14.85	475
2005	87.70	November-11	N	6.09	462.57
2005	87.70	June-14	N	8.92	276.5
2012	110.23	31-Jan-24	Y	11.56	1045
2019	99.48	31-Dec-25	N	6.61	77
2020	97.90		N		255
2019	99.48	31-Oct-24	N	5.35	87
2008	102.83	February-14	N	5.67	
2021	109.40	November-26	N	5.17	
2007	95.43	August-16	Y	9.09	
2017	97.29	September-23	N	6.56	
2014	108.25	June-21	N	7.33	
2003	79.62	January-12	N	8.09	
2006	89.93	June-16	Y	10.01	
2011	111.04	June-16	N	5.26	

Length Group	Voltage (kv)	Circuit	Circuit	Capital cost for transmission component (\$million) (nominal)
25 - 50 km	110 kv - 161 kv	Single Circuit	HVAC	\$4.00
50 - 100 km	110 kv - 161 kv	Double Circuit	HVAC	\$4.03
50 - 100 km	110 kv - 161 kv	Double Circuit	HVAC	\$3.64
25 - 50 km	110 kv - 161 kv	Double Circuit	HVAC	\$1.53
25 - 50 km	110 kv - 161 kv	Double Circuit	HVAC	\$1.49
50 - 100 km	110 kv - 161 kv	Double Circuit	HVAC	\$12.65
over 100 km	110 kv - 161 kv	Double Circuit	HVAC	\$58.00
over 100 km	110 kv - 161 kv	Double Circuit	HVAC	\$18.02
50 - 100 km	110 kv - 161 kv	Double Circuit	HVAC	\$5.83
over 100 km	110 kv - 161 kv	Double Circuit	HVAC	\$58.20
over 100 km	110 kv - 161 kv	Double Circuit	HVAC	\$16.00
over 100 km	110 kv - 161 kv	Double Circuit	HVAC	\$35.28
50 - 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$16.60
50 - 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$26.43
25 - 50 km	220 kv - 230 kv	Double Circuit	HVAC	\$13.30
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$62.12
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$45.65
50 - 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$39.02
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$79.20
25 - 50 km	220 kv - 230 kv	Double Circuit	HVAC	\$18.00
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$35.69
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$74.64
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$24.00
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$34.20
less than 25 km	220 kv - 230 kv	Single Circuit	HVAC	\$3.00
less than 25 km	220 kv - 230 kv	Single Circuit	HVAC	\$1.00
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$317.00
50 - 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$15.40
over 100 km	220 kv - 230 kv	Single Circuit	HVAC	\$397.00
over 100 km	220 kv - 230 kv	Single Circuit	HVAC	\$15.50
50 - 100 km	220 kv - 230 kv	Single Circuit	HVAC	\$18.60
less than 25 km	220 kv - 230 kv	Single Circuit	HVAC	\$1.80
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$178.00
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$292.80
less than 25 km	220 kv - 230 kv	Double Circuit	HVAC	\$2.23
less than 25 km	220 kv - 230 kv	Single Circuit	HVAC	\$2.48
over 100 km	220 kv - 230 kv	Double Circuit	HVAC	\$41.00
less than 25 km	220 kv - 230 kv	Double Circuit	HVAC	\$4.53
	220 kv - 230 kv		HVAC	\$27.40

over 100 km	220 kv - 230 kv	Single Circuit	HVAC	\$45.80
50 - 100 km	330 kv	Single Circuit	HVAC	\$34.90
over 100 km	330 kv	Double Circuit	HVAC	\$47.41
50 - 100 km	330 kv	Double Circuit	HVAC	\$22.92
25 - 50 km	330 kv	Double Circuit	HVAC	\$10.35
over 100 km	330 kv	Double Circuit	HVAC	\$31.69
over 100 km	330 kv	Double Circuit	HVAC	\$66.53
over 100 km	330 kv	Double Circuit	HVAC	\$38.44
less than 25 km	330 kv	Single Circuit	HVAC	\$6.53
over 100 km	330 kv	Double Circuit	HVAC	\$140.82
over 100 km	400 kv	Double Circuit	HVAC	\$101.50
over 100 km	400 kv	Double Circuit	HVAC	\$255.00
over 100 km	400 kv	Double Circuit	HVAC	\$60.00
25 - 50 km	400 kv	Double Circuit	HVAC	\$9.00
50 - 100 km	400 kv	Double Circuit	HVDC	\$27.78
25 - 50 km	400 kv	Double Circuit	HVDC	\$17.08
over 100 km	400 kv	Double Circuit	HVDC	\$85.74
over 100 km	400 kv	Double Circuit	HVAC	\$348.17
over 100 km	400 kv	Single Circuit	HVAC	\$160.86
over 100 km	400 kv	Double Circuit	HVAC	\$53.80
over 100 km	400 kv	Single Circuit	HVAC	\$46.50
less than 25 km	400 kv	Double Circuit	HVAC	\$5.42
over 100 km	400 kv	Double Circuit	HVDC	\$48.90
50 - 100 km	400 kv	Double Circuit	HVDC	\$23.06
	400 kv	Double Circuit	HVAC	\$82.60
over 100 km	500 kv		HVDC	\$39.62
25 - 50 km	500 kv	Double Circuit	HVAC	\$30.93
over 100 km	500 kv	Double Circuit	HVDC	\$296.00
over 100 km	500 kv	Double Circuit	HVAC	\$200.00
over 100 km	500 kv	Single Circuit	HVAC	\$113.80
over 100 km	500 kv	Double Circuit	HVAC	\$237.44
over 100 km	500 kv		HVDC	\$308.10
50 - 100 km	500 kv	Double Circuit	HVAC	\$60.00
over 100 km	765 kv	Double Circuit	HVAC	\$715.30
50 - 100 km	110 kv - 161 kv	Double Circuit		
	220 kv - 230 kv	Double Circuit		
	220 kv - 230 kv	Not available		
	220 kv - 230 kv	Double Circuit		
	220 kv - 230 kv	Not available		
	220 kv - 230 kv	Double Circuit		
	220 kv - 230 kv	Double Circuit		
	330 kv	Single Circuit		
	500 kv	Single Circuit		

Cost per km (million) (nominal)	Capital cost for transmission component (\$million) (constant: 2010)	Cost per km (million) (constant: 2010)	Discount Rate	O&M Rate
\$0.11	\$4.56	\$0.13	12%	2%
\$0.06	\$4.23	\$0.07	10%	2.5%
\$0.07	\$3.82	\$0.07	10%	2.5%
\$0.06	\$1.60	\$0.06	10%	2.5%
\$0.06	\$1.56	\$0.06	10%	2.5%
\$0.14	\$11.39	\$0.13	10%	2%
\$0.18	\$61.68	\$0.20	10%	2%
\$0.07	\$16.47	\$0.06	10%	
\$0.11	\$6.11	\$0.12	10%	2.5%
\$0.17	\$58.20	\$0.17	12%	3.5%
\$0.16	\$14.63	\$0.15	6.4%	2%
\$0.13	\$36.57	\$0.13	10%	
\$0.33	\$15.34	\$0.30	10%	2%
\$0.37	\$24.42	\$0.34	10%	2%
\$0.43	\$12.29	\$0.40	10%	2%
\$0.44	\$57.39	\$0.41	10%	2%
\$0.32	\$42.17	\$0.30	12%	2%
\$0.50	\$44.49	\$0.57	10%	2%
\$0.28	\$99.48	\$0.35	12%	2%
\$0.45	\$22.61	\$0.57	12%	2%
\$0.26	\$32.14	\$0.23	12%	2.5%
\$0.26	\$76.72	\$0.27	10%	2%
\$0.24	\$21.61	\$0.22	10%	
\$0.16	\$30.80	\$0.15	10%	3%
\$0.24	\$3.19	\$0.26	6%	5%
\$0.25	\$1.06	\$0.27	6%	5%
\$0.44	\$311.22	\$0.44	10%	3%
\$0.29	\$15.12	\$0.28	10%	3%
\$0.24	\$405.65	\$0.24	10%	2%
\$0.16	\$15.58	\$0.16	10%	2%
\$0.22	\$18.70	\$0.22	10%	2%
\$0.28	\$1.81	\$0.28	10%	2%
\$0.34	\$181.82	\$0.35	6%	2%
\$0.22	\$265.63	\$0.20	10%	
\$0.25	\$2.18	\$0.24	6%	1%
\$0.17	\$2.43	\$0.16	6%	1%
\$0.21	\$37.14	\$0.19		
\$0.41	\$4.41	\$0.40	12%	2.5%
	\$28.71		10%	

\$0.14	\$47.48	\$0.15	8%	2%
\$0.45	\$36.57	\$0.47	10%	
\$0.22	\$54.06	\$0.25	10%	
\$0.37	\$22.50	\$0.36	6%	1%
\$0.37	\$10.16	\$0.36	6%	1%
\$0.29	\$31.11	\$0.29	6%	1%
\$0.37	\$65.32	\$0.36	6%	1%
\$0.37	\$37.74	\$0.36	6%	1%
\$0.54	\$6.41	\$0.53	6%	1%
\$0.37	\$138.25	\$0.36	6%	1%
\$0.32	\$105.22	\$0.33	12%	
\$0.41	\$250.35	\$0.40	10%	2%
\$0.36	\$63.81	\$0.38	6%	5%
\$0.36	\$9.57	\$0.38	6%	5%
\$0.31	\$25.02	\$0.28	12%	2%
\$0.43	\$15.38	\$0.38	12%	2%
\$0.30	\$77.22	\$0.27	12%	2%
\$0.52	\$348.17	\$0.52	10%	1%
\$0.29	\$161.70	\$0.29	10%	4%
\$0.28	\$54.08	\$0.28	10%	2%
\$0.29	\$46.74	\$0.30	6%	
\$0.29	\$6.03	\$0.32	10%	
\$0.34	\$49.15	\$0.35	6%	2%
\$0.30	\$23.18	\$0.30	6%	2%
	\$86.56		12%	2.5%
\$0.40	\$39.83	\$0.40	10%	4%
\$0.77	\$28.57	\$0.71	10%	2%
\$0.39	\$273.45	\$0.36	10%	4%
\$0.42	\$184.76	\$0.39	10%	4%
\$0.25	\$129.75	\$0.28	10%	1%
\$0.86	\$270.73	\$0.98	10%	2%
\$0.29	\$279.51	\$0.27	10%	2%
\$0.78	\$60.31	\$0.78		
\$2.81	\$730.64	\$2.87	0%	
			6%	1%
			12%	3%
			8%	4%
			6%	2.5%
			10%	5%
			12%	4%
			12%	4%
			10%	
			10%	2%

Project type (if new construction)	Routing (Existing vs new)	Implementation Delays (Y/N)
	N/A	Y: supply (ii)
Construction	Existing	N/A
Construction	Existing	N/A
Construction	Existing	N/A
Construction	Existing	N/A
Construction	New + existing	Y:Procurement, management, re:
Construction	Existing	Y: COVID-19; (i) Right of Way (Ro
Construction	New	
Construction	Existing	Y: procurement
Construction	New	Y:(due to right-of-way,compensat
Construction and upgrade	Existing	N/A
Construction	New	Y: a slow procurement process, lc
Construction	New + existing	Y: resettlement matters
Construction	New + existing	Y: resettlement matters
Construction	New + existing	Y: resettlement matters
Construction	New + existing	Y: resettlement matters
Construction	N/A	Y: resettlement matters
Upgrade	Existing (mainly)	Y: Inflation, management
Construction	Existing	Y: DRC's post-conflict situation ar
Construction	Existing	Y: DRC's post-conflict situation ar
Construction	New	Y: land acquisition challenges; pr
Construction	Existing	Y:COVID, procurement, political s
Construction	Existing	N/A for the transmission compor
Construction	Existing (mainly)	Y: Procurement
Upgrade	N/A	Y: procurement, preparation and
Upgrade	N/A	Y: procurement, preparation and
Construction	N/A	N/A
Construction	N/A	N/A
Construction	New	Y: Financing, RAP, management
Construction and upgrade	N/A	N/A
Construction and upgrade	N/A	N/A
Construction and upgrade	N/A	N/A
Construction	New	N/A
Construction	New	Y: Covid, construction, procurem
New construction	New	Y: COVID on the hiring process of
New construction	New	Y: COVID on the hiring process of
New construction	Existing (mainly)	N/A
Construction	New	Y:in procurement and lengthy so
Construction and upgrade	New + existing	N/A for the transmission compor

Construction(upgrade of substation	New	Y:Procurement, construction
Construction and upgrade	New	Y:Construction - substation
Construction and upgrade	New + existing	Y: Procurement, rerouting, mana
New construction	New	Y: COVID on the hiring process of
New construction	New	Y: COVID on the hiring process of
New construction	New	Y: COVID on the hiring process of
New construction	New	Y: COVID on the hiring process of
New construction	New	Y: COVID on the hiring process of
New construction	New	Y: COVID on the hiring process of
Construction	Existing	Y:Procurement, studies
Construction	New	N
Upgrade	N/A	Y: procurement, preparation and
Upgrade	N/A	Y: procurement, preparation and
Construction	New	Y:procurement, management, lar
Construction	Existing	Y:procurement, management, lar
Construction	Existing	Y:procurement, management, lar
Construction	Existing	Y: Procurement, government mai
Construction	New	N/A
Construction and upgrade	N/A	N/A
Construction	New	N/A
Construction and upgrade	New	Y: additional financing
New construction	New	Y: Covid-19; Resettlement Actior
New construction	New	Y: Covid-19; Resettlement Actior
Construction and upgrade		Y: - difficulties in coordination be
Construction	New	Y: inadequate market responses
Construction	New + existing	Y: resettlement matters
Construction	New + existing	Y: Procurement
Construction	New	Y: Procurement
Construction	New	Y:Sustation
Upgrade	Existing (mainly)	Y: Inflation, management
New construction	New + existing	Y: Construction; right of way con
Construction	New	Y: rerouting;
Construction	New	Y:Land acquisition responsibility;
Construction		N/A
Construction (mainly substation)		Y:procurement, management
Upgrade		N/A
Upgrade (mainly substations)		Y: Procurement, restructuring prc
Upgrade and Construction (mainly the substations)		Y:Procurement, COVID, environm
Construction and upgrade		Y:Land acquisition
Construction and upgrade		Y:Procurement, management, lar
Upgrade (mainly substations)		Y : Procurement, disbursement,
Construction		Y: Management

ESMP (Y/N)	RAP (Y/N)
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Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	N/A
Y	Y
Y	Y
Y	N/A
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y

Procurement, h	Y
----------------	---

Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y
Y	Y

Main Graphs Tab

This tab summarizes the graphs needed for the discussions in the paper. As the tables a

1. Unit Cost per km by voltage

Unit cost per km by voltage (\$ million, 2010 value)

Assest Range	Average Unit Cost	World Bank Data		Data Sample Count
		Unit Cost Lower	Unit Cost Upper	
110 - 161 kV Single Circuit	0.131	0.131	0.131	1
110 - 161 kV Double Circuit	0.110	0.061	0.196	11
220 - 230 kV Single Circuit	0.216	0.147	0.278	8
220 - 230 kV Double Circuit	0.326	0.147	0.568	19
330 - 345 kV Single Circuit	0.504	0.475	0.534	2
330 - 345 kV Double Circuit	0.336	0.251	0.363	7
380 - 400 kV Single Circuit	0.292	0.287	0.296	2
380 - 400 kV Double Circuit	0.351	0.271	0.522	12
500 kV Single Circuit	0.281	0.281	0.281	1
500 kV Double Circuit	0.646	0.365	0.979	5
765 kV Double Circuit	2.865	2.865	2.865	1

2. Cost by length

Length Group	110 - 161 kv Double Circuit		220 - 230	
	Cost per km	Count	Rate	Cost per km
less than 25 km				0.322
25 - 50 km	0.062	2	1.00	0.480
50 - 100 km	0.095	4	1.53	0.373
over 100 km	0.142	5	2.29	0.282

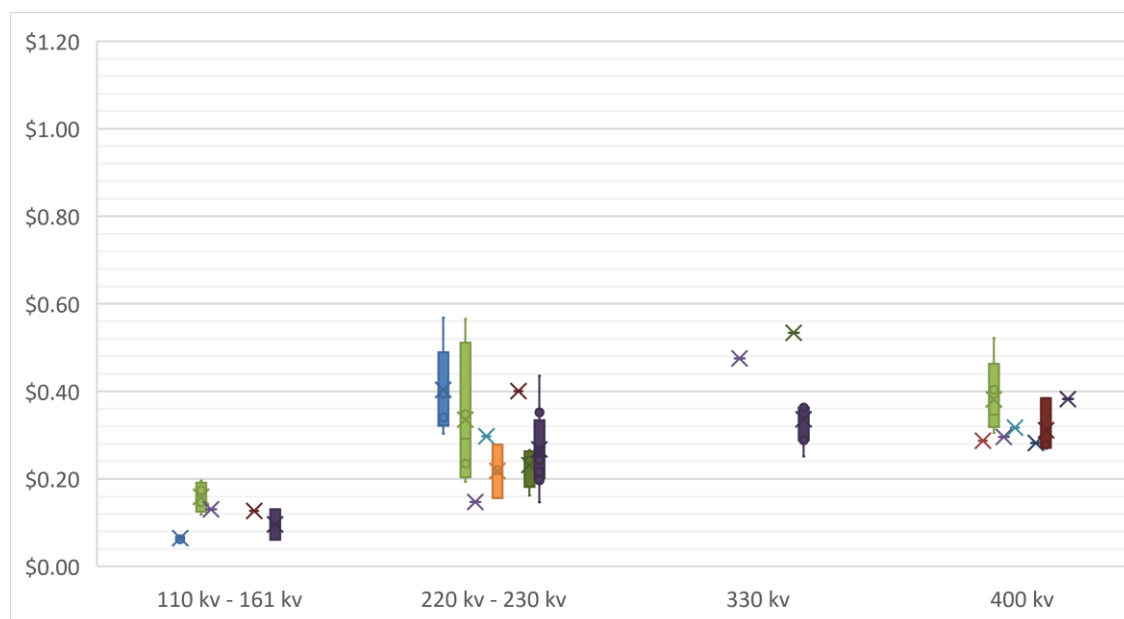
Average of Cost per km (million) (constant: 2010)

3. Regional comparison

Region	Voltage (kv)			
	110 kv - 161 kv	220 kv - 230 kv	330 kv - 340 kv	380 kv - 400 kv
East Asia and Pacific	0.065	0.403		

Eastern and Southern Africa	0.159	0.335		0.366
Europe and Central Asia	0.131	0.222	0.475	0.307
Middle East and North Africa		0.218		0.282
South Asia	0.127	0.400		0.311
Western and Central Africa	0.096	0.255	0.361	0.382
Total Average	0.112	0.293	0.374	0.342

4. Boxer chart for regional comparison

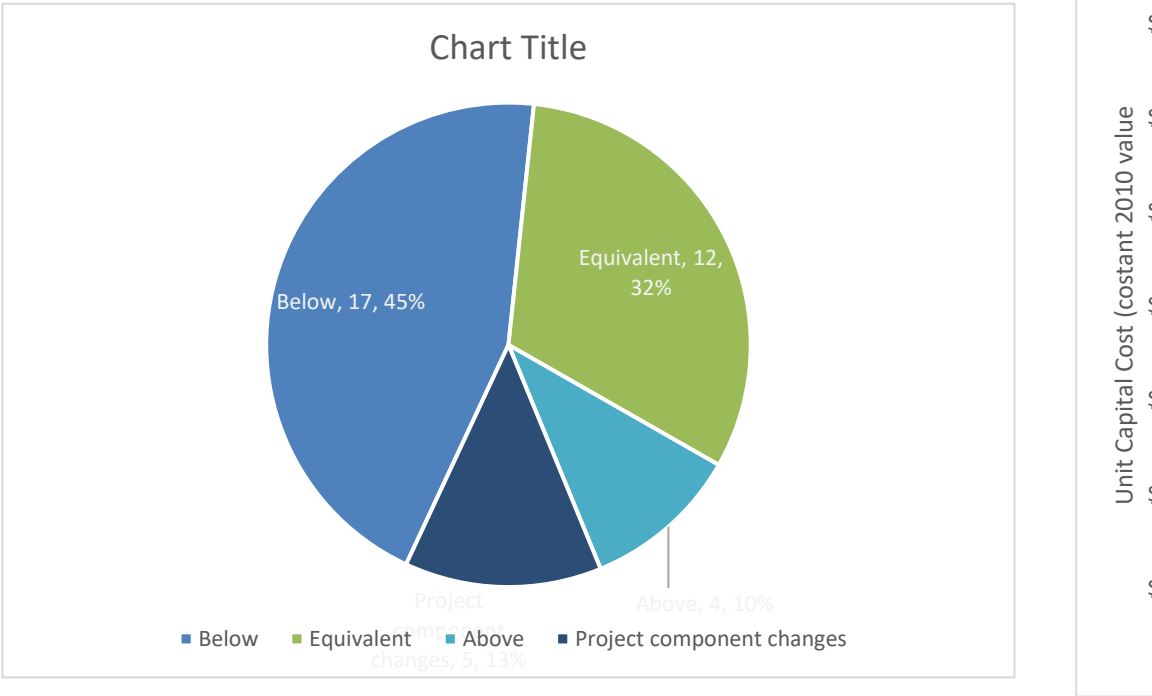


5. Decade comparison

2000s: Average of Cost per km (million) (constant 2010 value)				
Region	110 kv - 161 kv	220 kv - 230 kv	330 kv	400 kv
East Asia and Pacific	0.065	0.568		
Eastern and Southern Africa	0.146	0.457		0.374
Europe and Central Asia	0.131	0.147	0.475	0.317
South Asia		0.400		
Western and Central Africa	0.131		0.251	
Total Average	0.101	0.406	0.363	0.360

6. Appraisal vs Actual

Below	17
Equivalent	12
Above	4
Project component changes	5



Adjusted value (2010 value)

	Voltage	Region	Appraisal cos	Actual cost
P088867	110 kv - 161 kv	Europe and Cer	\$0.13	\$0.07

P103037	110 kv - 161	Eastern and Southern Africa	\$0.17	\$0.15
P131558	220 kv - 230	East Asia and Pacific	\$0.57	\$0.50
P114766	220 kv - 230	Europe and Central Asia	\$0.15	\$0.12
P069258	220 kv - 230	Eastern and Southern Africa	\$0.57	\$0.53
P119737	220 kv - 230	Eastern and Southern Africa	\$0.23	\$0.10
P113266	220 kv - 230	Western and Central Africa	\$0.20	\$0.17
P096207	330 kv	Europe and Central Asia	\$0.47	\$0.44
P111598	400 kv	Eastern and Southern Africa	\$0.52	\$0.29
P112516	400 kv	Eastern and Southern Africa	\$0.33	\$0.15
P082337	400 kv	Europe and Central Asia	\$0.30	\$0.24
P131558	500 kv	East Asia and Pacific	\$0.98	\$0.74
P126579	500 kv	Eastern and Southern Africa	\$0.27	\$0.22
P095155	500 kv	Europe and Central Asia	\$0.28	\$0.16

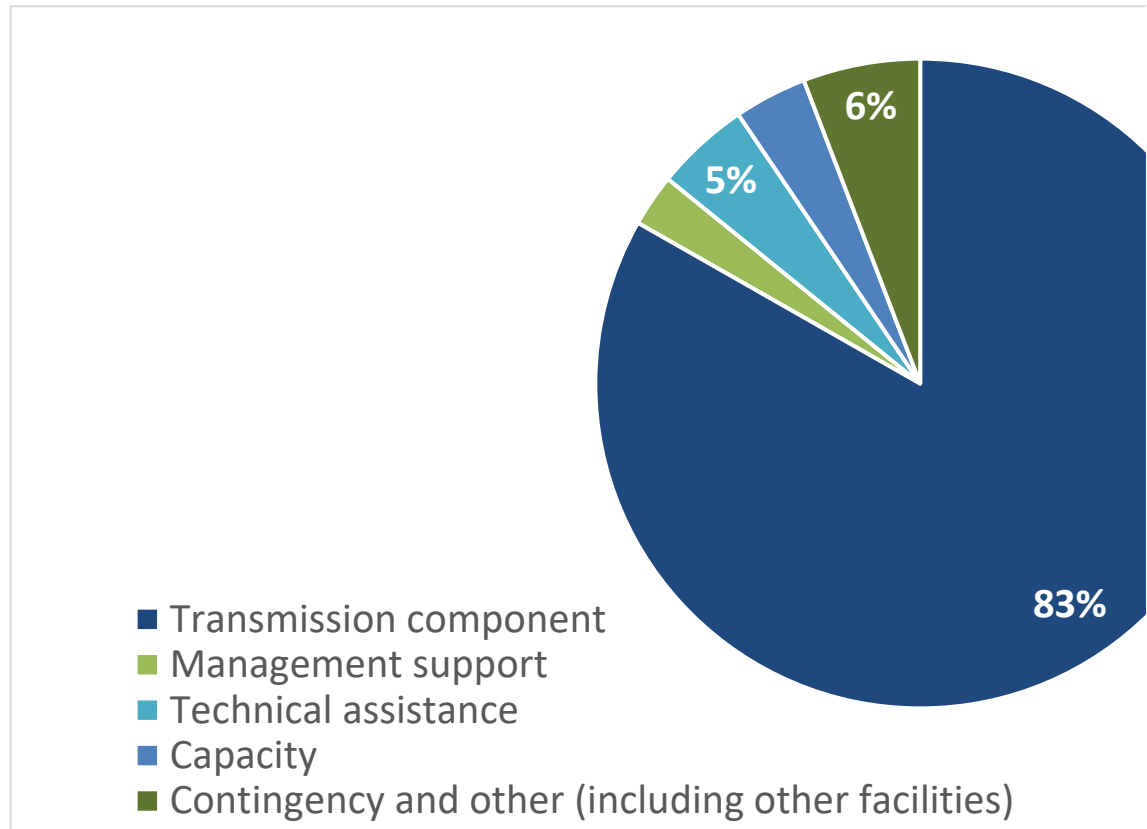
6. O&M Rate Data

Region	Voltage (kv)			
	110 kv - 161 kv	220 kv - 230 kv	330 kv	400 kv
East Asia and Pacific	2.50%	2.33%		
Eastern and Southern Africa	2.20%	2.25%		2.00%
Europe and Central Asia	2.00%	1.75%		
Middle East and North Africa		2.00%		2.00%
South Asia	1.50%	3.17%		1.50%
Western and Central Africa		2.86%	1.00%	5.00%
Total Average	2.23%	2.55%	1.00%	2.35%

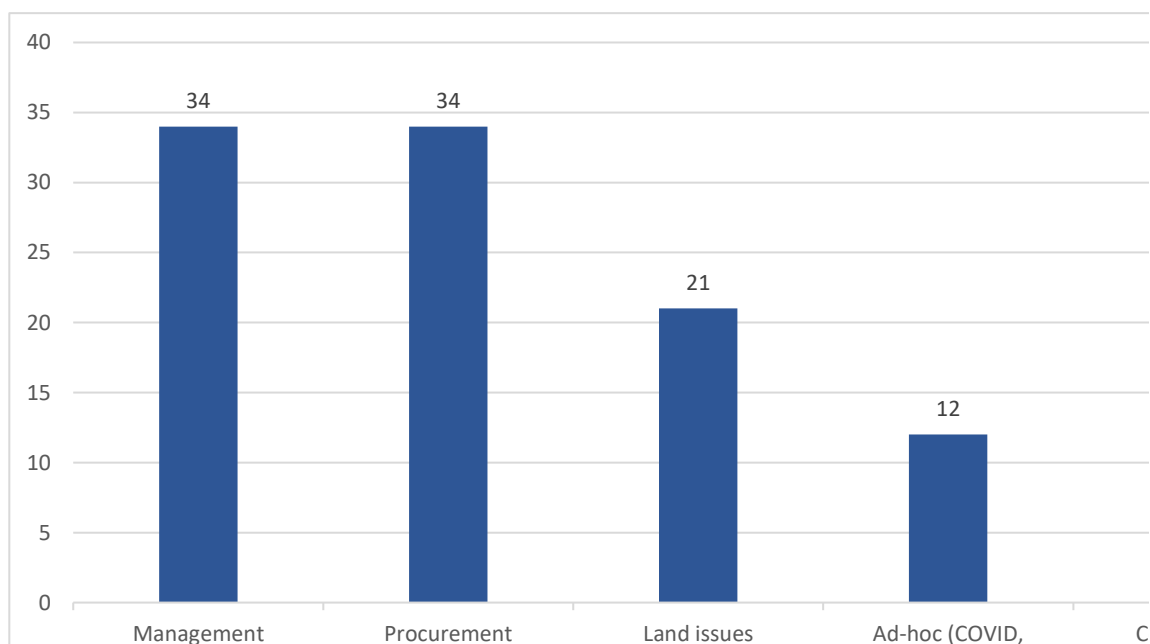
6. Applied Discount Rate Data

Region	Voltage (kv)			
	110 kv - 161 kv	220 kv - 230 kv	330 kv	400 kv
East Asia and Pacific	10.00%	10.29%		
Eastern and Southern Africa	8.88%	10.40%		9.50%
Europe and Central Asia	12.00%	10.00%	10.00%	8.00%
Latin America and Caribbean		8.00%		
Middle East and North Africa		10.00%		10.00%
South Asia	10.00%	12.00%		12.00%
Western and Central Africa	10.00%	8.46%	6.89%	6.00%
Total Average	9.72%	9.65%	7.45%	9.38%

7. WB Funding Breakdown



8. Project Restructure/delay reason analysis



9. Decade Comparison

2000s: Average of Cost per km (million) (constant 2005 US\$)			
Region	110 kv - 161 kv	220 kv - 230 kv	330 kv
East Asia and Pacific	0.065	0.568	
Eastern and Southern Africa	0.146	0.457	
Europe and Central Asia	0.131	0.147	0.475
South Asia		0.400	
Western and Central Africa	0.131		0.251
Total Average	0.101	0.406	0.363

2010s: Average of Cost per km (million) (constant 2010 US\$)			
Row Labels	110 kv - 161 kv	220 kv - 230 kv	330 kv
East Asia and Pacific		0.362	
Eastern and Southern Africa	0.196	0.235	
Europe and Central Asia		0.297	
Middle East and North Africa		0.218	
South Asia	0.127		
Western and Central Africa		0.255	0.377
Total Average	0.162	0.271	0.377

10. Interconnection

Project ID	Region	Country	Voltage (kv)
P084404	Eastern and Southern Africa	Mozambique	220 kv - 230 kv
P163752	Eastern and Southern Africa	Tanzania, Zanzibar	400 kv
P126579	Eastern and Southern Africa	Ethiopia; Kenya	500 kv
P173763	Eastern and Southern Africa	Djibouti; Ethiopia	220 kv - 230 kv
P167898	South Asia	Pakistan (mainland)	500 kv
P145054	South Asia	Kyrgyz Republic	500 kv
P162933	Western and Central Africa	Togo/Benin	330 kv
P168185	Western and Central Africa	Cameroon, Chad	220 kv - 230 kv
P146830	Western and Central Africa	Guinea, The Gambia	220 kv - 230 kv
P094919	Western and Central Africa	Ghana, Burkina Faso	220 kv - 230 kv
P166042	Western and Central Africa	Guinea, Mali	220 kv - 230 kv

P075994	Western and Central Africa	Ghana, Benin	330 kv
P113266	Western and Central Africa	Côte d'Ivoire	220 kv - 230 kv
P094917	Western and Central Africa	Ghana, Benin	330 kv

11. Interconnection

Voltage	Region	Unit Cost per km (million) (constant: 2010)	Average Discount Rate
220-230 kv	Eastern and Southern Africa	\$0.34	10.40%
	Individual	\$0.29	11.33%
	Interconnection	\$0.38	9.00%
	Western and Central Africa	\$0.26	8.46%
	Individual	\$0.23	7.71%
	Interconnection	\$0.28	9.33%
330 kv	Western and Central Africa	\$0.36	6.89%
	Individual	\$0.35	6.00%
	Interconnection	\$0.39	8.67%
400 kv	Eastern and Southern Africa	\$0.37	9.50%
	Individual	\$0.36	9.43%
	Interconnection	\$0.40	10.00%

11. New Construction

Voltage	Region	Unit Cost per km (million) (constant: 2010)	Average Discount Rate
220-230 kv	Western and Central Africa	\$0.22	9.00%
	Existing Routes	\$0.21	10.00%
	New Routing	\$0.24	8.00%

12. LCOE (Source: IEA)

Generation Source	LCOE - 2020 (\$/kwh)	LCOE - 2030 (\$/kwh)
Coal	\$0.046 - \$0.07	\$0.07 - \$0.148
Gas	\$0.025 - \$0.08	\$0.037 - \$0.109
Solar PV	\$0.04 - \$0.095	\$0.022 - \$0.049
Wind onshore	\$0.042 - \$0.1	\$0.039 - \$0.089

13. LCOT (Data Presentation)

Voltage Range	Transmission Capacity (MW)	Utilization Factor	MWh transferred
220/230	300	0.25	657000
		0.5	1314000
		0.8	2102400
330	600	0.25	1314000
		0.5	2628000
		0.8	4204800
400	1000	0.25	2190000
		0.5	4380000
		0.8	7008000

ire presented in one tab, space/format adjustment is necessary when needed.

Literature Benchmark Data				Average Unit Cost Ratio (World Bank vs Benchmark)
Difference	Average Unit Cost	Unit Cost Lower	Unit Cost Upper	
0.001	0.304	0.304	0.304	2.33
0.135	0.369	0.318	0.421	3.35
0.131	0.596	0.403	0.854	2.76
0.422	0.996	0.519	1.403	3.06
0.059	0.828	0.676	1.220	1.64
0.111	1.208	0.561	1.952	3.59
0.009	0.455	0.455	0.455	1.56
0.251	1.234	1.234	1.234	3.52
0.000	1.221	1.064	1.542	4.35
0.615	1.427	0.915	1.771	2.21
0.000				

110 kv Double Circuit			220 - 230 kv Single Circuit			330 kv Double Circuit
Count	Rate	Cost per km	Count	Rate	Cost per km	Count
2	0.67	0.240	4	1.09		
2	1.00				0.363	1
4	0.78	0.220	1	1.00	0.363	1
11	0.59	0.182	3	0.83	0.326	5

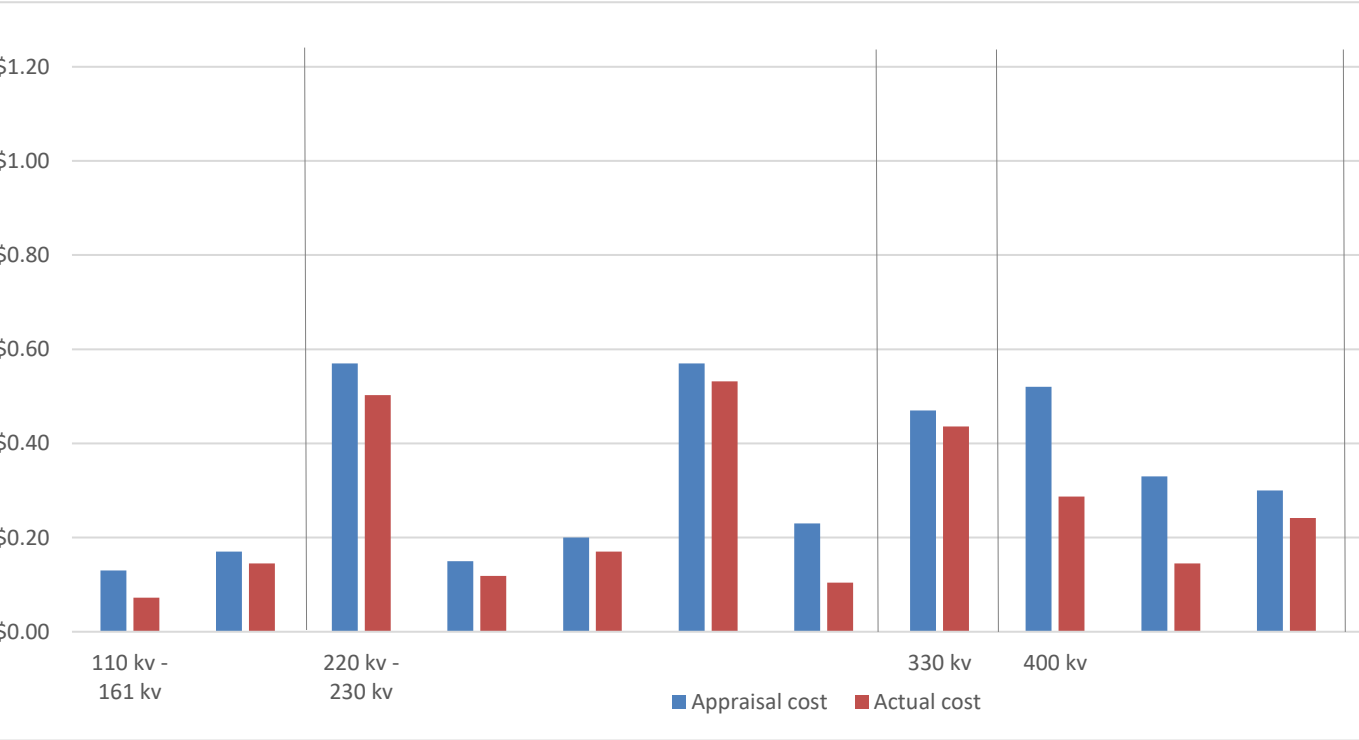
500 kv	765 kv
0.847	

0.267	
0.532	
0.384	2.865
0.522	2.865

Data is presented so for the visualization		
Voltage (kv)	East Asia and Pa	Eastern and Southern Africa
110 kv - 161 k	\$0.07	\$0.20
110 kv - 161 k	\$0.07	\$0.12
110 kv - 161 k	\$0.06	\$0.17
110 kv - 161 k	\$0.06	\$0.15
220 kv - 230 k	\$0.30	\$0.35
220 kv - 230 k	\$0.34	\$0.57
220 kv - 230 k	\$0.40	\$0.23
220 kv - 230 k	\$0.41	\$0.19
220 kv - 230 k	\$0.57	
220 kv - 230 kv		
220 kv - 230 kv		
220 kv - 230 kv		
330 kv		
330 kv		
330 kv		
330 kv		
330 kv		
330 kv		
400 kv	\$0.29	\$0.33
400 kv		\$0.40
400 kv		\$0.52
400 kv		\$0.35
400 kv		\$0.30
500 kv	\$0.71	\$0.27
500 kv	\$0.98	
500 kv		

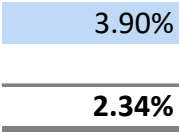
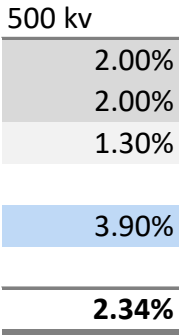
		2010s: Average of Cost per km (million) (constant 2010 value)			
500 kv	Row Labels	110 kv - 161 kv	220 kv - 230 kv	330 kv	400 kv
	0.979	East Asia and Pacific	0.362		
		Eastern and S	0.196	0.235	0.336
	0.281	Europe and Central Asia	0.297		0.296
		Middle East and North Africa	0.218		0.282
		South Asia	0.127		0.311
		Western and Central Africa	0.255	0.377	0.382
	0.630	Total Average	0.162	0.271	0.329

Graph comparison between actual vs appraisal values (adjusted 2010 value)



Percentage		Nominal Value			
		Voltage	Region	Appraisal cost	Actual c Difference
56%		110 kv - 161 kv	Europe and Central Asia	\$0.11	\$0.08
					\$0.03

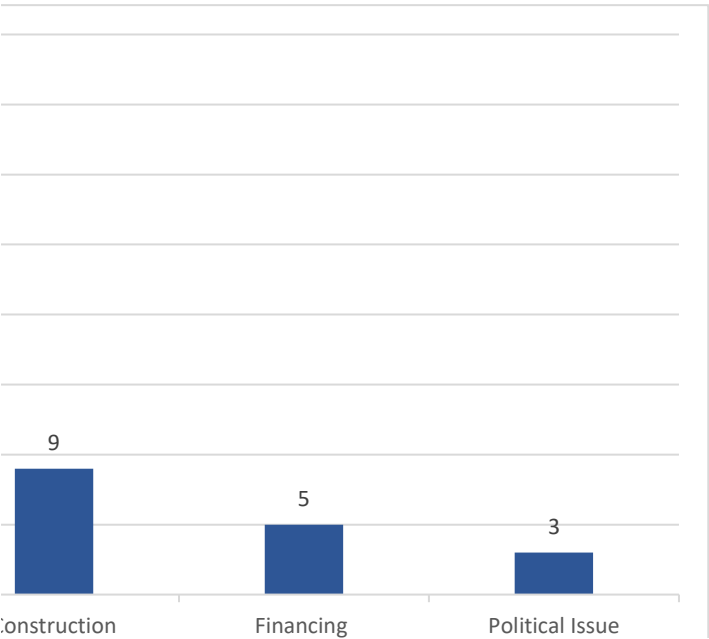
85%	110 kv - 161 Eastern and S	\$0.17	\$0.16	\$0.01
88%	220 kv - 230 East Asia and	\$0.50	\$0.55	\$0.05
79%	220 kv - 230 Europe and C	\$0.14	\$0.13	\$0.02
93%	220 kv - 230 Eastern and S	\$0.45	\$0.50	\$0.05
45%	220 kv - 230 Eastern and S	\$0.26	\$0.10	\$0.16
85%	220 kv - 230 Western and (	\$0.22	\$0.19	\$0.03
93%	330 kv Europe and C	\$0.45	\$0.41	\$0.04
55%	400 kv Eastern and S	\$0.52	\$0.27	\$0.25
44%	400 kv Eastern and S	\$0.32	\$0.16	\$0.16
80%	400 kv Europe and C	\$0.29	\$0.23	\$0.06
76%	500 kv East Asia and	\$0.86	\$0.81	\$0.05
81%	500 kv Eastern and S	\$0.29	\$0.25	\$0.04
58%	500 kv Europe and C	\$0.25	\$0.18	\$0.07



500 kv	Region	
10.00%	South Asia	10.18%
10.00%	East Asia and	10.14%
9.50%	Middle East a	10.00%
	Europe and C	9.64%
	Eastern and S	9.60%
10.00%	Latin America	8.00%
	Western and	7.85%
9.82%	Total Average	9.24%



Transmi	8068.07
Manage	252.27
Techni	
cal	
assista	
nce	455.13
Capacit	
y	351.41
Conting	565.58
Sum	9692.46



If delayed in operation:	
Total	70
No	19
Yes	51

Reasons breakdown	
Manage	34
Procure	34
Land iss	21
Ad-hoc	12
Constru	9
Financir	5
Political	3

nt 2010 value)	
400 kv	500 kv
	0.979
0.374	
0.317	0.281
0.360	0.630

nt 2010 value)	
400 kv	500 kv
	0.715
0.336	0.267
0.296	0.783
0.282	
0.311	0.384
0.382	
0.329	0.486

<V

<V

<V

<V

<V

<V

Average O&M Rate	Number of data points
2.25%	4
2.25%	2
2.25%	2
2.86%	12
3.17%	6
2.50%	6
1.00%	8
1.00%	6
1.00%	2
2.00%	6
2.00%	5
2.00%	1

Average O&M Rate	Number of data points
1.83%	7
2.50%	3
1.50%	4



Capital Cost			Levelized cost		
UIC (\$ million/km)	500 km	1000 km	r: 6%, 500km	r: 10%, 500km	r: 6%, 1000km
	(\$ million)	(\$ million)	(\$/kwh)	(\$/kwh)	(\$/kwh)
\$0.255	\$128	\$255		\$0.013	\$0.020
				\$0.007	\$0.010
				\$0.004	\$0.006
\$0.361	\$181	\$361		\$0.009	\$0.014
				\$0.005	\$0.007
				\$0.003	\$0.005
\$0.382	\$191	\$382		\$0.006	\$0.009
				\$0.003	\$0.005
				\$0.002	\$0.003

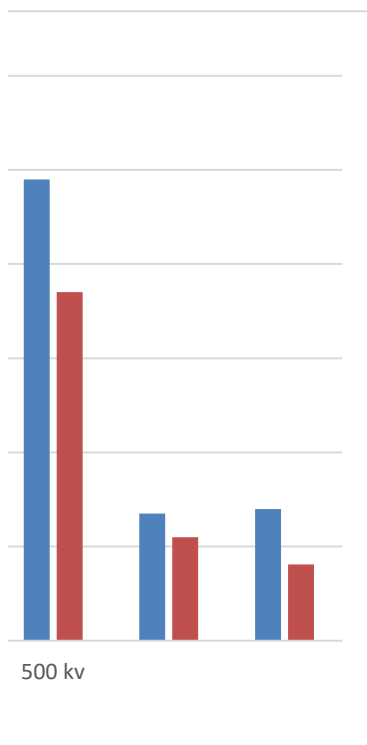


:			400 kv Double Circuit			500 kv Double Circuit		
Rate	Cost per km	Count	Rate	Cost per km	Count	Rate		
	0.317	1	0.83					
1.00	0.383	1	1.00	0.715	1	1.00		
1.00				0.783	1	1.10		
0.90	0.384	5	1.00	0.684	2	0.96		



Europe and Cen Europe and Cen Middle I Middle I South Asia Dou Western ar Western and Central Afri					
\$0.13			\$0.13		\$0.06
					\$0.13
\$0.15	\$0.30	\$0.16	\$0.40	\$0.24	\$0.35
		\$0.22		\$0.26	\$0.20
		\$0.28		\$0.27	\$0.24
				\$0.16	\$0.27
					\$0.15
					\$0.44
					\$0.28
					\$0.22
\$0.47				\$0.53	\$0.36
					\$0.25
					\$0.36
					\$0.36
					\$0.29
					\$0.36
					\$0.36
\$0.30	\$0.32	\$0.28	\$0.28		\$0.38
			\$0.38		\$0.38
			\$0.27		
\$0.28	\$0.78		\$0.40		
			\$0.36		
			\$0.39		

500 kv
0.715
0.267
0.783
0.384
0.486



Original data information

Project Project Name Start date Closing date Region

P13155	Transmission Ef	Aug-14	Dec-21	East Asia and
P11476	Kazakhstan Moi	Sep-09	Apr-13	Europe and C
P11973	Electricity Secto	Jun-11	Jan-19	Eastern and S
P10303	Electricity Expar	May-10	Dec-17	Eastern and S
P11159	Tanzania - Back	Aug-10	Dec-16	Eastern and S
P09620	Power Transmis	Aug-07	Jun-16	Europe and C
P08233	ECSEE APL 3 - F	Jan-06	Jul-16	Europe and C
P11251	Botswana - Mor	Oct-09	Aug-14	Eastern and S
P08886	Energy Commu	Jun-05	Jun-12	Europe and C
P09515	North-South Ele	Oct-05	Nov-11	Europe and C
P12657	AFR RI-The East	Jul-12	Jan-24	Eastern and S
P11326	WAPP-APL4 (Ph	May-12	Mar-23	Western and
P06925	AFCC2/RI-3A-So	Nov-03	Sep-16	Eastern and S
P14734	Transmission Gi	May-14	Mar-23	Europe and C
P07201	Transmission D	Jul-01	Dec-08	Western and
P07401	Ethiopia/Nile B	Dec-07	Jun-13	Eastern and S
P10426	MA-Support to	Jun-08	Dec-15	Middle East a
P10532	GMS Power Tra	Jun-07	Dec-11	East Asia and
P09596	Siddhirganj Pow	Oct-08	Dec-19	South Asia
P10164	TZ-Energy Deve	Dec-07	Sep-17	Eastern and S
P08440	Mozambique Tr	Jul-07	Aug-16	Eastern and S
P11732	Indonesia Powe	Jul-10	Oct-19	East Asia and
P14792	OMVS - TRANS	May-17	Dec-22	Western and
P15658	Modernization	Nov-16	Jun-24	Europe and C
P08487	Second Transmi	Jul-05	Jun-14	East Asia and
P08334	Power Transmis	May-05	Dec-11	Europe and C
P12992	Bangladesh: Rui	Feb-14	Jun-21	South Asia

P09491	WAPP: The First	Jun-11	Dec-18	Western and
P11289	Kabeli Transmis	May-11	Dec-16	South Asia
P11691	Alma Transmiss	Nov-10	Apr-15	Europe and C
P09598	Electricity Distri	Jun-08	Feb-14	South Asia
P09491	WAPP APL 1 (2r	Jun-06	Jun-16	Western and
P11576	Nepal-India Elec	Jun-11	Oct-21	South Asia
P09720	AFCC2/RI-Regio	May-07	Jun-18	Eastern and S
P11017	Cameroon - Krik	Nov-11	May-13	Western and
P11007	Increased Acces	Jun-09	Dec-18	Western and
P13318	Lusaka Transmi	May-13	Feb-22	Eastern and S
P11993	Talimarjan Tran	Mar-11	Jun-16	Europe and C
P06484	Rural Electrifica	Dec-03	Jan-12	East Asia and
P07401	Ethiopia/Nile B	Dec-07	Jun-13	Eastern and S

83%

3%

5%

4%

6%





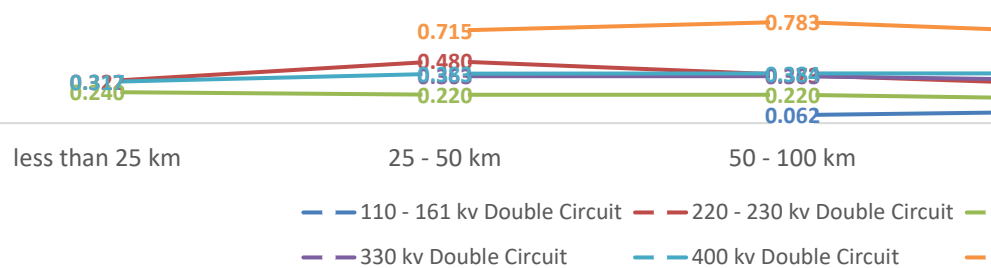


r: 10%, 1000km
(\$/kwh)

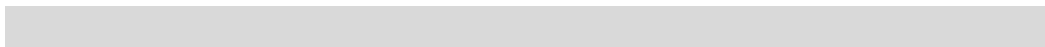
\$0.041
\$0.020
\$0.013

\$0.029
\$0.014
\$0.009

\$0.018
\$0.009
\$0.006



[illegible]



Country	Voltage	Appraisal (nominal)	Actual (nominal)
Vietnam	220/500 (i)	220kv: 0.5 500kv: 0.86	220kv: 0.55 500kv: 0.81
Kazakhstan	220 kv	\$0.14	\$0.13
Uganda	220 kv	\$0.26	\$0.10
Kenya	132 kv	\$0.17	\$0.16
Tanzania	400 kv	\$0.52	\$0.27
Ukraine	330 kv	\$0.45	\$0.41
Macedonia	400 kv	\$0.29	\$0.23
Botswana	400	\$0.32	\$0.157
Serbia	110 kV	\$0.11	\$0.08
Kazakhstan	500 kv	\$0.25	\$0.18
Ethiopia; Ken	500 kv	\$0.29	\$0.25
Côte d'Ivoire,	225 kv (i)	\$0.22	\$0.19
DRC, Zambia	220 kv	\$0.45	\$0.50
Georgia	220 kv (i)		
Nigeria	330/132 kV (i)		
Ethiopia	230/220 kV (i)		
Morocco	400/225 kV (i)		
Cambodia	115 kv		
Bangladesh	230 kV (i)		
Tanzania	132 kv (i)		
Mozambique	220 kv (i)		
Indonesia	500/275/150 kv (i)		
Senegal	225 kv (i)		
Uzbekistan	220/500 (i)		
Vietnam	500/220/110 kv (i)		
Azerbaijan	330/220/110 kv (i)		
Bangladesh	230/132 kv (i)		

Ghana, Burkina 225 kv (i)	
Nepal	132 kv (i)
Kazakhstan	500/220 kv (i)
Pakistan	220/132 kv (i)
Ghana, Benin 330/161 kv (ii)	
Nepal	400 kv (i)
DRC	400/220 kv (i)
Cameroon	225 kV (i)
Benin	161 kv (i)
Zambia	132/88 kv (i)
Uzbekistan	500 kv (i)
Cambodia	230/115 kV (i)
Ethiopia; Sudan	230/220 kV (i)

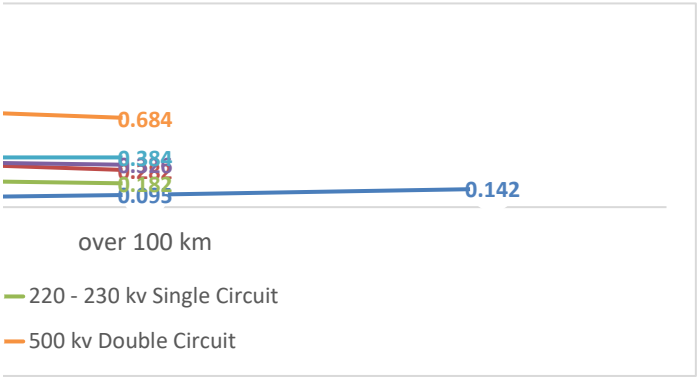




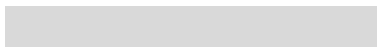












Differences	Reasons
Equivalent	competitive price and effective project management
Equivalent: 96%	
Below: 47%	outcomes of the bidding for the engineering, procurement, and cons
Below: 94.1%	lower than estimated prices for transmission and distribution lineanc
Below: 51.27%	(i) joint bidding process, which despite causingdelays in the project ir
Below: 91.1%	Reasons not found
Below: 80%	Didn't mention the reason
Below: 49%	Competitively bid and awarded at significantly lower prices than estir
Below: 70%	Not specified
Below: 74%	Savings in taxes and efficiently administered procurement
Structural Change	[added 2/2025] - due to canceling of component; line component de
Above - 104%	[added 2/2025] - total cost increased due to combination of factors,
Above: 111%	Around 2006-2008 the price of power transmission equipment increa
Above: 114%	[added 2/2025] - Total project cost increased to 70.229 from 61.72 d
Equivalent: 98%	
Equivalent - 100%	
Equivalent - 100%	
Equivalent	The ICR indicates 100% of expenditure
Equivalent	difficult to measure; changes in project components
Equivalent - close to 100%	
Equivalent	
Equivalent - close to 100%	
Equivalent: 97%	[added 2/2025] - restructured thrice mainly because of a lower cont
Equivalent	[added 2/2025] - delayed due to COVID; no changes to the final cost
Below - close to 90%	
Below - close to 90%	ICR not available
Below - 92.11%	

Below: 66%	overall - bidding and procurement
Below	Total project at 65%
Below	Total: smaller; 80.21%; transmission segments: smaller; 53.5% significant
Below	74.3% smaller: (i) the NPT's planning and budgeting practice, which in
Below - close to 90%	
Below	smaller: 86%. Changes in project components and incompleteness of work
Above: 135%	Rerouting; changes in financial procedures
Structural Change	Components not implemented: This is attributed to the procurement
Structural Change	Partially achieved. About 37.4 km of transmission lines were completed
Structural Change	smaller: 62%. Changes in project components and cancellation of full
Structural Change	smaller: 74.9%; Change in project components and cancellation of full





[REDACTED]

[REDACTED]

[REDACTED]



truction (EPC) contracts for the construction of the transmission line and associated substations.

plementation, nevertheless indeed resulted in lower prices for the Project, and (ii) in addition to

includes estimates that were conservatively drawn (contingencies accounted for 10 percent of proj

t delays in the early stages of the project arising from shortcomings in inter-agency arrangements
ed, achieving 65 percent of the target. 13 km for Ubungo-Kurasini portion cannot be implemente

This bidding process attracted lower-priced bids than had been estimated, largely because the cc
bidding-related economies of scale, historically low commodity prices,combined with favorable c

ect cost (or about US\$50 million), and they were not required for project implementation; (ii)the

and coordination; Vietnam's decision to not export power to Cambodia, which necessitated a re
ed now due to a road expansion project by TANROADS, which affects the way leave. 1.3 km for Mz

Contractors were also the equipment manufacturers, hence the cheaper bids. In addition, the low exchange rates also contributed to significant savings. Since savings materialized only during the l

downward pressure on steel prices during the economic downturn that followed the global finan

structuring and re -design of the project; and the Bank's decision to not extend the project durati
zinga Creek, procurement and installation works are progressing with expected time of completio

or bids, to a larger extent, were because of the rapid reduction in global prices of steel and aluminum. In the last months of the Project's implementation, the decision was made by the team and TANESCO to

financial crisis of 2008 (prices went from a precrisis level of over US\$1,000/metric ton to an average of

on as requested by the borrower due to the slow pace of progress as well as the Bank's holding
on 31st March 2018. Total: smaller: 72.6%; Broader transmission component: 85.6%

ium to almost 55 percent between 2008 and 2010. Almost two-thirds of the price of transmissior
partially cancel the undisbursed saved funds in the amount ofUS\$18.76 million equivalent, and tl

f US\$400/metric ton during project implementation) that further saved about 5 percent of project

back on all Board presentations for Cambodia due to an Inspection Panel case relating to Cambodia

Infrastructure constitute the price of towers and conductors, whose costs are directly linked to the price of the equipment. The costs of the equipment were recommitted to other country priorities (See section 3.3 for additional detail).

t cost (or US\$30 million); (iii) the increasing competition among contractors during the same period

nd resulted in a lower cost (of about US\$70million); and (iv) effective contract management practi

ices by implementing agencies (IAs), for example, finding the most cost-efficient trucking route, in

crease of mass excavation, use of excavation materials for fillings, and others that saved about US

\$10 million These together have created a cost saving of US\$160 million for

Calculation of the Levelized Cost of Transn

$$LCOT = \left[\frac{(\text{CAPEX} + \text{O\&M}) * r}{[1 - (1 + r)^{-n}]} \right] \div [K *$$

Select Region:

World_Bank_Data_Average

Insert Line Length (km)

250

CAPEX

Voltage (kv)	Unit Investment Cost (\$/km)
Line_220KV	293,000
Line_330KV	374,000
Line_400KV	342,000
Line_500KV	522,000

CAPEX + O&M Cost (OM = CAPEX * O&M rate)

Voltage (kv)	O&M Rate Scenario
220	3%
330	
400	
500	

Discount Rate Factor (r)

Discount Rate Scenario	Project Length (Year)
6%	40

Incremental Capacity of Transmission (K [MW])*

Voltage (kv)	Capacity Lower (MW)
220	300
330	600
400	1000
500	1200

Utilization Factor (UF)

Generation	UF for Reference
Solar	0.2
Hydro	0.4
Gas	0.6
Coal	0.8

* The transmission capacity here presents a general range; numb

** The selection of the capacity factor corresponds roughly to dif

mission (LCOT)

UF * 8760]

- Instructions:
- The color coding below indicates sections that require
 - To start, select a region; the unit investment cost (UI
 - User may change other inputs, including line length,
 - A compiled data based on specific assumptions is shc

Selection area based on World
Area requires inputs.

Total CAPEX (\$/km)	
	73,250,000
	93,500,000
	85,500,000
	130,500,000
CAPEX + O&M (\$, 500 km)	
	75,447,500
	96,305,000
	88,065,000
	130,500,000
Factor Value	
	0.90277781
Capacity Higher (MW)	
	500
	800
	1200
	1500
UF for Calculation**	
	0.25
	0.5
	0.8

Results Dashboard	
Voltage Range	Transmission Capacity (MW)
220	300
330	600
400	1000
500	1200

ers used to calculate the LCOT is the "lower" category. User may change the numbers as fit.
ferent generation sources.

e selection (through drop-down menu), sections that require user inputs, and the results dashbo
 C) will be automatically updated based on collected data. User may also insert a number as adju
 O&M rate, discount rate, project length, and utilization factor as fit.
 own in Tab 5.2.

d Bank data.

Utilization Factor	MWh Transferred	UIC (\$/km)	Total CAPEX (\$/km)	
0.25	657000	293,000	73,250,000	
0.5	1314000			
0.8	2102400			
0.25	1314000	374,000	93,500,000	
0.5	2628000			
0.8	4204800			
0.25	2190000	342,000	85,500,000	
0.5	4380000			
0.8	7008000			
0.25	2628000	522,000	130,500,000	
0.5	5256000			
0.8	8409600			

ard.
investment for the UIC.

Levilized cost (\$/kwh)	
	0.008
	0.004
	0.002
	0.005
	0.002
	0.002
	0.003
	0.001
	0.001
	0.003
	0.002
	0.001

LCOT Data Summary

List of Assumptions:

- O&M = 3%.
- Discount rate (r) is calculated and presented under two scenarios: 6% and 10%.
- Project life (n) = 40 years
- Incremental Transmission Capacity (k) is indicated as below:
 - 220/230 kV: 300 MW | 330 kV: 600 MW | 400 kV: 1000 MW | 500 kV: 1200 MW
- Utilization Factor (UF) adopts the following scenarios: 0.25, 0.5, 0.8

Note: The result display is based on available data.

Assumptions					UIC (\$ million/km)
Voltage Range (kV)	Transmission Capacity (MW)	Utilization Factor (UF)	MWh transferred		
Region: World Bank Data Average					\$0.293
220/230	300	0.25	657000		
		0.5	1314000		
		0.8	2102400		
330	600	0.25	1314000	\$0.374	
		0.5	2628000		
		0.8	4204800		
400	1000	0.25	2190000	\$0.342	
		0.5	4380000		
		0.8	7008000		
500	1200	0.25	2628000	\$0.522	
		0.5	5256000		
		0.8	8409600		
Region: East Asia and Pacific					\$0.847
500	1200	0.25	2628000		
		0.5	5256000		
		0.8	8409600		
Region: Eastern and Southern Africa					\$0.335
220/230	300	0.25	657000		
		0.5	1314000		
		0.8	2102400		
400	1000	0.25	2190000	\$0.366	
		0.5	4380000		
		0.8	7008000		

500	1200	0.25	2628000	\$0.267
		0.5	5256000	
		0.8	8409600	
Region: Middle East and North Africa				
220/230	300	0.25	657000	\$0.218
		0.5	1314000	
		0.8	2102400	
400	1000	0.25	2190000	\$0.282
		0.5	4380000	
		0.8	7008000	
Region: South Asia				
220/230	300	0.25	657000	\$0.400
		0.5	1314000	
		0.8	2102400	
330	600	0.25	1314000	\$0.311
		0.5	2628000	
		0.8	4204800	
400	1000	0.25	2190000	\$0.384
		0.5	4380000	
		0.8	7008000	
Region: Western and Central Africa				
220/230	300	0.25	657000	\$0.255
		0.5	1314000	
		0.8	2102400	
330	600	0.25	1314000	\$0.361
		0.5	2628000	
		0.8	4204800	
400	1000	0.25	2190000	\$0.382
		0.5	4380000	
		0.8	7008000	

Capital Cost		LCOT		
250 km (\$ million)	500 km (\$ million)	r: 6%, 250km (\$/kwh)	r: 10%, 250km (\$/kwh)	r: 6%, 500km (\$/kwh)
\$73	\$146	\$0.008	\$0.012	\$0.015
		\$0.004	\$0.006	\$0.008
		\$0.002	\$0.004	\$0.005
		\$0.005	\$0.007	\$0.010
		\$0.002	\$0.004	\$0.005
		\$0.002	\$0.002	\$0.003
		\$0.003	\$0.004	\$0.005
		\$0.001	\$0.002	\$0.003
		\$0.001	\$0.001	\$0.002
\$131	\$261	\$0.003	\$0.005	\$0.007
		\$0.002	\$0.003	\$0.003
		\$0.001	\$0.002	\$0.002
\$212	\$423	\$0.005	\$0.008	\$0.011
		\$0.003	\$0.004	\$0.005
		\$0.002	\$0.003	\$0.003
\$84	\$167	\$0.009	\$0.013	\$0.017
		\$0.004	\$0.007	\$0.009
		\$0.003	\$0.004	\$0.005
		\$0.003	\$0.004	\$0.006
		\$0.001	\$0.002	\$0.003
		\$0.001	\$0.001	\$0.002
\$92	\$183	\$0.003	\$0.004	\$0.006
		\$0.001	\$0.002	\$0.003
		\$0.001	\$0.001	\$0.002

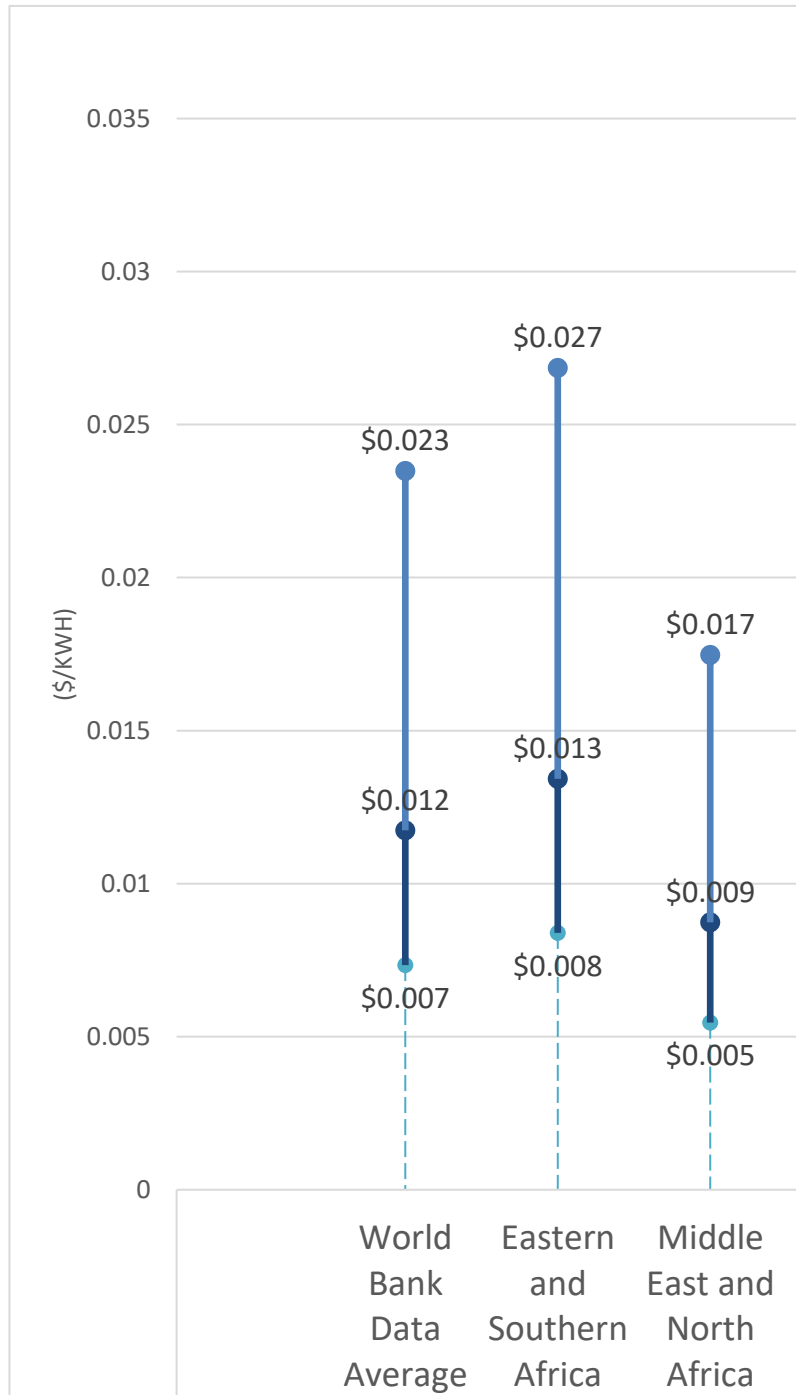
\$67	\$135		\$0.002	\$0.003	\$0.003
			\$0.001	\$0.001	\$0.002
			\$0.001	\$0.001	\$0.001
\$55	\$109		\$0.006	\$0.009	\$0.011
			\$0.003	\$0.004	\$0.006
			\$0.002	\$0.003	\$0.004
	\$141		\$0.002	\$0.003	\$0.004
			\$0.001	\$0.002	\$0.002
			\$0.001	\$0.001	\$0.001
\$100	\$200		\$0.010	\$0.016	\$0.021
			\$0.005	\$0.008	\$0.010
			\$0.003	\$0.005	\$0.007
	\$156		\$0.002	\$0.004	\$0.005
			\$0.001	\$0.002	\$0.002
			\$0.001	\$0.001	\$0.002
	\$192		\$0.002	\$0.004	\$0.005
			\$0.001	\$0.002	\$0.002
			\$0.001	\$0.001	\$0.002
\$64	\$127		\$0.007	\$0.010	\$0.013
			\$0.003	\$0.005	\$0.007
			\$0.002	\$0.003	\$0.004
	\$180		\$0.005	\$0.007	\$0.009
			\$0.002	\$0.004	\$0.005
			\$0.001	\$0.002	\$0.003
	\$191		\$0.003	\$0.005	\$0.006
			\$0.001	\$0.002	\$0.003
			\$0.001	\$0.001	\$0.002

LCOT Data Presentation

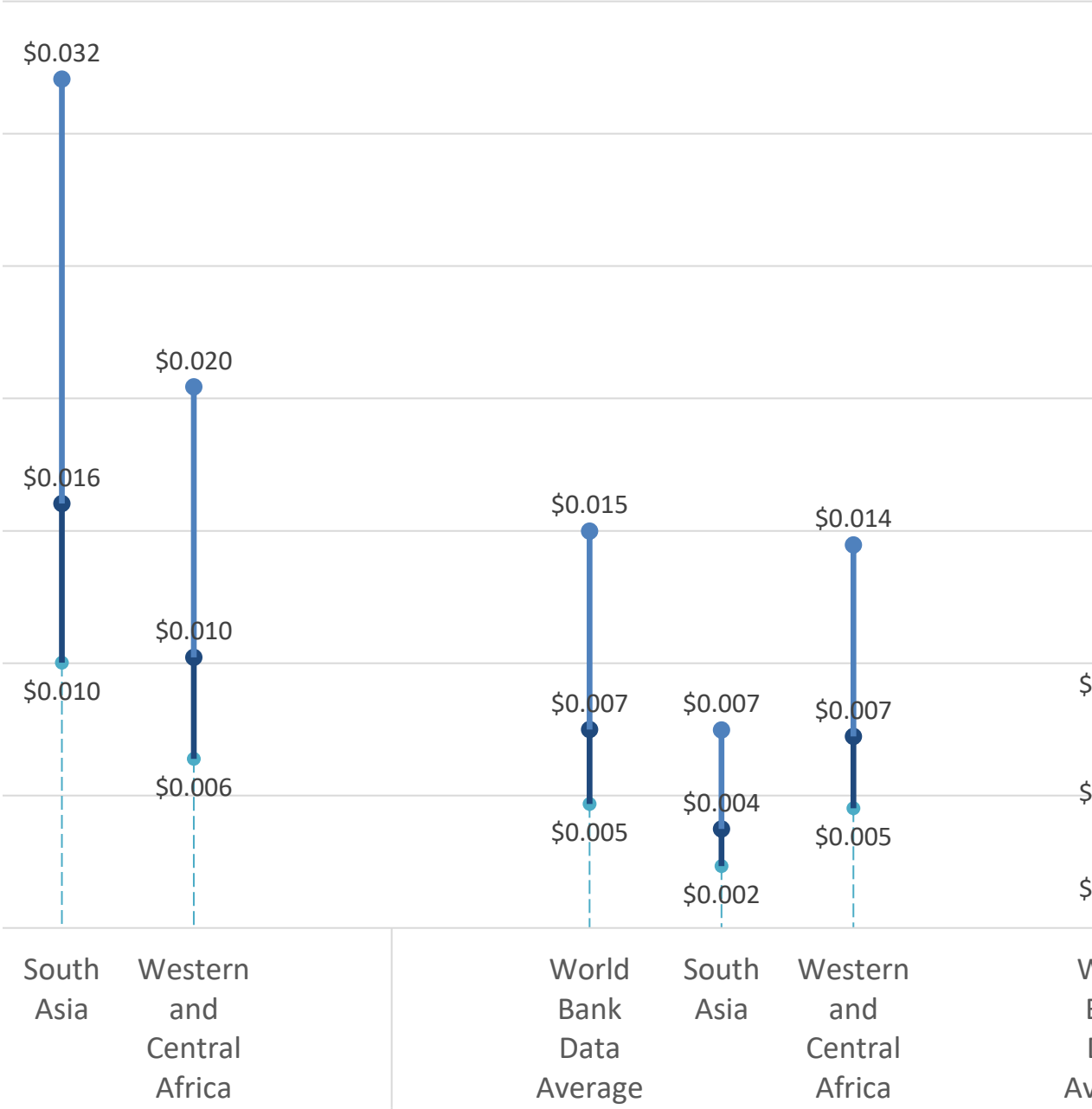
Selection:

- $r = 10\%$ as the most commonly adopted discount rate
- length = 500 km

r: 10%, 500km (\$/kwh)	
	\$0.023
	\$0.012
	\$0.007
	\$0.015
	\$0.007
	\$0.005
	\$0.008
	\$0.004
	\$0.003
	\$0.010
	\$0.005
	\$0.003
	\$0.016
	\$0.008
	\$0.005
	\$0.027
	\$0.013
	\$0.008
	\$0.009
	\$0.004
	\$0.003



Levelized Cost of Transmission (LCOE)



● UF = 0.25 ● UF = 0.5 ● UF = 0.8

* Selection: $r = 10\%$; length = 500

COT)*



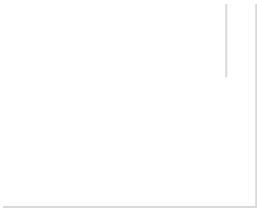
30 kv

500 kv

km

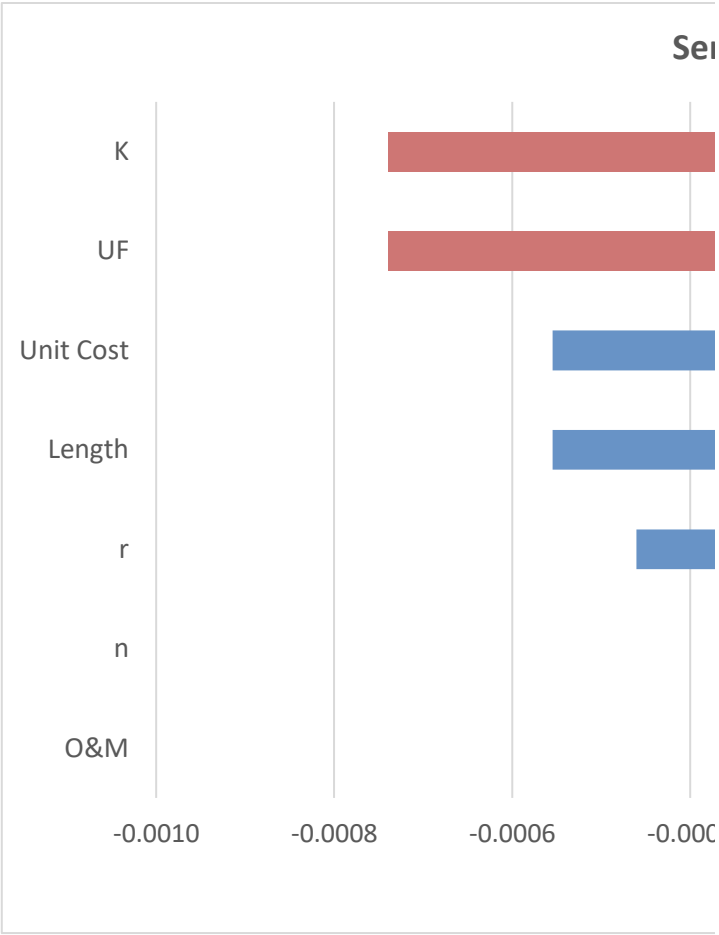


Utilization Factor		UF1		UF2	
330 kv	Eastern and Southern Africa	\$0.008	\$0.005	\$0.013	
	Middle East and North Africa	\$0.005	\$0.003	\$0.009	
	South Asia	\$0.010	\$0.006	\$0.016	
	Western and Central Asia	\$0.006	\$0.004	\$0.010	
500 kv	World Bank East Asia and Pacific	\$0.003	\$0.003	\$0.007	
	South Asia	\$0.002	\$0.001	\$0.004	
	Western and Central Asia	\$0.005	\$0.003	\$0.007	
	World Bank Eastern and Southern Africa	\$0.003	\$0.002	\$0.004	
765 kv	World Bank Eastern and Southern Africa	\$0.003	\$0.002	\$0.004	
	Middle East and North Africa	\$0.002	\$0.001	\$0.003	
	South Asia	\$0.004	\$0.001	\$0.004	
	Western and Central Asia	\$0.003	\$0.002	\$0.005	



LCOT Sensitivity

UF3
\$0.027
\$0.017
\$0.032
\$0.020
\$0.013
\$0.007
\$0.014
\$0.008
\$0.009
\$0.007
\$0.007
\$0.009
\$0.010
\$0.016
\$0.005



Variable

O&M
n
r
Length
Unit Cost
UF
K

Variable

CAPEX
Length
O&M
r
n
UF
K

Variable

CAPEX

O&M
r
n
UF
K

Length

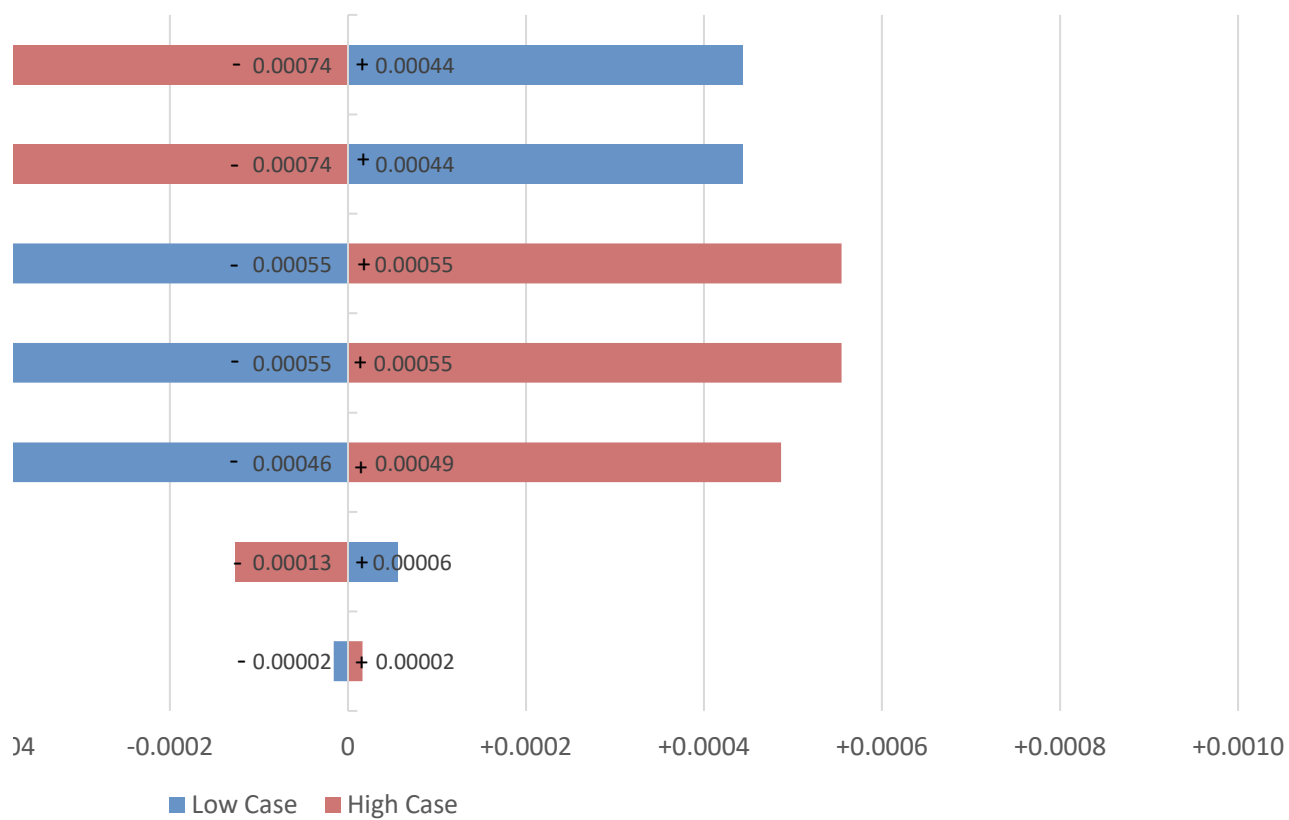
Variables
CAPEX (\$/km)
O&M (%)
r (%)
n (year)
K (MW)
UF
LCOT (\$/kwh)

$$\frac{(CAPEX+OM)*r}{(1-(1+r)^{-n})}$$

$$K*UF*8760$$

LCOT

Sensitivity of LCOT to Key Input Parameters



Low Case

0.00002
0.00013
0.00046
0.00055
0.00055
0.00074
0.00074

LCOT (Base)

0.00222
0.00222
0.00222
0.00222
0.00222
0.00222
0.00222

High Case

0.00002
0.00006
0.00049
0.00055
0.00055
0.00044
0.00044

LCOT (Low)

-0.00055
-0.00222
-0.00002
-0.00046
0.00013
0.00074
0.00074

LCOT (Base)

0.00222
0.00222
0.00222
0.00222
0.00222
0.00222
0.00222

LCOT (High)

0.00055
-0.00222
0.00002
0.00049
-0.00006
-0.00044
-0.00044

LCOT (Low)

0.00166
0.00220
0.00176
0.00235
0.00296
0.00296

LCOT (Base)

0.00222
0.00222
0.00222
0.00222
0.00222
0.00222

LCOT (High)

0.00277
0.00223
0.00271
0.00216
0.00177
0.00177

Range

0.00111
0.00000
0.00003
0.00095
-0.00019
-0.00118
-0.00118

300

Low	Base	High
225,000	\$300,000	375,000
2.25%	3%	3.75%
6%	8%	10.00%
30	40	50
600	800	1,000
0.38	0.50	0.63
	\$0.0022	

7,416,000.00

0.95

3504000

0.00222

$$LCOT = \left[\frac{(\text{CAPEX} + \text{O\&M})}{[1 - (1 + r)^{-N}]}\right]$$

1/0	1/0
6%	3%
21%	22%
25%	25%
25%	25%
33%	20%
33%	20%

$$\frac{1) * r}{\cdot n} \Bigg] \div [K * UF * 8760]$$

World Bank Projects, data as of 12/21/2022 12:37:13 A

Project Id	Region	Country	Project Status
P127974	South Asia	Republic of Ir	Active
P166796	Western and	Republic of M	Active
P106172	Western and	Federal Repu	Closed
P116854	Middle East a	West Bank an	Closed
P104034	Latin America	Republic of H	Closed
P098742	Western and	Republic of G	Closed
P095593	East Asia and	Democratic R	Closed
P163592	Europe and C	Central Asia	Active
P116748	Europe and C	Republic of A	Active
P078091	Western and	Burkina Faso	Closed
P092218	East Asia and	Republic of Ir	Active
P112340	Eastern and S	Republic of U	Closed
P070530	Western and	Republic of S	Closed
P162279	Eastern and S	Republic of M	Closed
P096648	Western and	Federal Repu	Closed
P104752	Latin America	Federative R	Closed
P101879	Latin America	Federative R	Closed
P113415	Western and	Republic of T	Closed
P159052	Eastern and S	Republic of A	Active
P165313	Europe and C	Central Asia	Active
P157118	Other	World	Pipeline
P165400	South Asia	People's Rep	Active
P108440	Western and	Republic of M	Closed
P166785	Western and	Burkina Faso	Active
P097974	Eastern and S	Republic of B	Closed
P057761	Eastern and S	Republic of M	Closed
P077452	Eastern and S	Republic of Z	Closed
P166805	Eastern and S	Republic of A	Active
P098949	Europe and C	Kyrgyz Repu	Closed
P151075	South Asia	Islamic Repu	Active
P151618	East Asia and	Solomon Isl	Closed
P124715	Western and	Republic of C	Closed
P090110	Latin America	Republic of P	Closed
P090116	Latin America	Republic of P	Closed
P071794	South Asia	People's Rep	Closed
P162111	Western and	Republic of L	Closed
P148797	Western and	Republic of G	Active
P104456	Western and	Republic of C	Closed
P129097	Western and	Republic of L	Closed
P083131	Eastern and S	Republic of K	Closed
P133445	Western and	Republic of L	Closed

P149683	Western and Republic of Li	Active
P133005	Eastern and S Republic of U	Closed
P171742	Eastern and S Federal Demc	Active
P173416	Western and Republic of Li	Active
P164885	Western and Central Africa	Active
P162086	South Asia Republic of Ir	Active
P163881	Western and Republic of C	Active
P153124	Western and Republic of Li	Closed
P161630	Western and Republic of G	Active
P112798	South Asia Republic of Ir	Closed
P167278	Western and Republic of B	Closed
P101653	South Asia Republic of Ir	Closed
P106899	Europe and C Montenegro	Closed
P162245	Western and Central Africa	Active
P161234	Western and Burkina Faso	Closed
P086775	Latin America Republic of H	Closed
P154805	East Asia and Republic of Ir	Closed
P152936	East Asia and Republic of th	Closed
P120014	Eastern and S Republic of K	Active
P145104	Eastern and S Republic of K	Active
P128768	Western and Burkina Faso	Active
P092211	Middle East a Republic of Y	Closed
P117177	East Asia and Lao People's I	Closed
P086379	Middle East a Republic of D	Closed
P112253	Middle East a Republic of D	Closed
P110978	East Asia and Lao People's I	Closed
P153268	East Asia and Republic of th	Active
P162151	East Asia and Republic of th	Active
P090104	Western and Federal Repu	Closed
P119715	East Asia and Lao People's I	Closed
P075531	East Asia and Lao People's I	Closed
P096400	Europe and C Republic of T	Closed
P173749	Western and Republic of B	Active
P168273	Middle East a Republic of T	Active
P170397	Eastern and S Southern Afri	Active
P160427	Eastern and S Southern Afri	Active
P148102	Europe and C Republic of A	Active
P171059	Western and Republic of Si	Active
P145649	Middle East a Kingdom of N	Closed
P126421	Eastern and S Eastern and S	Closed
P096217	Europe and C Macedonia, R	Closed
P110841	Europe and C Republic of T	Closed
P105654	Africa Africa	Closed

P096598	Europe and C Montenegro	Closed
P090656	Europe and C Republic of A	Closed
P040558	Europe and C Republic of M	Closed
P172594	Eastern and S Republic of R	Active
P166936	Eastern and S Kingdom of L	Active
P164625	Middle East a Republic of T	Active
P173088	Eastern and S Somali Demo	Active
P163033	Western and Western and	Active
P164331	Eastern and S Republic of M	Active
P163870	Eastern and S Republic of M	Active
P164225	Western and Republic of G	Active
P165497	Eastern and S Somali Demo	Active
P173918	Western and Republic of S	Active
P177871	Europe and C Kyrgyz Repub	Active
P165183	East Asia and Federated St	Active
P170366	Western and Republic of S	Closed
P169912	Western and Republic of M	Active
P164090	Western and Republic of N	Active
P162760	Eastern and S Republic of Z	Active
P170590	South Asia Republic of Ir	Active
P170552	Europe and C Georgia	Active
P165453	Eastern and S Republic of M	Active
P156208	Eastern and S Democratic R	Active
P153743	Western and Republic of N	Active
P147878	Western and Republic of G	Closed
P127837	Latin America Plurinational	Closed
P173882	Eastern and S Republic of R	Closed
P126489	Eastern and S Republic of R	Closed
P127203	Latin America Republic of H	Closed
P169040	Eastern and S Republic of R	Closed
P161885	Western and Federal Repu	Active
P160395	Eastern and S Federal Dem	Active
P160377	Western and Republic of T	Active
P156719	Latin America Republic of H	Active
P159023	Western and Republic of S	Closed
P161015	Western and Republic of B	Active
P162022	South Asia Islamic Repu	Active
P159426	East Asia and Socialist Rep	Closed
P158505	Middle East a Republic of D	Active
P159397	East Asia and Socialist Rep	Active
P120016	Western and Republic of G	Closed
P159112	Eastern and S Republic of U	Active
P158655	Western and Republic of S	Closed

P155563	Eastern and S Federal Demc Active
P120172	Eastern and S Federal Demc Closed
P111567	Eastern and S Republic of R Closed
P125565	Western and Republic of S Closed
P119893	Eastern and S Federal Demc Active
P153805	Western and Republic of Si Closed
P074191	Western and Republic of G Closed
P152975	South Asia Islamic Repu Closed
P101556	Eastern and S Federal Demc Closed
P145138	Europe and C Ukraine Closed
P097271	Eastern and S Federal Demc Closed
P113495	East Asia and Socialist Repu Closed
P049395	Eastern and S Federal Demc Closed
P077380	Eastern and S Federal Demc Closed
P173506	Eastern and S Democratic R Active
P074688	East Asia and Socialist Repu Closed
P057929	Eastern and S State of Eritre Closed
P178042	Western and Republic of B Active
P096801	Europe and C Republic of T Closed
P083908	South Asia Islamic Repu Closed
P073367	Latin America Plurinational Closed
P078619	Western and Republic of G Closed
P178914	Eastern and S Republic of M Active
P150066	South Asia Nepal Active
P114782	Eastern and S Eastern and S Closed
P169143	Europe and C Republic of T Active
P122944	Europe and C Republic of A Closed
P161794	East Asia and Republic of K Closed
P162783	Africa East Union of the Active
P155605	Europe and C Republic of A Closed
P174736	Latin America Republic of H Active
P145657	Western and Republic of S Closed
P107940	Western and Islamic Repu Closed
P121507	South Asia Islamic Repu Active
P131084	Western and Republic of M Active
P166458	Eastern and S Republic of R Closed
P147277	Latin America Dominican R Closed
P149599	East Asia and Lao People's I Closed
P110405	Eastern and S Republic of M Closed
P116398	East Asia and Socialist Repu Closed
P088619	Eastern and S Democratic R Closed
P149410	South Asia Islamic Repu Closed
P176620	Western and Republic of S Active

P162580	Western and Western and Active
P162671	Eastern and S Republic of R Closed
P177140	Eastern and S United Repu Active
P175456	Europe and C Republic of T Active
P166413	East Asia and Republic of th Closed
P117864	Latin America Republic of P Closed
P147818	South Asia Republic of Ir Dropped
P157372	South Asia Islamic Repu Active
P118935	East Asia and Kingdom of C Dropped
P110617	Latin America Federative Re Closed
P115893	South Asia Islamic Repu Active
P110614	Latin America Federative Re Closed
P093787	Latin America Federative Re Closed
P172788	South Asia Republic of M Active
P102316	Latin America Argentine Re Closed
P149239	South Asia Nepal Active
P144324	Latin America Republic of H Closed
P170132	Europe and C Republic of T Active
P154351	Latin America Republic of H Active
P178347	Middle East a Republic of Y Active
P148355	Western and Republic of M Active
P176683	Western and Central Africa Active
P166685	Eastern and S Republic of U Active
P167569	Western and Western and Active
P164044	Western and Western and Active
P108444	Eastern and S Republic of M Closed
P167381	East Asia and Solomon Islar Active
P162902	East Asia and Solomon Islar Active
P075174	South Asia Republic of Ir Closed
P160009	Eastern and S Republic of K Active
P160658	East Asia and Republic of V Closed
P110051	South Asia Republic of Ir Closed
P074011	Eastern and S Federal Dem Closed
P079633	Western and Republic of B Closed
P043311	South Asia Nepal Closed
P172256	East Asia and Republic of Ir Active
P086414	South Asia Republic of Ir Closed
P115566	South Asia Republic of Ir Closed
P160344	Western and Burkina Faso Active
P145664	Western and Republic of M Closed
P169561	Eastern and S United Repu Active
P166170	Eastern and S Kingdom of E Active
P110177	Western and Republic of C Closed

P110075	Western and Republic of B	Closed
P119737	Eastern and S Republic of U	Closed
P097201	Africa Africa	Closed
P160771	Western and Republic of G	Closed
P159807	South Asia People's Repu	Active
P157224	South Asia Republic of Ir	Closed
P126182	Western and Federal Repu	Closed
P146696	Western and Republic of G	Closed
P115874	East Asia and Socialist Repu	Closed
P122773	Europe and C Republic of U	Closed
P116410	Eastern and S Republic of S	Closed
P174650	South Asia People's Repu	Active
P169983	Eastern and S Republic of A	Active
P167820	East Asia and The Indepenc	Active
P172891	Western and Federal Repu	Active
P157827	South Asia Islamic Repu	Closed
P169867	Western and Republic of T	Closed
P166390	Western and Republic of Si	Active
P162454	Middle East a Republic of Ir	Active
P167330	East Asia and Kingdom of T	Closed
P158249	Eastern and S Republic of M	Active
P152343	East Asia and Mongolia	Active
P155038	South Asia Republic of Ir	Closed
P099211	East Asia and Socialist Repu	Closed
P084461	Middle East a West Bank an	Closed
P151785	Eastern and S Republic of M	Active
P150634	Eastern and S Republic of R	Closed
P107797	South Asia People's Repu	Closed
P154283	South Asia Republic of Ir	Active
P157060	Western and Burkina Faso	Closed
P149522	Latin America Belize	Active
P152411	Middle East a West Bank an	Closed
P157096	Eastern and S Democratic R	Active
P133446	Europe and C Kyrgyz Repub	Closed
P106262	East Asia and Republic of th	Closed
P120304	Western and Republic of Si	Active
P074624	Eastern and S United Repu	Closed
P116199	Middle East a West Bank an	Closed
P131228	South Asia Islamic Repu	Closed
P099626	Eastern and S Republic of M	Closed
P125374	Western and Republic of G	Closed
P116166	East Asia and Mongolia	Closed
P116198	Middle East a Arab Republi	Closed

P110643	Europe and C Republic of Tı	Closed
P149115	Western and Burkina Faso	Active
P112573	Western and Republic of Cı	Closed
P125996	East Asia and Socialist Repu	Closed
P146287	Western and Republic of M	Active
P105279	Western and Republic of Sı	Closed
P092055	East Asia and Democratic R	Closed
P111018	Western and Republic of M	Closed
P089866	Latin Americaı Dominican Rı	Closed
P090666	Europe and C Bosnia and Hı	Closed
P114111	Western and Central Africa	Closed
P174553	South Asia Islamic Repuı	Closed
P078707	South Asia People's Repuı	Closed
P098531	Latin Americaı Republic of H	Closed
P086865	Middle East a Republic of Yı	Closed
P069126	Western and Burkina Faso	Closed
P159183	East Asia and Republic of Fi	Dropped
P120540	East Asia and Socialist Repuı	Closed
P117651	Europe and C Republic of Tı	Closed
P066397	East Asia and Republic of tı	Closed
P063913	East Asia and Republic of Ir	Closed
P159969	Western and Republic of N	Closed
P133107	Western and Republic of Si	Closed
P127208	Latin Americaı Republic of H	Closed
P118239	Latin Americaı Republic of H	Closed
P121056	Western and Republic of Si	Closed
P145254	Eastern and S United Repuı	Closed
P107335	Western and Republic of Si	Closed
P170248	South Asia Nepal	Closed
P170230	South Asia Islamic Repuı	Active
P115464	Western and Republic of Cı	Closed
P112242	Europe and C Republic of A	Closed
P110173	Eastern and S Republic of Kı	Closed
P089244	Europe and C Republic of Tı	Closed
P110202	Eastern and S State of Eritreı	Closed
P111019	Western and Republic of Bı	Closed
P175874	Latin Americaı Dominican Rı	Active
P161319	East Asia and Solomon Islar	Active
P164001	Western and Federal Repuı	Active
P114204	Latin Americaı Federative Rı	Closed
P040907	East Asia and Mongolia	Closed
P173161	Western and Republic of Tı	Active
P109932	Latin Americaı Dominican Rı	Closed

P110110	South Asia People's Rep	Closed
P108905	Eastern and S Democratic R	Closed
P177563	Europe and C Republic of T	Active
P174708	Middle East a West Bank an	Active
P168211	Europe and C Republic of T	Active
P163568	Western and Republic of T	Active
P160910	East Asia and Republic of t	Active
P152659	Western and Republic of T	Closed
P112164	Latin America Republic of H	Closed
P164318	Eastern and S Republic of M	Active
P122141	Europe and C Republic of T	Closed
P170928	Middle East a West Bank an	Active
P157943	Eastern and S Republic of Z	Closed
P178628	Latin America Argentine Re	Active
P170329	Latin America Argentine Re	Active
P076234	Europe and C Republic of A	Closed
P069996	Eastern and S Republic of U	Closed
P169498	Western and Republic of S	Closed
P159010	Eastern and S Democratic R	Closed
P089382	South Asia People's Rep	Closed
P174786	Latin America Republic of C	Closed
P172749	Western and Republic of B	Closed
P157343	Western and Republic of G	Closed
P163586	Latin America Jamaica	Closed
P161167	Middle East a Republic of Ir	Closed
P154819	Latin America Republic of P	Closed
P148957	Europe and C Romania	Closed
P127411	Western and Republic of C	Closed
P143242	East Asia and Solomon Islar	Closed
P125866	Western and Republic of M	Closed
P128284	Western and Republic of S	Closed
P122941	Eastern and S Union of the	Closed
P122669	Western and Republic of C	Closed
P123679	Western and Republic of T	Closed
P126349	Eastern and S Republic of Z	Closed
P123437	Middle East a West Bank an	Closed
P127314	Western and Republic of G	Closed
P118593	Middle East a West Bank an	Closed
P117282	Western and Republic of T	Closed
P107218	Eastern and S Republic of Z	Closed
P113456	Western and Republic of T	Closed
P107365	Europe and C Ukraine	Closed
P111078	Middle East a West Bank an	Closed

P110618	Western and Republic of T	Closed
P112133	Latin America Republic of H	Closed
P104774	Middle East a Lebanese Rep	Closed
P098867	South Asia Islamic Repu	Closed
P156651	Western and Republic of Si	Closed
P174388	Eastern and S Republic of M	Closed
P088181	East Asia and Democratic R	Closed
P176137	South Asia Islamic Repu	Closed
P159844	Western and Republic of T	Closed
P163283	Western and Burkina Faso	Closed
P120660	Western and Republic of Li	Closed
P145443	South Asia Islamic Repu	Dropped
P165639	Western and Republic of Si	Closed
P116666	Eastern and S United Repu	Closed
P154693	South Asia Nepal	Closed
P157079	Europe and C Kyrgyz Repub	Active
P146816	East Asia and Solomon Islar	Closed
P121651	Europe and C Republic of T	Closed
P115642	Eastern and S Democratic R	Closed
P110555	Europe and C Republic of T	Closed
P066396	East Asia and Socialist Repu	Closed
P171851	Europe and C Macedonia, R	Closed
P148099	Europe and C Kyrgyz Repub	Closed
P118814	East Asia and Lao People's	Closed
P089116	Europe and C Republic of S	Closed
P176983	Latin America Republic of E	Active
P155920	Latin America Republic of H	Closed
P155732	Middle East a Republic of Ir	Active
P151803	Latin America Republic of H	Closed
P130051	Europe and C Romania	Closed
P119324	Eastern and S Republic of B	Closed
P117370	Eastern and S Republic of Z	Closed
P099011	Western and Burkina Faso	Closed
P096305	Eastern and S Democratic R	Closed
P174027	South Asia Republic of Ir	Active
P118531	East Asia and Republic of Ir	Closed
P115897	East Asia and Socialist Repu	Closed
P111905	East Asia and Republic of Ir	Closed
P074594	Middle East a West Bank an	Closed
P165509	Europe and C Montenegro	Active
P153986	East Asia and Solomon Islar	Active
P169196	Eastern and S Democratic R	Active
P163984	Western and Republic of G	Active

P116190	South Asia	Nepal	Closed
P169259	East Asia and	Republic of Ir	Active
P162299	East Asia and	People's Repu	Active
P160379	South Asia	Republic of Ir	Active
P161869	South Asia	People's Repu	Active
P164894	East Asia and	Republic of Ki	Closed
P109885	Europe and C	Republic of A	Closed
P095229	Eastern and S	Republic of A	Closed
P115145	Latin America	Dominican R	Closed
P173324	Middle East a	Republic of Ti	Closed
P155268	Middle East a	West Bank an	Closed
P152523	Middle East a	West Bank an	Closed
P127163	Middle East a	West Bank an	Closed
P113621	Middle East a	West Bank an	Closed
P175251	Eastern and S	Republic of Ki	Closed
P129257	Other	World	Closed
P107163	East Asia and	Republic of Ir	Closed
P160829	Europe and C	Republic of M	Active
P125824	Eastern and S	United Repu	Closed
P101645	Eastern and S	United Repu	Closed
P103037	Eastern and S	Republic of Ki	Closed
P104265	Middle East a	Kingdom of N	Closed
P082337	Europe and C	Macedonia, R	Closed
P069258	Eastern and S	Eastern and S	Closed
P174812	Latin America	Republic of P	Active
P171683	Europe and C	Republic of U	Active
P167898	South Asia	South Asia	Active
P146330	Western and	Federal Repu	Active
P154987	South Asia	Islamic Repu	Active
P147921	Western and	Western and	Active
P152755	Western and	Republic of C	Active
P156584	Europe and C	Republic of U	Active
P146199	Europe and C	Republic of A	Active
P146788	Europe and C	Ukraine	Active
P131558	East Asia and	Socialist Repu	Closed
P147348	Europe and C	Georgia	Active
P145054	South Asia	South Asia	Active
P129920	South Asia	People's Repu	Closed
P123994	East Asia and	Republic of Ir	Closed
P132631	South Asia	Nepal	Closed
P133184	Eastern and S	Republic of Zi	Closed
P124351	Africa	Africa	Closed
P115767	South Asia	Nepal	Closed

P114875	East Asia and Socialist Repu	Closed
P119939	Europe and C Republic of U	Closed
P116919	Europe and C Republic of K	Closed
P111598	Eastern and S United Repu	Closed
P117323	East Asia and Republic of Ir	Closed
P116784	Eastern and S Republic of B	Closed
P112516	Eastern and S Republic of B	Closed
P114766	Europe and C Republic of K	Closed
P095982	South Asia Islamic Repu	Closed
P096207	Europe and C Ukraine	Closed
P084404	Eastern and S Republic of M	Closed
P084871	East Asia and Socialist Repu	Closed
P083341	Europe and C Republic of A	Closed
P064844	East Asia and Kingdom of C	Closed
P072018	Western and Federal Repu	Closed
P110729	South Asia South Asia	Dropped
P169117	Europe and C Georgia	Active
P163752	Eastern and S Eastern and S	Active
P094919	Western and Western and	Closed
P173763	Middle East a Middle East a	Active
P166042	Western and Western and	Active
P133305	Eastern and S Republic of U	Active
P146830	Western and Western and	Active
P126579	Eastern and S Eastern and S	Active
P113266	Western and Western and	Active
P105329	East Asia and Kingdom of C	Closed
P105331	East Asia and Lao People's	Closed
P164354	Eastern and S Eastern and S	Active
P171248	Europe and C Republic of T	Active
P159974	South Asia People's Repu	Active
P095965	South Asia People's Repu	Closed
P162933	Western and Western and	Active
P088867	Europe and C Republic of S	Closed
P167843	South Asia Islamic Repu	Active
P112893	South Asia Nepal	Closed
P094917	Western and Western and	Closed
P095155	Europe and C Republic of K	Closed
P075994	Western and Western and	Closed
P168185	Western and Western and	Active
P157055	Western and Republic of C	Active
P174034	Western and Republic of N	Active

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Project Name	Project Development Implementing Consultant Selection
North Eastern Region Power System Improvement Project	The proposed POWERGRID Corporation Limited
Mali Electricity Sector Improvement Project (MESIP)	The Project Development Energie du Mali (EDM)
Nigeria Electricity and Gas Improvement Project (NEGIP)	The development TCN - Project Management
GZ-Electric Utility Management Additional Financing	The project development objective is to improve
Power Sector Efficiency Enhancement Project (PROMEF)	The project development The National Electric Power
Guinea- GEF- Electricity Sector Efficiency Improvement Project	The project seeks to improve the electricity
TP: Energy Services Delivery Project	The objectives are to stabilize the power
CASA1000 Community Support Project - Kyrgyz Republic	To engage community Development
Electricity Supply Reliability Project	The project development High Voltage Electric Network
Burkina Faso Energy Access Project	The project development Unite d'Execution de la Reforme
Indonesia Infrastructure Finance Facility	To strengthen PT Indonesia Infrastructure
UG: Energy for Rural Transformation APL-2 (GEF)	The objective of the proposed project is to
SN-GEF Elec Svc for Rural Areas (FY05)	The project's development objective is to
MG - Inclusive and Resilient Growth DPO	Project Development Central Bank of Madagascar
Commercial Agriculture Development Project	The Project Development Federal Ministry of Agriculture
Paraíba Second Rural Poverty Reduction	The COOPERAR II project would support
Additional Financing for the Rural Poverty Reduction Project	(i) Improve wellbeing and incomes of the
Emergency Infrastructure Rehabilitation and Energy Project	The Project Development AGETUR, Ministry of Urban
Angola Commercial Agriculture Development Project (PC)	The proposed Ministry of Agriculture Yes
CASA1000 Community Support Project for Tajikistan	The Project Development OJSHC Barqi Tojik, National
Private Sector Competitiveness (P130091)	The development Financiera Nacional de Desarrollo
Additional Financing II for Rural Electrification and Renewable	Increase access Power Cell, Ir Yes
Mali Energy Support Project	The objective Energie du Mali (EDM-SA)
Solar Energy and Access Project	The objective ABER, Ministry of Energy, Science
Multisectoral Water and Electricity Infrastructure Project	The proposed project development objective
INFRASTRUCTURE SERVICES	The objective of the Project is to improve
Zm: Increased Access to Electricity Services	The objective Zambia Electric Supply Corporation
Angola - Electricity Sector Improvement and Access Project	The Project Development Ministerio da Energia Yes
Second Village Investment Project	The objective Community Development and
Pakistan Community Support Project (CASA-1000)	The Project Development Sustainable Development Yes
ELECTRICITY ACCESS EXPANSION PROJECT	The project development Ministry of Mines, Energy &
Cote d'Ivoire - Emergency Infrastructure Renewal Project	The objective PRICI PCU-Abidjan
PE Rural Electrification	The objective of the proposed Project is to
PE Rural Electrification	The objective of the proposed Project is to
Rural Electrification and Renewable Energy Development	Increase access to electricity in rural areas
Liberia Fourth Poverty Reduction Support Development	The objective Ministry of Finance and Development
Emergency Water and Electricity Services Upgrading Project	The objective PIU
CM-Energy Sector Development SIL (FY08)	The overall project ARSEL, EDC, MINEE, AER
Additional Financing Liberia Electricity System Enhancement	The objective of the project is to improve
Energy Sector Recovery Project	(a) enhance the policy, institutional and regulatory
Liberia Accelerated Electricity Expansion Project (LACEEP)	The project development Ministry of Lands, Mines and

Liberia Renewable Energy Access Project	<p>The Project Ministry of Mines, Lands, and
Uganda Energy for Rural Transformation IPF Phase2 Additional Financing	The objective of the proposed project is to
Access to Distributed Electricity and Lighting in Ethiopia	The development Development Bank of Ethiopia
Liberia Electricity Sector Strengthening and Access Project	Increase access to electricity in Liberia. Yes
CAR Emergency Electricity Supply and Access Project	The PDO is to increase electricity supply. No
Jharkhand Power System Improvement Project	The project is in Jharkhand. Yes
CM- Rural Electricity Access Project for Underserved Regions	The Project is in the Ministry of Economy, Planning and
Liberia: LACEEP Additional Financing	The project development objectives are to
Emergency Water and Electricity Services - Additional Financing	The objective is PHIMP
Power System Development IV - Additional Financing	The objective of the project is to strengthen
Benin Second Fiscal Reform and Growth PBG	The program development objectives of the
POWER SYSTEM DEVELOPMENT PROJECT IV	The objective of the project is to strengthen
Energy Community of South East Europe APL 3 - Montenegro	The objective of ECSEE APL is the development
Water and Electricity Upgrading Project	The Project is in the Ministry of Development and
Burkina Faso - Bagre Growth Pole Project Additional Financing	To increase private investment. Yes
HN Rural Infrastructure Project	The project is in the Fondo Hondureno de Inversión
Power Distribution Development Program-for-Results	The Project is PT Perusahaan Listrik Negara
National Electrification Project	The objective is Ministry of Cooperative and
KE Electricity Modernization Project	The proposed Ministry of Energy and Petroleum
KE Electricity Modernization Project	The proposed project development objective
Burkina Faso Electricity Sector Support Project	The project is in the Societe Nationale d'Electricite
RY-Rural Energy Access Project	The development Ministry of Electricity and Energy
GEF Project: Lao Rural Electrification Phase II Project	a) To increase electricity access. Ministry of Energy and Mining
Djibouti Power Access and Diversification Project	The revised project is Primature
DJ- Power Access and Diversification, Additional Financing	The project development objectives are to
Rural Electrification Phase II Project of the Rural Electrification	a) To increase electricity access. Ministry of Energy and Mining
Access to Sustainable Energy Project	The Project is in the National Power Corporation
Power System Efficiency and Resilience Project	The project is in the Electric Power No
Nigeria National Energy Development Project	1. To continue support for the Borrower's
Lao PDR: AUSAID Grant Additional Financing of the Rural	The objective of REP I Project are to: (i) increase
Rural Electrification Phase I Project of the Rural Electrification	The objective of REP I Project are to: (i) increase
Energy Community of South East Europe APL 3 Project	The development objective of Turkey EC
Benin Electricity Access Scale-up (BEAS) Project	The PDO is to increase electricity access. Yes
Tunisia Energy Sector Improvement Project	The Project is in STEG. TBD
Temane Regional Electricity Project	The Project Development Objective is to
Temane Regional Electricity Project	The Project is in the Ministry of Mineral Resources
Electricity Supply Reliability Project: Additional Financing	The project development objective is to increase
Enhancing Sierra Leone Energy Access	The Project is in the Electricity Dis No
ONEE Support Project (formerly, ONE Support Project)	The objective of the Project is to increase
Second Additional Financing for Southern African Power	To expand transmission capacity to better
Energy Infrastructure Improvement Project	The countries of South East Europe include
Energy Community of South East Europe APL6 Project - T	Increase the investment. TEIAS
Additional Financing for Southern Africa Power Market P	To expand transmission capacity to better

Energy Community of South East Europe APL 3 - Monten	The objective of ECSEE APL is the develop
Energy Community of South East Europe (ECSEE) Project	The project would provide investment su
Energy 2 Project	Improve the security and reliability of the
Rwanda - Energy Access and Quality Improvement Proje	Improve acce Development Bank of Rwar
Lesotho Renewable Energy & Energy Access Project	The Project D Ministry of Energy and Met
Tunisia-Italy Power Interconnector - Project Preparation	The PDO is to Elmed Etudes Yes
Somali Electricity Sector Recovery Project	The Project D Ministry of Fi Yes
WAPP-APL4 (Phase 1): Côte d'Ivoire Liberia Sierra Leone	The revised PDOs for the W Yes
Malawi - Electricity Access Project	The developn Electricity Supply Corporati
Madagascar - Least-Cost Electricity Access Development	The Project D JIRAMA, Mini Yes
Guinea Electricity Access Scale Up Project	The project d Electricité de Guinée, Agen
Somali Electricity Access Project	The Project D Ministry of Finance, Somali
THIRD MULTI-SECTORAL STRUCTURAL REFORMS DEVELOP	Support Gove Ministry of Finance and Bu
Electricity Sector Modernization and Sustainability Proje	To improve tl Ministry of Energy
SUSTAINABLE ENERGY DEVELOPMENT AND ACCESS PRO	To improve r Department of Resources a
Senegal - Third Multi-Sectoral Structural Reforms Develop	Support Gove Ministry of Finance and Bu
Additional Financing to the Mali Rural Electrification Hyb	The objective AMADER Yes
Additional Financing Electricity Access Expansion Project	<p>The Project Development Objective (
Electricity Service Access Project	The project d Rural Electrification Author
West Bengal Electricity Distribution Grid Modernization	The project's West Bengal State Electricit
Energy Supply Reliability and Financial Recovery Guarant	The objectives of the Proje No
Mozambique Energy For All (ProEnergia)	The Project D Fundo de Energia (FUNAE),
DRC Electricity Access & Services Expansion (EASE)	The project d Unite de Coordination et de
Electricity Access Expansion Project	<p>The Proje NIGELEC, Ministry of Energy
Additional Financing for Energy Development and Access	To improve the operational efficiency of
Access and Renewable Energy Project	The proposc Ministerio de Energia, Vice
Rwanda Energy Supplemental DPO	The Program MINECOFIN, MININFRA
Rwanda Electricity Access additional Financing	The project development objective is to i
Rebuilding Energy Infrastructure and Access	The objective EDH, MTPTC, UCP Energie
Third Rwanda Energy Sector Development Policy Operat	The Program MINECOFIN, MININFRA
Nigeria Electrification Project	The developn Federal Ministry of Finance
Ethiopia Electrification Program (ELEAP)	The developn Ethiopian Ele Yes
Energy Sector Support and Investment Project	The objective MInistère de l'Energie et de
Haiti: Renewable Energy for All	The Project D MTPTC Energ Yes
Senegal Structural Reform Development Policy Operatio	The Program Ministry of Communication
Energy Service Improvement Project	The project d Ministry of Energy, Water a
Herat Electrification Project	Provide acces Da Afghanistan Breshna Sh
Vietnam - Additional Financing for Medium Cities Develop	The developn Project Management Unit (
Djibouti: Sustainable Electrification Program	The proposc Electricite de Djibouti
Vietnam Scaling up Urban Upgrading Project	To improve a Construction-investment P
Additional Financing for the Ghana Energy Development	To improve the operational efficiency of
Uganda Rural Electrification	The Project D Ministry of Energy and Min
Additional Financing to the Senegal Electricity Sector Sup	The project development objectives are t

ENREP ADDITIONAL FINANCING	The Development Objectives of the Elect
Additional Financing for Energy Access Project	The proposed additional financing does r
Rwanda Electricity Access Scale-up and Sector Wide App	The project d REG-EDCL, Energy Developi
Electricity Sector Support Project	The project d Ministry of Energy and Min
Electricity Network Reinforcement and Expansion Project	The Developr Development Bank of Ethic
SL: Western Area Power Generation Project	The Project Development Objective is to
Ghana: Energy Development and Access Project	To improve tl VRA (TF090542), Electricity
Afghanistan Power System Development Project Additio	The development objective of the projec
Electricity Access (Rural) Expansion Project Phase II	The Project's Ethiopia Electric Power
Ukraine - Smart Grid Project – PPG	The objective of the assignment is to pre
Accelerated Electricity Access (Rural) Expansion	The project's development objective is to
Rural Energy II-Additional Financing	The overall development objective of the
ENERGY ACCESS PROJECT	The proposed additional financing does r
ENERGY ACCESS PROJECT	The proposed additional financing does r
Access Governance & Reform for the Electricity and Wat	The objective Ministère des Ressources H
Second Rural Energy Project	The overall d Vietnam Electricity (EVN), C
ERITREA POWER DISTRIBUTION AND RURAL ELECTRIFICA	Restructuring and additional financing w
Benin Second Unlocking Human and Productive Potentia	The proposec Ministry of Economy and Fi
Electricity Distribution Rehabilitation Project	To help improve the reliability of power s
Emergency Power Rehabilitation Project	The project objective is to provide an imp
Decentralized Infrastructure for Rural Transformation	The project development objective is to
GH Third Poverty Reduction Support Credit	PROMOTING GROWTH, INCOME AND EM
Emergency Power Restoration Project	To rehabilitat Electricity Sup Yes
Nepal: Power Sector Reform and Sustainable Hydropowe	The Project D Nepal Electricity Authority,
Additonal Financing for the Regional and Domestic Powe	Project development objective Ref. PAD I
Renewable Energy Integration Project Additional Financi	The PDO is "T TEIAS- Turkisl TBD
Second Rural Investment Project	The Project D State Service on Agricultura
Fourth Economic Reform Development Policy Operation	The program Ministry of Finance & Econ
Comoros Solar Energy Integration Platform	Improve the (Ministry of Pr TBD
Albania Competitiveness Development Policy Loan	Project Devel Ministry of Urban Developr
Additional Financing Haiti Renewable Energy for All	The Project D MTPTC Energy Cell, Ministr
Senegal Banda Gas to Power Guarantee	This PRG is linked to the Mauritania Gas t
MR-Banda Gas to Power Guarantee	The Projectâ Development Objective (PI
Dasu Hydropower Stage I Project	The overall pi Water and Power Developr
Mali Rural Electrification Hybrid System Project	The objective AMADER
Second Rwanda Energy Sector Development Policy Oper	The Program MINECOFIN, MININFRA
Distribution Grid Modernization and Loss Reduction Proj	<p>The proje CDEEE, SIE
Power Grid Improvement Project	The project d ELECTRICITE DU LAOS
MG - Integrated Growth Poles Additional Financing Cred	The overall purpose of the proposed proj
Medium Cities Development Project	The developn Ha Nam Provincial People's
Emergency Living Conditions Improvement Support Proj	The project's objective is to assist the Go
CASA-1000 Community Support Program (P149410)	The Project D MRRD
Senegal Energy Access Scale Up Project	Increase acce Senegal National Electricity

Solar Development in Sub-Saharan Africa - Phase 1 (Sahel)	The Series of West African Power Pool (V
Rwanda Energy Sector Development Policy Loan	The Program MININFRA
Additional Financing: TZ-Rural Electrification Expansion P	The Program Rural Energy Agency
Additional Financing to Rural Electrification Project	The objective Pamir Energy Company, Ba
Myanmar RBF for Off-grid Solar	The objective Ministry of A
Second Rural Electrification	Yes
Greening the Energy Mix in DVC	The developn MINEM - Ministerio de Ene
Additional Financing to PK: Tarbela 4th Extension Hydro	The proposec Damodar Valley Corporatio
Cambodia Rural Electrification Project	The Objective PMU T4HP, WAPDA
Bahia State Integrated Rural Poverty Additional Financin	The project development objectives are t
Tarbela Fourth Extension Hydropower Project	The proposed project would increase soc
Sergipe State Integrated Project: Rural Poverty	The Objective National Transmission and
Bahia State Integrated Project: Rural Poverty	xThe project is an integral part of the Go
Accelerating Renewable Energy Integration and Sustaina	The Project would increase social and eco
Additional Financing Argentina Provincial Agricultural De	The developn Ministry of Er
Nepal: Private Sector-Led Mini-Grid Energy Access Proje	Yes
HN RURAL INFRASTRUCTURE PROJECT (AF)	1. To increase and diversify agricultural p
Rural Electrification Project	The PDO is to Alternative E
Haiti Modern Energy Services For All	Yes
Yemen Emergency Electricity Access Project-Phase II	The GEF project development objectives
FCS RE Segou Solaire Mali Project	The objective Pamir Energy
CAR-Electricity Sector Strengthening and Access Project	Yes
Electricity Access Scale-up Project (EASP)	The Project D MTPTC - Mini
Regional Electricity Access and BEST Project	Improve acce UNOPS for the benefit of Y
ECOWAS-Regional Electricity Access Project	The Project D Ministry of Economy and Fi
MZ-Energy Development and Access Project (APL-2)	Increase supx Ministry of Development o
GEF Electricity Access and Renewable Energy Expansion	The Project D Ministry of Energy and Min
GEF Electricity Access and Renewable Energy Expansion	The Project D NIGELEC, SON
Third Andhra Pradesh Economic Reform Loan/Credit	Yes
Kenya: Off-grid Solar Access Project for Underserved Cou	The Developr Energie du Mali (EDM SA),
Rural Electrification Project Stage II	The overall P
HARYANA POWER SYSTEM IMPROVEMENT PROJECT	Funco Nacional de Energia
Ethiopia/Nile Basin Initiative: Ethiopia-Sudan Interconne	The project d Solomon Islands Electricity
Electricity Services Delivery Project	The project d Solomon Islar
Nepal Power Development Project	Yes
Development of Pumped Storage Hydropower in Java Ba	The proposed operation is the third in a s
POWER SYSTEM DEVELOPMENT PROJECT -III	The Project D Ministry of Energy and Petr
Fifth Power System Development Project	The Project D Ministry is Ministry of Clim
Burkina Faso AF to Electricity Sector Support Project	The developn DHBVN, HVPN
Banda Gas to Power Guarantee	The Project's long-term development obj
Zanzibar Energy Sector Transformation and Access Proje	The development objectives are to: (a) ac
Network Reinforcement and Access Project	To increase a Department of Electricity D
Cameroon - Kribi Gas Power Project	The objective PT Perusahaa
	Yes
	The objective of the Project is to strengt
	The project d Power Grid Corporation of
	The project development objectives are t
	The Gas To Power Project will use natura
	The objective Zanzibar Elec
	Yes
	To improve tl Eswatini Elec
	Yes
	The project development objective is to (

Increased Access to Modern Energy Electricity Sector Development Project	The project's Project Coordination Unit h
AFCC2/RI-Regional and Domestic Power Markets Develop	The project d Uganda Electricity Transmis
Power Sector Recovery Project - Additional Financing	Project devel CDP-SNEL-DRC
Power System Reliability and Efficiency Improvement Pro	The Project Development C No
First Programmatic Electricity Distribution Reform Devel	Improve the r Power Grid Company of Ba
Nigeria Electricity and Gas Improvement Project (add. fir	The propossec Department of Energy, Gov
Power Sector Recovery Project	The development objectives of the Proje
Vietnam Power Sector Reform Development Policy Oper	The Project D Electricite de Guinee (EDG)
Advanced Electricity Metering Project	To support th Electricity Regulatory Autho
Eskom Investment Support Project	The project d UzbekEnerg
Electricity Distribution Modernization Program	The project d ESKOM
Third Angola Growth and Inclusion Development Policy F	The PDO is to Bangladesh Rural Electrifica
Energy Utility Performance and Reliability Improvement	The developn Ministry of Finance of the F
Nigeria Distribution Sector Recovery Program	To improve tl PNG Power Limited (PPL)
Mazar-e-Sharif Gas-to-Power Project	The Program Bureau of Public Enterprise
Togo First Fiscal Management and Energy Reform DPF	The project d Afghan Power Plant Compa
Sierra Leone Energy Sector Utility Reform Project Addit	The Program Ministry of Economy and Fi
IQ: Basra Electricity Dissemination and Development Pro	The Project D Ministry of Er No
Tonga Second Inclusive Growth Development Policy Ope	To improve tl South Electric Yes
Power Efficiency and Reliability Improvement Project (PE	Project Devel Ministry of Finance and Na
Second Energy Sector Project	The Project D Ministry of M Yes
Andhra Pradesh 24X7 Power for All	The developn Ministry of Energy
Rural Distribution Project	The developn Andhra Pradesh Southern F
GZ-ELECTRIC UTILITY MANAGEMENT PROJECT	The objective of the proposed project is t
MG-Electricity Sec Operations & Governance Improveme	The project d PETL, PENRA
Rwanda Electricity Sector Strengthening Project	<p>The Proje JIRAMA, Ministry of Water
Bangladesh Power Sector Development Policy Credit	<p>The Proje Energy Utility Corporation I
Shared Infrastructure for Solar Parks Project	Project Development Objective (Note: wi
BF - DPO series on Energy and PFM	The propossec Indian Renewable Energy D
Belize Energy Resilience for Climate Adaptation (GEF/SC	The Program Ministry of Economy, Finan
Gaza Emergency Response for Electricity Network Rehab	The developn Belize Electricity Limited (B
STP Power Sector Recovery Project	The Project objective is to restore and im
Electricity Supply Accountability and Reliability Improver	The project d Ministério das Infraestrutu
Bicol Power Restoration Project	The propossec Severelectro OJSC
Sierra Leone Energy Sector Utility Reform Project	The projects's development objective is t
Tanzania Emergency Power Supply	The Project D Ministry of Energy, Electrici
GZ-Gaza Electricity Network Rehabilitation Project	The development objectives of this proje
DABS Planning And Capacity Support	The Project o Palestinian Energy and Nati
Energy Sector Support Project	<p>The Proje Da Afghanistan Breshna Sh
Emergency Electricity and Water Rahabilitation Project -	To increase tl ESCOM, Ministry of Natura
MN-Energy Sector Additional Financing	The development objective of the Emerg
EG-Giza North Additional Financing	The objective of the project is to reduce :
	The project development objective is to c

PROGRAMMATIC ELECTRICITY SECTOR DEVELOPMENT PROJECT	The PEDPLs will support the implementation of the
Burkina Faso - Additional Financing Electricity Sector Support Project	
Urgent Electricity Rehabilitation Project	The project is managed by Cote d'Ivoire Energies, Direction
Distribution Efficiency Project	The project is managed by Electricity Regulatory Authority
Mali Rural Electrification Hybrid System Project - GPOBA	
Senegal Energy Sector Recovery Development Policy Financing	The primary objective of the operation is to improve the
Gas Sheep Harvesting project	The development objective of the project is to increase the
Additional Financing Mali Household Energy and Universal Access	The development objectives are: (i) accelerate the
DO Electricity Distribution Rehabilitation Project	The Project development objective is to improve the
Energy Community of South East Europe (ECSEE) APL3-B	The countries of South East Europe, including Albania, Bulgaria, Croatia, Czech
CAR - Emergency Power Response Project	The project is managed by AGETIP CAF
Pakistan Program for Affordable and Clean Energy (PACE)	To reduce carbon emissions, the Government of Pakistan has
Power Sector Development Technical Assistance Project	The project Development Objectives are: (i) Improve the
HT Electricity Project	Contribute to the sustainable improvement of the power
RY-Power Sector	Increase electricity supply and improve the efficiency of the
Power Sector Development Project	(i) Enhance the policy and institutional environment for
Utility Services for the Development of Housing in Informal Settlements	The project is managed by Department of Housing, Ministry of
SEIER Additional Financing	No change to PDO following restructuring of the project
Second Environmental Sustainability and Energy Sector Development	The development objective of the Government of the Republic of
Rural Power Project	The objective of the project is to assist in the development of
Java-Bali Power Sector Restructuring and Strengthening	The development objective of the Project is to improve the
Second Public Investment Reform Support Credit	Project Development Ministry of Planning
Sierra Leone - Sixth Governance Reform and Growth Grant	Ministry of Finance and Economic Development
Economic Reconstruction and Growth Development Policy	The goal of the operation is to support the Ministry of Economy and Finance
Emergency Development Policy Operation	The objective of the proposed operation is to improve the
GRGC-3 Supplemental Credit	The project development objectives have been revised to
TZ-Second Power and Gas Sector DPO	The objective of the project is to improve the
Governance Reform and Growth Grant - 3	The project development objectives have been revised to
Second Programmatic Energy Sector Development Policy	The Energy Sector Ministry of Energy, Water and Power
Electricity Distribution Efficiency Improvement Project	The project is managed by Multan Electric Power Company
CAPE VERDE - Recovery and Reform of the Electricity Sector	The proposed main development objective is to improve the
Privatization of the Power Distribution System Operator	The objective of the proposed PRG is to support the
Kenya - Energy Sector Recovery Project Additional Financing	
Energy Loss Reduction Project	The development objective of the project is to improve the
Power Distribution and Rural Electrification	
Additional Financing for the Benin Energy Services Delivery Project	
Electricity Reform for Sustainable Growth Development	The Development Ministry of Finance
Tina River Hydropower Development Project	The project is managed by Solomon Islands Ministry of
Power Sector Recovery Performance Based Operation	The objective of the project is to improve the
ELETROBRAS Distribution Rehabilitation	The Project is managed by Eletrobras
Energy Sector Project	The objective of the project is to reduce the
Gambia Electricity Restoration and Modernization Project	The Project is managed by National Water and Electricity
DO Emergency Recovery & Disaster Mgmt	The project is managed by National Institute of Hydraulics

Bangladesh DSC IV-Supplemental Financing II	DSC IV, just like the previous three DSC o
Emergency MS Rehabilitation & Recovery ERL Second Ad	The overall purpose of the EMRRP is to ir
Additional Financing to Power Utility Financial Recovery	The Program Ministry of Energy and Wat
Phase 2 Advancing Sustainability in Performance, Infrast	To improve o Palestinian Energy and Nat
Tajikistan Power Utility Financial Recovery	The Program Ministry of Energy and Wat
Gambia Electricity Restoration and Modernization Project	The Project D National Water and Electric
Sustainable Energy Development Project	The objective Marshalls Enç Yes
Gambia Electricity Support Project	The Project D National Water and Electric
HT (AF) Electricity Loss Reduction Project	Contribute to the sustainable improveme
Madagascar Electricity Sector Operations and Governanc	<p>The Proje Ministry of Energy and Hyd
AF ENERGY LOSS REDUCTION	The objectives of the Project are to assist
Advancing Sustainability in Performance, Infrastructure,	To improve o Palestinian Energy and Nat
Zambia Scaling Solar Energy Guarantee Project	To increase solar electricity generation c
Additional Financing for the Social Safety Nets for Vulne	Support the t Provincial Directorate of M
Social Safety Nets for Vulnerable Populations in the Prov	Support the t Provincial Dir Yes
Rural Investment Project (AZRIP)	Improved living standards and increased
Energy for Rural Transformation Project	(Phase 1 only) Put in place a functioning
Third Productivity and Transparency Support Grant	The main obj Anti Corruption Commissio
STP Strengthening Growth and Fiscal Policy DPO	Project Devel Directoria do Orçamento
Investment Promotion and Financing Facility	To accelerate Bangladesh Bank
Costa Rica Second Fiscal and Decarbonization Managem	The Program Ministry of Environment ar
Benin Second Fiscal Management and Structural Transfo	The Program Ministry of Economy and Fi
Second Macroeconomic Stability for competitiveness an	Project Devel Ministry of Energy
Second Competitiveness and Fiscal Management Progra	Project Devel Ministry of Finance and Pul
IRAQ: SECOND EXPENDITURE RATIONALIZATION, ENERG	The proposc Prime Minister's Office
Panama Second Shared Prosperity Development Policy Loan	Ministry of Economy and Fi
First Fiscal Effectiveness and Growth Development Polic	This is the firs Ministry of Public Finance
CV-DPL 4-PRSC VIII	Project Devel Ministry of Finance and Pla
Solomon Islands Development Policy Operation-2	The Program Ministry of Finance & Treas
Mali RRSC	The proposed stand-alone Development
SN- First Governance and Growth Support Project	The proposc Ministry of Economy and Fi
Comoros Development Policy Grant 2	The overarching objective of the propose
CV-DPL 3-PRSC VII	The main development objectives of the
GM-Budget Support -DPL (First in the series)	The main dev Ministry of Finance and Ecc
Zambia Third Poverty Reduction Support Credit	This PRSC operation supports two object
WBG - PRDP Support IV	The proposed operation will help the PA
Poverty Reduction Support Grant (PRSG-8)	Project Development Objective (Note: wi
WBG Support for Fiscal Sustainability and Public Financ	The proposed operation will help the PA
Economic Recovery and Gov. Grant 3	The objective is to support the policy ref
Poverty Reduction Support Credit (PRSC1)	This PRSC operation supports two object
Togo: Economic Recovery and Gov. Grant 2	To support Government-owned reforms
Development Policy Lending 3 (DPL 3)	The DPL-III supported the Government's
WBG Support for Fiscal Sustainability and Public Financ	This operation helps the PA to implement

TG-Economic Recovery and Governance Grant	The objective of the grant is to support g
Supplemental EGRO-II Development Policy Grant	This DPL supports the Government's poli
LB-EMERGENCY POWER SECTOR REFORM CAPACITY REIN	The objective of the project is to accelera
PK PRSC I - Supplemental Financing	
Productivity and Transparency Support Credit	Improve the quality, transparency and ac
Madagascar Covid-19 Response DPO	The developn Ministry of Economy and Fi
Consolidation Support Program Policy Grant	The First Consolidation Support Program
2021 Afghanistan Incentive Program Development Policy	Program Dev Ministry of Finance
Togo Fiscal Reform Credit	Project Development Objective (Note: wi
BF - DPO2 series on Energy and PFM	The Program Ministry of Economy, Finan
LIBERIA Electricity System Enhancement Project (LESEP)	To reduce gre LIBERIA ELECTRICITY CORPC
Afghanistan Resource Corridor Project	The Project D Ministry of Mines, Ministry
SL Supplemental Financing for the PTSC-I	Project Development Objective (Note: wi
Tanzania Poverty Reduction Support Credit 8	This PRSC series supports the implement
Nepal Energy Sector Development Policy Credit	Project Development Objective (Note: wi
Heat Supply Improvement Project	The Project D Community D Yes
Solomon Islands Sustainable Energy, Additional Financi	The objective of the project is to improve
Environmental Sustainability and Energy Sector (ESES) DI	The development objective of the Goveri
DRC Emergency Project to Mitigate Impact of Financial C	To help mitigate the immediate and seve
Energy Emergency Project	The development objective of the projec
System Efficiency Improvement, Equitization & Renewal	No change to PDO following restructuring
North Macedonia Public Finance and Competitiveness DI	The proposc Ministry of Labor and Socia
Governance and Competitiveness Programmatic DPO	Project Devel Ministry of Finance
Lao Sixth Poverty Reduction Support	Project Development Objective (Note: wi
Programmatic Private and Financial Development Policy Credit (PPFDPC-1) (Serbia)	
Ecuador Green and Resilient Recovery (EGARR) DPF	This operation seeks to assist the governi
Honduras First Fiscal Sustainability and Enhanced Compe	The Program EMPRESA NACIONAL DE EN
Emergency Operation for Development	The Project is Ministry of Education, C/O
Honduras Fiscal Sustainability and Enhanced Social Prote	Project Devel Secretaría de Desarrollo e I
Development Policy Operation - DDO	The objective Ministry of Public Finance
Burundi Fifth Economic Reform Support Grant	Project Development Objective (Note: wi
Second Poverty Reduction Support Credit (PRSC 2)	The objectives of the PRSCs will contribut
Poverty Reduction Support Credit (8)	DON'T PUB OLD
DRC Emergency Multi Sectoral Rehab & Recovery ERL Su	The overall purpose of the EMRRP is to ir
Second Accelerating India's COVID-19 Social Protection F	The proposc Ministry of Finance Depart
Fourth Infrastructure Development Policy Loan	The objectives are to: (a) increase the qu
VN Urban Upgrading Project Additional Finance	To alleviate poverty in urban areas by im
Second Infrastructure Development Policy Loan	
Emergency Municipal Services (Rehab. II)	The objectives of EMSRP II are to (a) prov
Montenegro Second Energy Efficiency Project	The project d Ministry of Health, Ministry
Solomon Islands Community Benefit Sharing Pilot Project	The project d Ministry of Mines, Energy &
Power Sector Recovery Project Additional Financing	The project d Agencia Fiduc No
Ghana Energy Sector Transformation Initiative Project	The Project d Ministry of Er Yes

NP Power Add Financing

Indonesia Sustainable Least-cost Electrification Technical Assistance	The Project D PT PLN (Persero)	Yes
China Distributed Renewable Energy Scale-Up Project	Promote the National Energy	Yes
Innovation in Solar Power and Hybrid Technologies	The Project D Solar Energy Corporation of India	Yes
Bangladesh Scaling-up Renewable Energy Project	The Project D Infrastructure Development	Yes
Scaling-Up Renewable Energy Program Investment Plan	The Project D The World Bank	Yes
Rural Investment Additional Financing	Improved living standards and increased	
ANGOLA EMERGENCY MULTI-SECTOR RECOVERY - PHASE 1	The overall purpose of the program is to	
Public Finance and Social Sector Development Policy Loan	1. Improvement of the quality and efficiency	
TUNISIA FIRST RESILIENCE AND RECOVERY EMERGENCY INTERVENTION	The program Ministry of Development, Planning and Economic Affairs	
GZ-Second Municipal Development Project AF	The Revised Objective of the Project is to	
GZ Emergency Response Second Municipal Development Project	The Revised Objective of the Project is to	
GZ-Second Municipal Development Project	The Revised Objective of the Project is to	
WBG Support for Fiscal Sustainability and Public Financial Management	The objective is to support the PA as it in	
Accelerating Reforms for an Inclusive and Resilient Recovery	Accelerating Reforms for an Inclusive and Resilient Recovery	
The Governance of Basic Local Public Services - 3rd Global Infrastructure Anti-Crisis Facility	Improve the provision of urban basic services	
ID-Infrastructure DPL (IDPL)	The objective is to support a subset of the	
Moldova Power System Development Project	The Project D Moldova Energy Projects Inc	
Additional Financing for the Tanzania Energy Development Project	The objective of the project is to improve	
TZ-Energy Development & Access Expansion Project	The objective Ministry of Energy and Mineral Resources	
Electricity Expansion	The project h Ministry of Energy and Petroleum	
MA-Support to ONE (Office National de l'Electricité)	The objective Office National de l'Electricité	
ECSEE APL 3 - FYR Macedonia	The objective Macedonian Transmission Company	
AFCC2/RI-3A-Southern Africa Power Market APL 1 (FY04)	To expand transmission capacity	
Transmission Investment Plan (PIT) to support Post-COVID-19	To increase electricity supply	
Electricity Sector Transformation and Resilient Transmission	The Project d JSC "National Electric Grid of Kyrgyzstan"	Yes
Additional Financing for the Central Asia South Asia Electricity	The objective National Electric Grid of Kyrgyzstan	
NG-Electricity Transmission Project	The project d Transmission Company of Mongolia	
National Transmission Modernization I Project	The Project D National Transmission Company	Yes
OMVS - TRANSMISSION EXPANSION PROJECT	The project d Mali - SOGEM - Societe de Gestion	
Electricity Transmission and Reform Project	The project d Société Nationale de Transmission	
Modernization and Upgrade of Transmission Substations	The project d JSC "Uzbekistan National Power Grid"	
Electricity Transmission Network Improvement Project	The proposed Yerevan Thermal Power Plant	
Second Power Transmission Project	To improve transmission capacity	
Transmission Efficiency Project (TEP)	The objective National Power Transmission Company	
Transmission Grid Strengthening Project	The project d Georgian State Electrosystem	
Central Asia South Asia Electricity Transmission and Trade	The objective National Electric Grid of Kyrgyzstan	
Bangladesh: Rural Electricity Transmission and Distribution	The proposed Bangladesh Rural Electrification	
Indonesia Second Power Transmission Development Project	The development PT PLN	
Additional Financing for Nepal-India Electricity Transmission and Trade	Project	
Lusaka Transmission and Distribution Rehabilitation Project	The objective ZESCO	
Kafue-Muzuma-Victoria Falls Regional Transmission Line	To improve transmission capacity	
Nepal-India Electricity Transmission and Trade Project	The development Nepal Electricity Authority	

Second Transmission and Distribution Project Additional	The main objective of the project is to su
Talimarjan Transmission Project	Project Devel Uzbekenergo
Alma Transmission Project	The objective Kazakhstan Electricity Tran
Tanzania - Backbone Transmission Investment Project	The Project D TANESCO
Indonesia Power Transmission Development Project	The developn PT PLN
Morupule B Generation and Transmission Project	The PDOs are to support Botswana in: (i)
Botswana - Morupule B Generation and Transmission Pr	The PDOs are Ministry of Minerals, Energ
Kazakhstan Moinak Electricity Transmission Project	The primary project objective is to increa
Electricity Distribution and Transmission Improvement P	The objectives of the APL1 project are to
Power Transmission Project in Support of the Energy Sec	The overall pi National Power Company L
Mozambique Transmission Upgrade Project	To reduce thç Electricidade de Moçambiq
Second Transmission and Distribution Project	The objective of the project is the efficien
Power Transmission Project	The primary ç Azerenerji
Rural Electrification and Transmission Project	The main development objectives of the
Transmission Development Project	Support the FGN's overall program of po
Central Asia South Asia Electricity Transmission and Trad	8. The objective of the project is to facilit
Energy Supply Reliability and Financial Recovery	The objective Georgian State Electrosyste
AFR RI-3A Tanzania-Zambia Transmission Interconnector	The Developr Tanzania Electric Supply Co
WAPP: The First Phase of the Inter-Zonal Transmission H	The developn Ghana Grid Company Limit
Second Djibouti-Ethiopia Power System Interconnection	The Project D Electricité de Djibouti
Guinea – Mali Interconnection Project	The Project D EDG, EDM TBD
Uganda Grid Expansion and Reinforcement Project (GER)	The Project D Uganda Electricity Transmis
OMVG Interconnection Project	The project d OMVG
AFR RI-The Eastern Electricity Highway Project under the	Project objec Ethiopian Electric Power, K
WAPP APL4 (Phase 1) - Côte d'Ivoire, Sierra Leone, Liber	The revised P The CLSG Regional Transmi
GMS Power Trade (Cambodia) Project	The GMS-PT Projects have the developm
GMS Power Trade (Laos) Project	To enhance p Ministry of Energy and Min
Mozambique - Malawi Regional Interconnector Project	The project d Electricity Supply Corporati
Support for Preparation of the Rural Electrification, Seb	The project d Ministry of Er No
Enhancement and Strengthening of Power Transmission	The proposec Power Grid Company of Ba
Siddhirganj Power Project	Increase supç Gas Transmission Comapny
North Core/Dorsale Nord Regional Power Interconnecto	The Project D WAPP Secretariat, Project I
Energy Community of South East Europe (APL) Program -	Within the overall ECSEE APL objectives/
Additional Financing For Dasu Hydropower Stage I Proje	The overall pi Water and Power Developr
Kabeli Transmission Project	The project d Nepal Electricity Authority,
WAPP APL 1 (2nd Phase - COASTAL TRANSMISSION BAC	The goal of V Volta River Authority, Comi
North-South Electricity Transmission Project	The primary development objective is to
WAPP APL 1 (1st Phase - COASTAL TRANSMISSION BACK	The developn Ghana Grid Company (GRIC
Cameroon - Chad Power Interconnection Project	The Project D SONATREL, SI No
Cote d'Ivoire - Electricity Transmission and Access Projec	The project d CI-Energies
Niger Accelerating Electricity Access Project (Haské)	Increase acce Société Niger No

Project URL	Board Approval Date	Project Closing Date	Last Stage Review Date	Financing Type	Current Project	IBRD Commitment
https://projects.worldbank.org/en/projects-operations/project-detail/179000000	2018-04-10T00:00:00Z	2024-12-31T12:00:00AM	2023-12-31T12:00:00AM	000	0	179000000
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https://projects.worldbank.org/en/projects-operations/project-detail/607700000	2021-06-21T00:00:00Z	2028-12-31T12:00:00AM	2028-12-31T12:00:00AM	000	0	607700000
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https://projects.worldbank.org/en/projects-operations/project-detail/413000000	2006-06-27T00:00:00Z	2012-12-31T12:00:00AM	2012-12-31T12:00:00AM	000	0	413000000
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https://projects.worldbank.org/en/projects-operations/project-detail/417000000	2021-02-18T00:00:00Z	2026-09-30T12:00:00AM	2026-09-30T12:00:00AM	000	0	417000000
https://projects.worldbank.org/en/projects-operations/project-detail/44122463	2006-08-03T00:00:00Z	2014-10-31T12:00:00AM	2014-10-31T12:00:00AM	000	0	44122463
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https://projects.worldbank.org/en/projects-operations/project-detail/2225000	2016-07-20T00:00:00Z	2020-03-31T12:00:00AM	2020-03-31T12:00:00AM	000	0	2225000
https://projects.worldbank.org/en/projects-operations/project-detail/200000000	2012-06-26T00:00:00Z	2020-09-30T12:00:00AM	2020-09-30T12:00:00AM	000	0	200000000
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https://projects.worldbank.org/en/projects-operations/project-detail/24670000	2018-01-24T00:00:00Z	2018-12-31T12:00:00AM	2018-12-31T12:00:00AM	000	0	24670000
https://projects.worldbank.org/en/projects-operations/project-detail/225000000	2014-05-29T00:00:00Z	2023-06-30T12:00:00AM	2023-06-30T12:00:00AM	000	0	225000000
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IDA Commitment	Total IDA and Grant Amount	Borrower	Lending Instrument	Environment	Environment
952200000		Department of	Investment P A		
0		Republic of M	Investment P B		
200000000		Federal Minis	Investment P B		
2500000			Specific Inves B		
41220000		Honduras Mi	Specific Inves B		
11700000			Specific Inves C		
2800000			Emergency R B		
11000000	1,000,000	Ministry of Fi	Investment P B		
52000000		Republic of A	Investment P B		
41100000		GOVERNMENT	Specific Inves B		
100000000		Republic of Ir	Investment P F		
93000000			Adaptable Pr B		
5000000			Specific Inves B		
45000000		Ministry of Fi	Development Policy Lending		
185000000		The Federal R	Specific Inves B		
28160000			Specific Inves B		
40000000			Specific Inves B		
26820000		Government	Emergency R B		
230000000		Ministry of Fi	Investment P B		
0	2,000,000	Republic of T	Investment P B		
0		Economic Aff	Investment Project Financing		
179000000	20,000,000	PEOPLE'S REF	Investment Project Financing		
120000000		Cellule de Co	Investment P B		
607700000	93,700,000	Burkina Faso	Investment Project Financi	Substantial	
50000000			Specific Inves B		
41300000			Sector Invest B		
71000000		Ministry of M	Specific Inves B		
417000000	167,000,000	Republic of A	Investment Project Financi	Substantial	
44122463			Specific Inves F		
15000000	15,000,000	Islamic Repuk	Investment P B		
2225000	2,230,000	Solomon Islar	Investment Project Financing		
200000000		Ministry of Ec	Investment P A		
111450000			Specific Inves B		
100000000			Specific Inves B		
654520000			Specific Inves B		
24670000	4,670,000	Republic of Li	Development Policy Lending		
22500000		Ministry of Ec	Investment P B		
50000000		REPUBLIC OF	Specific Inves A		
23454540			Emergency R B		
225520000	100,000,000		Specific Inves B		
35000000		Republic of Li	Investment P B		

27000000	0	Republic of Li Investment P B
12000000		Adaptable Pr B
500000000		Federal Demc Investment Project Financi Substantial
64200000	5,200,000	Republic of Li Investment Project Financi Substantial
0		Central Africa Investment P B
465200000		Government Investment P B
0		Republic of C Investment P B
60000000		Investment P B
0		Ministry of Ec Investment P B
2114000000		Specific Inves A
450000000		Development Policy Lending
2114000000		Specific Inves A
16500000		Adaptable Pr B
20000000		Ministry of Ec Investment P B
60000000		Investment P A
54800000		Honduras Mi Specific Inves B
1450000000	420,000,000	PT. Perusahaan Program-for-Results Financing
567000000		The Republic Investment P B
562000000	7,500,000	Republic of K Investment P B
200000000		Investment P B
52870000		Burkina Faso Investment P B
115300000	0	Ministry of PI Specific Inves B
10448000		Specific Inves B
7300000		EDD Specific Inves B
6000000		Specific Inves B
34820000		Ministry of Fi Adaptable Pr B
23240000	23,240,000	National Pow Investment P F
0		Republic of th Investment P B
181000000		Specific Inves B
0		Adaptable Pr B
50880000	10,000,000	Adaptable Pr B
195100000		Adaptable Pr B
200000000		Republic of B Investment Project Financi Moderate
0		STEG Investment P B
1426000000	24,000,000	Investment Project Financing
1426000000	24,000,000	Ministry of Ec Investment P A
50000000		Investment P B
52700000	2,700,000	Republic of Si Investment Project Financi Substantial
40500000		Investment P B
201500000		Adaptable Pr A
27300000		Specific Inves F
227130316		Republic of T Adaptable Pr B
252500000		Adaptable Pr A

16000000		Adaptable Pr	B
52140000		Adaptable Pr	B
39925095		Specific Inves	B
288000000	50,000,000	Republic of R	Investment Project Financi
0	12,900,000	Kingdom of L	Investment P B
13402000	7,000,000	Republic of T	Investment P A
150000000		Federal Repu	Investment Project Financi
126981628	4,600,000		Investment Project Financing
0		Ministry of Fi	Investment P B
0		REPUBLIC OF	Investment P B
108500000		Republic of G	Investment P B
5500000	5,750,000	The Federal R	Investment P B
0		Republic of S	Development Policy Lending
58000000		Kyrgyz Repub	Investment Project Financi
0		Department c	Investment P B
0		Ministry of Fi	Development Policy Lending
0	2,700,000	Government	Investment Project Financing
162000000	92,000,000		Investment P B
36800000	7,000,000	Republic of Z	Investment P B
385700000	135,000,000	INDIA	Investment Project Financi
160000000			Investment Project Financing
0	66,000,000	Ministry of Ec	Investment P B
147000000		Republique D	Investment P B
65000000		Government	Investment P B
60000000			Investment P B
59150000		Ministry of D	Investment P B
0		MINECOFIN	Development Policy Lending
60000000			Specific Inves B
100000000	10,000,000	Republic of H	Investment P B
0		MINECOFIN	Development Policy Lending
765000000		Federal Repu	Investment P B
677000000		Ministry of Fi	Program-for-Results Financing
36000000		Republic of T	Investment P B
19620000	19,620,000	Ministry of Fi	Investment P B
60000000		Ministry of th	Development Policy Lending
61000000		Government	Investment P B
60000000		Islamic Repuk	Investment P B
64650000		State Bank of	Investment P A
28150000		Ministry of Ec	Investment P B
330000000		Socialist Repl	Investment P A
77000000			Specific Inves C
13700000	13,700,000	Republic of U	Investment P B
168000000		Ministry of Er	Investment P B

248000000	5,000,000		Investment P B
245000000			Specific Inves B
80000000		MINECOFIN	Investment P B
187500000		Republic of S	Investment P B
275000000		Federal Demc	Investment P B
40000000			B
210550000	85,500,000	Ministry of Er	Investment P B
15000000			Investment P B
160000000		FEDERAL DEN	Specific Inves B
500000			Technical Ass C
184210000			Specific Inves B
250600000			Specific Inves B
199120000	15,420,000		Specific Inves B
15300000	4,930,000		Specific Inves B
939000000	34,000,000	DEMOCRATIC	Investment Project Financi High
324250000		State Bank of	Specific Inves B
57200000			Specific Inves B
0		Republic of B	Development Policy Lending
344400000			Specific Inves B
125000000	20,000,000		Emergency R B
22400000			Adaptable Pr B
125000000			Development Policy Lending
60000000		Republic of N	Investment Project Financi High
24000000		Nepal	Investment P A
380000000			Adaptable Pr B
0		TEIAS- Turkisl	Investment Project Financing
53600000		Republic of A	Investment P B
5000000		Republic of K	Development Policy Lending
0	0	Government	Investment P B
70000000			Development Policy Lending
6900000	2,900,000		Investment Project Financing
99000000			A
130000000			A
4694750000		Islamic Repuk	Investment P A
44900000	19,900,000	Republic of N	Investment P B
0		MINECOFIN	Development Policy Lending
120000000		DOMINICAN	Investment P B
30000000		Lao People's	Investment P B
40000000			Specific Inves C
258900000		Socialist Rept	Investment P A
82000000			Specific Inves B
40000000	30,000,000	Ministry of Fi	Investment P B
160000000		Republic of S	Investment Project Financi Substantial

0		West African Investment P A
125000000		MINECOFIN Development Policy Lending
341000000	6,000,000	UNITED REPU Program-for-Results Financing
100000000		Ministry of Er Investment Project Financing
0	3,450,000	Ministry of PI Investment P B
82705000		Republic of P Specific Inves B
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823500000		Investment P A
0		Specific Inves B
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27050000		Specific Inves B
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107400000	50,000,000	Republic of N Investment Project Financi Moderate
448000000		Specific Inves A
8000000	7,610,000	Nepal Investment P F
200000000		Specific Inves B
31700000		Ministry of Er Investment Project Financi Substantial
15650000	15,650,000	Republic of H Investment P F
120400000		Yemen, Repu Investment Project Financi Substantial
48400000		Republic of N Investment Project Financing
138000000	30,000,000	Central Africa Investment Project Financi Substantial
638000000	40,000,000	Republic of U Investment Project Financi Substantial
465000000		Republic of N Investment Project Financi Substantial
0		Republic of N Investment P B
80000000		Ministry of Er Adaptable Pri B
0	950,000	Solomon Islar Investment P B
0	8,700,000	Solomon Islar Investment P B
225000000		Development Policy Lending
150000000		The National Investment P B
12500000		Ministry of Fi Investment P B
410000000		Government Investment Project Financing
70950000		Specific Inves B
95700000		Adaptable Pri B
150141936	0	Specific Inves A
755000000	230,000,000	Republic of Ir Investment Project Financi High
1043000000		Specific Inves A
1562000000		Power Grid C Investment P A
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32000000		A
142000000	25,000,000	United Repuk Investment Project Financi Substantial
0		Kingdom of E Investment Project Financi Moderate
350000000		A

163190000	47,090,000	Societe Benin Investment P B
153200000		Republic of U Investment P A
883000000	173,300,000	Ministry Of Fi Investment P B
250000000		Investment P B
770000000		Peoples Repu Investment P B
250000000		Republic of Ir Development Policy Lending
100000000		Specific Inves B
500000000		The Republic Investment P B
311800000		Ministry of In Development Policy Lending
246100000		UZBEKISTAN Specific Inves C
13861820000	5,375,050,000	ESKOM Holdi Investment P A
4115000000	28,000,000	People's Rept Program-for-Results Financing
0		REPUBLIC OF Development Policy Lending
314000000		Papua New G Investment P A
1743000000	700,000,000	Federal Repu Program-for-Results Financ Low
108600000	43,800,000	Afghan Powe Investment P B
0		Ministère de Development Policy Lending
0		Government Investment Project Financing
0		Republic of Ir Investment P B
0		Ministry of Fi Development Policy Lending
150000000		Republic of N Investment P B
55400000	12,400,000	Mongolia Investment P B
570000000	160,000,000	INDIA Investment P B
206280000		Specific Inves B
181300000	156,200,000	Ministry of Fi Specific Inves B
65000000		Ministry of Ec Investment P B
95000000		The Republic Investment P B
120000000		Development Policy Lending
400000000	25,000,000	Indian Renew Investment P F
100000000		Ministry of Ec Development Policy Lending
11975000	8,000,000	Government Investment P B
		Investment P B
41000000	13,000,000	Ministry of Fi Investment P B
25470000	470,000	Kyrgyz Repub Investment P B
21600000		Emergency R B
40000000		Government Investment P B
110800000		Emergency R C
16000000		Palestine Libe Investment P B
6000000	6,000,000	Ministry of Fi Investment P B
84700000		Department c Investment P B
2200000		Emergency R B
12500000		Specific Inves B
764000000		Specific Inves A

800000000			Development Policy Lending
37000000			Investment P B
59130316			Specific Inves B
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6790000	5,000,000		Investment P B
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1847000			Specific Inves B
38860000			Specific Inves B
152100000			Specific Inves B
272410000			Adaptable Pr B
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0		THE ISLAMIC	Development Policy Lending
20500000			Technical Ass C
7474760			Specific Inves C
57130316		Ministry of PI	Specific Inves B
130340200			Specific Inves B
4000000		Republic of Fi	Investment P B
26510000			Specific Inves B
700000000			Development Policy Lending
16700000			Adaptable Pr F
211640000.03			Specific Inves B
50000000		Republic of N	Development Policy Lending
25000000			Development Policy Lending
20000000		Ministry of Ec	Development Policy Lending
30000000			Development Policy Lending
7000000			Development Policy Lending
100000000		MINISTRY OF	Development Policy Lending
10000000		Ministry of Fi	Development Policy Lending
0		Nepal	Development Policy Lending
209080000		The Islamic R	Investment Project Financi Moderate
58500000		Ministry of Fi	Investment P B
78000000			C
81000000			Specific Inves B
30000000		Ministry of Fi	Specific Inves A
17500000			Specific Inves B
7000000			Adaptable Pr B
0		Dominican R	Development Policy Lending
240480000	162,600,000	Solomon Islar	Investment P A
0		Federal Repu	Program-for-Results Financing
709300000		Centrais Eletr	Investment P B
36200000			Specific Inves B
0		Ministry of Fi	Investment Project Financing
80000000		Ministry of Fi	Emergency R B

100000000		Development Policy Lending
12000000		Emergency R A
80000000		Ministry of Fi Program-for-Results Financing
23500000	23,500,000	Ministry of Fi Investment P B Substantial
0		Republic of T Program-for-Results Financing
121500000		Ministry of Fi Investment P B
34000000		Republic of th Investment P B
18500000		Ministry of Fi Investment P B
5000000		Specific Inves C
40000000		Ministry of B Investment Project Financing
18000000		Specific Inves A
0	63,000,000	Ministry of Fi Investment Project Financi Substantial
6300000		B
50000000		Province of B Investment Project Financing
0		Province of B Investment Project Financi Moderate
21100000		Specific Inves F
111240000	6,900,000	Adaptable Pr F
0		Republic of Si Development Policy Lending
5000000		Ministry of Fi Development Policy Lending
557000000		Government Financial Intermediary Loan
0		Republic of C Development Policy Lending
0		Republic of B Development Policy Lending
200000000		Ministry of Fi Development Policy Lending
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20000000		Development Policy Lending
20000000		Development Policy Lending
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40000000		Development Policy Lending

175000000		Development Policy Lending
10000000		Development Policy Lending
8000000		Emergency R C
150000000		Development Policy Lending
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0		Republic of N Development Policy Lending
10000000	3,020,000	Development Policy Lending
0	300,000,000	ISLAMIC REPL Development Policy Lending
51900000		Development Policy Lending
75000000		Ministry of Ec Development Policy Lending
53000000	43,000,000	Ministry of La Emergency R B
0		Investment P A
10000000		Development Policy Lending
115000000		Development Policy Lending
250000000	150,000,000	Development Policy Lending
41000000		Kyrgyz Repub Investment P B
13000000		Investment P B
600000000		Development Policy Lending
100000000		Emergency R C
6500000		Emergency R B
347900000		Specific Inves B
0		Ministry of Fi Development Policy Lending
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50000000		Secretaria de Development Policy Lending
750000000		Ministry of Fi Investment P B
55000000		Development Policy Lending
1333000000		Ministry of P Development Policy Lending
35000000		Development Policy Lending
30000000		Development Policy Lending
100000000		Development Policy Lending
143600000		Emergency R A
0		Republic of Ir Development Policy Lending
200000000		Development Policy Lending
410000000		Specific Inves A
200000000		Development Policy Lending
40200000		Emergency R B
8997800		Montenegro Investment P B
2800000	2,800,000	Solomon Islar Investment Project Financing
0		Ministry of Fi Investment Project Financing
20000000		The Republic Investment P C

104200000		Specific Inves B
2150000	2,150,000	PT PLN (Perse Investment Project Financi
0	7,280,000	People's Rep Investment P B
400000000	49,810,000	India Investment P B
0	88,250,000	People's Rep Investment P B
300000	300,000	Ministry of Pt Investment P B
25560000		Specific Inves F
134000000		Emergency R B
150000000		Development Policy Lending
0		REPUBLIC OF Development Policy Lending
		Investment P B
		Investment P B
74850000	32,250,000	Palestine Libe Investment P B
40000000		Development Policy Lending
0		The National Development Policy Lending
260000		Technical Ass C
700000000	500,000,000	Development Policy Lending
45000000		Ministry of Fi Investment P A
43380000		Specific Inves B
147959950		United Repuk Specific Investment Loan
1390650000	800,000,000	The Republic Investment P A
150000000		Office Nation Specific Inves B
39710000		MEPSO Adaptable Pr F
200190250	3,360,000	Ministry Of Fi Adaptable Pr A
0		Ministry of Ec Investment Project Financi
427000000	47,000,000	Republic of U Investment Project Financi
0		Kyrgyz Repub Investment Project Financing
490000000		Federal Repu Investment P B
536330000		Islamic Repuk Investment P B
103000000		The Republic Investment P B
375000000		Republic of C Investment P A
196300000		Republic of U Investment P B
0		Republic of A Investment P B
318425000	48,430,000	Ministry of Fi Investment P B
731250000		Socialist Repl Investment P B
61875000		Ministry of Fi Investment P A
1126500000	296,000,000	Kyrgyz Repub Investment P A
837000000		Government Investment P B
159030000		Republic of Ir Investment P B
39000000		Specific Inves B
210000000	0	Ministry of Er Investment P B
100000000	30,000,000	Ministry of Er Investment P B
182300000		Nepal Investment P B

372400000		Specific Inves B
203040000		Government Specific Inves B
200900000		Kazakhstan E Specific Inves B
468450000		MINISTRY OF Specific Investment Loan
254100000		Republic of Ir Investment P B
242660000		A
1662000000		Republic of B Specific Inves A
69000000.01		Specific Inves B
309900000		Adaptable Pri B
238000000		Ministry of Fi Specific Inves B
13800000		Republic of N Specific Inves B
324020000		Specific Inves B
55400000		Specific Inves B
67920000		Specific Inves B
113350000		Specific Inves B
0		Specific Inves A
0		Ministry of Fi Investment Project Financi High
605000000	130,000,000	Eastern Africa Investment P A
111000000		Ministry of Fi Investment P B
66940000	11,940,000	Republic of D Investment Project Financi Substantial
0		Republic of N Investment P A
127300000		REPUBLIC OF Investment P B
711000000	495,000,000	The Republic Investment P B
1262500000	472,300,000	Ministry of Er Investment P A
472500000		Republic of Li Investment P A
20000000		Specific Inves B
14000000		Specific Inves B
0	24,000,000	Ministry of Fi Investment P A
500000	500,000	Ministry of Fi Investment Project Financi Substantial
718060000		People's Rep Investment P B
662890000		The People's Investment P A
640220044	169,830,000	Federal Repu Investment P B
30750000		Adaptable Pri B
0		Ministry or W Investment Project Financing
47690000		Ministry of Fi Specific Inves B
75000000	0	Ministry of Fi Adaptable Pri B
160840000	13,140,000	Specific Inves B
83000000	29,500,000	Ministry of Fi Adaptable Pri B
0	265,500,000	Republic of C Investment P A
325000000		Republic of C Investment P B
317500000	7,500,000	Republic of N Investment Project Financi Substantial

Sector 1	Sector 2	Sector 3	Theme 1	Theme 2
Energy Transi				
Energy Transi	Public Admin	ICT Infrastruc		
Energy Transi	Public Admin			
Energy Transi				
Energy Transi	Other Energy	Public Admini		
Energy Transi	Central Gove	Other Energy		
Other Energy	Central Gove	Energy Transi		
Energy Transi	Sub-National	Early Childho		
Energy Transi				
Energy Transi	Central Gove	Renewable Ei		
Energy Transi	Rural and Inte	Railways		
Energy Transi	Renewable Ei	Renewable Ei		
Energy Transi	Central Gove			
Central Gove	Social Protect	Energy Transi		
Fisheries	Crops	Public Admini		
Other Agricul	Energy Transi	Water Supply		
Other Agricul	Social Protect	Energy Transi		
Other Water	Urban Transp	Water Supply		
Public Admin	Irrigation and	Agricultural n		
Public Admin	Sub-National	Energy Transi		
Public Admin	Energy Transi	Mining		
Renewable Ei	Renewable Ei	Energy Transi		
Energy Transi	Public Admin	Other Energy		
Other Energy	Energy Transi	Renewable Ei		
Energy Transi	Water Supply	Other Energy		
Rural and Inte	Other Water	Energy Transi		
Other Energy	Energy Transi	Renewable Ei		
Public Admin	Energy Transi	Other Energy		
Social Protect	Other Educat	Sub-National		
Sub-National	Energy Transi	Other Agricul		
Public Admin	Energy Transi			
Urban Transp	Water Supply	Other Water		
Energy Transi	Renewable Ei	Renewable Ei		
Energy Transi	Renewable Ei	Renewable Ei		
Energy Transi	Renewable Ei	Renewable Ei		
Central Gove	Banking Instit	Health		
Central Gove	Energy Transi	Water Supply		
Central Gove	Energy Transi			
Non-Renewal	Energy Transi	Public Admini		
Non-Renewal	Other Energy	Energy Transi		
Oil and Gas	Public Admin	Energy Transi		

Renewable E	Energy Transi	Non-Renewal
Energy Transi		
Renewable E	Public Admin	Energy Transi
Renewable E	Energy Transi	Other Energy
Renewable E	Energy Transi	Public Admini
Energy Transi	Public Admin	
Renewable E	Energy Transi	Renewable E
Public Admin	Energy Transi	
Public Admin	Water Supply	Energy Transi
Energy Transi		
Agricultural E	Other Educat	Energy Transi
Energy Transi		
Energy Transi	Renewable E	
Renewable E	Renewable E	Water Supply
Rural and Inte	Energy Transi	Irrigation and
Rural and Inte	Other Water	Energy Transi
Energy Transi		
Energy Transi		
Energy Transi		
Energy Transi	Renewable E	Renewable E
Energy Transi	Non-Renewal	Other Energy
Energy Transi	Public Admin	Renewable E
Energy Transi	Renewable E	Renewable E
Energy Transi	Renewable E	Renewable E
Other Energy	Energy Transi	
Energy Transi	Other Energy	Renewable E
Renewable E	Renewable E	Renewable E
Non-Renewal	Energy Transi	
Other Energy	Energy Transi	Central Gover
Energy Transi	Renewable E	Renewable E
Energy Transi	Other Energy	Renewable E
Energy Transi		
Energy Transi	Public Admin	
Energy Transi	Renewable E	Public Admini
Energy Transi	Oil and Gas	
Energy Transi	Public Admin	
Energy Transi		
Energy Transi	Renewable E	Public Admini
Energy Transi		
Energy Transi		
Energy Transi	Public Admin	
Energy Transi	Other Energy	

Energy Transi	Renewable Ei
Energy Transi	
Energy Transi	Other Energy Central Gover
Energy Transi	Renewable Ei Renewable Ei
Energy Transi	Renewable Ei Renewable Ei
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Energy Transi	Renewable Ei Public Admini
Energy Transi	Other Energy Renewable Ei
Energy Transi	Public Admin Renewable Ei
Energy Transi	Renewable Ei Public Admini
Energy Transi	Renewable Ei Public Admini
Energy Transi	Renewable Ei Public Admini
Energy Transi	ICT Infrastruc Oil and Gas
Energy Transi	Public Admin ICT Services
Energy Transi	Renewable Ei Public Admini
Energy Transi	ICT Infrastruc Oil and Gas
Energy Transi	Renewable Ei Public Admini
Energy Transi	Public Admin
Energy Transi	Renewable Ei Public Admini
Energy Transi	ICT Infrastruc Public Admini
Energy Transi	Public Admin
Energy Transi	Renewable Ei Public Admini
Energy Transi	Renewable Ei Banking Instit
Energy Transi	Public Admin
Energy Transi	
Energy Transi	Renewable Ei Public Admini
Energy Transi	Other Energy Renewable Ei
Energy Transi	Renewable Ei Public Admini
Energy Transi	Public Admin Other Energy
Energy Transi	Other Energy Renewable Ei
Energy Transi	Renewable Ei
Energy Transi	Renewable Ei Public Admini
Energy Transi	Public Admin
Energy Transi	Public Admin Renewable Ei
Energy Transi	ICT Infrastruc
Energy Transi	Public Admin Renewable Ei
Energy Transi	Public Admin
Energy Transi	Sub-National Water Supply
Energy Transi	Public Admin
Energy Transi	Other Water Urban Transp
Energy Transi	
Energy Transi	
Energy Transi	Public Admin

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 Energy Transi Social Protect Other Transp
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 Energy Transi Other Public , Forestry
 Other Energy Energy Transi Central Gove
 Public Admin Energy Transi Water Supply
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 Energy Transi Other Energy Central Gove
 Energy Transi Other Energy
 Central Gove Health Primary Educ
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 Renewable Er Energy Transi Other Energy
 Renewable Er Energy Transi Public Admini
 Renewable Er ICT Infrastruc Energy Transi
 Other Public , Energy Transi Social Protect
 Central Gove ICT Services Fisheries
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 Other Energy Public Admin Energy Transi
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 Other Industr Other Public , Urban Transp
 Urban Transp Other Water Sub-National
 Other Transp Other Water Energy Transi
 Energy Transi Social Protect Irrigation and
 Energy Transi Public Admin

Public Admin Energy Trans Renewable Er
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 Other Agricul Water Supply Sub-National
 Renewable Er Other Public Other Energy
 Other Agricul Energy Trans Water Supply
 Other Agricul Sub-National Social Protect
 Renewable Er Energy Trans
 Other Public Energy Trans Irrigation and
 Public Admin Renewable Er Renewable Er
 Other Water Rural and Inte Sub-National
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 Renewable Er Energy Trans Public Admin
 Oil and Gas Non-Renewal Energy Trans
 Energy Trans Renewable Er Other Energy
 Energy Trans Public Admin
 Non-Renewal Energy Trans Social Protect

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 Energy Transi Public Admin
 Central Gove Renewable Er Energy Transi
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 Energy Transi Renewable Er ICT Infrastruc
 Energy Transi Other Energy Public Admini
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 Energy Transi Oil and Gas
 Energy Transi Central Gove Other Energy
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 Energy Transi Public Admin
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 Energy Transi Central Gove
 Energy Transi Public Admin Non-Renewal
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 Energy Transi Other Energy
 Energy Transi Renewable Er
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 Non-Renewal Oil and Gas Energy Transi

Energy Transi	Other Energy	Central Gove
Energy Transi		
Energy Transi	Other Energy	Public Admini
Other Energy	Energy Transi	Public Admini
Energy Transi	Renewable Ei	Renewable Ei
Energy Transi	Oil and Gas	Other Energy
Energy Transi	Oil and Gas	Central Gove
Energy Transi	Public Admin	Other Energy
Energy Transi		
Energy Transi	Renewable Ei	Non-Renewal
Renewable Ei	Other Energy	Energy Transi
Social Protect	Other Energy	Energy Transi
Energy Transi	Central Gove	Oil and Gas
Energy Transi		
Energy Transi	Other Energy	
Energy Transi	Non-Renewal	Central Gove
Energy Transi	Sanitation	Water Supply
Energy Transi	Public Admin	
Energy Transi	Other Water	Other Energy
Energy Transi	Other Energy	Renewable Ei
Energy Transi	Central Gove	
Central Gove	Energy Transi	
Central Gove	Energy Transi	Other Educat
Central Gove	Energy Transi	Other Public ,
Central Gove	Energy Transi	
Central Gove	Sub-National	Energy Transi
Oil and Gas	Other Energy	Energy Transi
Central Gove	Sub-National	Energy Transi
Forestry	Renewable Ei	Other Energy
Other Energy	Public Admin	Energy Transi
Non-Renewal	Public Admin	Energy Transi
Energy Transi	Other Energy	
Energy Transi	Public Admin	
Oil and Gas	Energy Transi	Other Energy
Energy Transi	Central Gove	
Energy Transi	Other Energy	Renewable Ei
Other Energy	Social Protect	Energy Transi
Public Admin	Renewable Ei	Energy Transi
Other Energy	Public Admin	Energy Transi
Other Energy	Energy Transi	
Other Energy	Energy Transi	Central Gove
Other Water	Public Admin	Energy Transi
Other Water	Energy Transi	Irrigation and

Energy Transi	Banking Instit	Other Industr
Energy Transi		
Public Admin	Energy Transi	Renewable Er
Public Admin	Renewable Er	Energy Transi
Public Admin	Renewable Er	Energy Transi
Public Admin	Renewable Er	Energy Transi
Public Admin	Energy Transi	Renewable Er
Public Admin	Non-Renewal	Energy Transi
Public Admin	Energy Transi	
Renewable Er	Energy Transi	Public Admini
Renewable Er	Public Admin	Energy Transi
Renewable Er	Public Admin	Energy Transi
Renewable Er	Energy Transi	
Social Protect	Energy Transi	
Social Protect	Energy Transi	Public Admini
Energy Transi	Rural and Inte	Sanitation
Energy Transi		
Agricultural n	Central Gove	Other Educat
Banking Instit	Energy Transi	Central Gove
Banking Instit	Energy Transi	Rural and Inte
Central Gove	Other Energy	Energy Transi
Central Gove	ICT Services	Other Energy
Central Gove	Other Public	Social Protect
Central Gove	Trade	Energy Transi
Central Gove	Sub-National	Energy Transi
Central Gove	Other Educat	Social Protect
Central Gove	Energy Transi	Capital Marke
Central Gove	Other Transp	Energy Transi
Central Gove	Mining	Water Supply
Central Gove	Other Agricul	Other Industr
Central Gove	Energy Transi	Health
Central Gove	Irrigation and	Energy Transi
Central Gove	Other Industr	Aviation
Central Gove	ICT Infrastruc	Primary Educ
Central Gove	Other Industr	Other Energy
Central Gove	Public Admin	Energy Transi
Central Gove	Oil and Gas	Energy Transi
Central Gove	Public Admin	Sub-National
Central Gove	Crops	Energy Transi
Central Gove	Energy Transi	Public Admini
Central Gove	Crops	Mining
Central Gove	Capital Marke	Banking Instit
Central Gove	Energy Transi	Social Protect

Central Government	Agriculture	Energy Transition
Central Government	Other Transport	Energy Transition
Central Government	Energy Transition	Other Energy
Central Government	Health	Other Education
Crops	Central Government	Energy Transition
Health	Central Government	Social Protection
Law and Justice	Central Government	Primary Education
Oil and Gas	Energy Transition	Insurance and
Oil and Gas	Central Government	ICT Infrastructure
Oil and Gas	Central Government	Other Energy
Oil and Gas	Public Administration	Energy Transition
Oil and Gas	Other Transport	Other Industries
Other Agriculture	Central Government	Energy Transition
Other Agriculture	Other Public	Energy Transition
Other Energy	Renewable Energy	Energy Transition
Other Energy	Energy Transition	Public Administration
Other Energy	Energy Transition	Renewable Energy
Other Energy	Other Industries	Energy Transition
Other Energy	Water Supply	Energy Transition
Other Energy	Oil and Gas	Central Government
Other Energy	Energy Transition	Central Government
Other Industries	Central Government	Social Protection
Other Industries	Energy Transition	Central Government
Other Industries	Banking Institutions	Central Government
Other Industries	Other Non-bank	Mining
Other Public	Non-Renewable	Energy Transition
Other Public	Other Industries	Energy Transition
Other Public	Health	Energy Transition
Other Public	Social Protection	Energy Transition
Other Public	Energy Transition	Health
Other Public	Other Agriculture	Other Industries
Other Public	Central Government	Energy Transition
Other Public	Sub-National	Agriculture
Other Transport	Water Supply	Other Education
Other Water	Social Protection	Energy Transition
Other Water	Energy Transition	Other Transport
Other Water	Urban Transport	Social Protection
Other Water	Energy Transition	Other Transport
Other Water	Energy Transition	Urban Transport
Public Administration	Energy Transition	Health Facilities
Public Administration	Water Supply	Energy Transition
Renewable Energy	Energy Transition	
Renewable Energy	Other Energy	Oil and Gas

Renewable En	Energy Transi	Public Admini
Renewable En	Energy Transi	
Renewable En	Energy Transi	ICT Infrastruc
Renewable En	Renewable En	Energy Transi
Renewable En	Energy Transi	
Renewable En	Energy Transi	Renewable En
Rural and Inte	Social Protect	Sub-National
Rural and Inte	Other Water	Energy Transi
Social Protect	Energy Transi	Other Energy
Sub-National	ICT Services	Energy Transi
Sub-National	Energy Transi	Urban Transp
Sub-National	Energy Transi	Urban Transp
Sub-National	Energy Transi	Urban Transp
Sub-National	Central Gove	Energy Transi
Water Supply	Sanitation	Irrigation and
Water Supply	Energy Transi	Urban Transp
Water Supply	Energy Transi	Rural and Inte
Energy Transi	Public Admin	
Energy Transi	Public Admin	
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Energy Transi	Renewable En	Renewable En
Energy Transi	Other Energy	Renewable En
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Energy Transi	Public Admin Mining
Energy Transi	Public Admin
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Energy Transi	Renewable Ei
Energy Transi	
Energy Transi	Other Energy
Energy Transi	ICT Infrastruc Other Energy
Energy Transi	Public Admin Other Energy
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Energy Transi	Public Admin
Energy Transi	Public Admin
Energy Transi	Public Admin
Energy Transi	
Energy Transi	Other Energy Central Gover
Energy Transi	Public Admin
Energy Transi	
Energy Transi	Renewable Ei
Public Admin Renewable Ei	Renewable Ei
Renewable Ei Public Admin	Energy Transi
Renewable Ei Energy Transi	Public Admini
Non-Renewal Oil and Gas	Energy Transi
Energy Transi	Public Admin Other Energy
Energy Transi	
Energy Transi	Renewable Ei Public Admini
Energy Transi	Renewable Ei
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Energy Transi	
Energy Transi	
Energy Transi	Renewable Ei Public Admini
Energy Transi	Public Admin
Energy Transi	Renewable Ei Public Admini

World Bank Commodity Price Data (The Pink Sheet)

annual indices, 2010=100, 1960 to present, real 2010 US dollars

(annual series are available in nominal and real dollars)

Updated on April 04, 2023

Source: <https://thedocs.worldbank.org/en/doc/5d903e848db1d1b83e0ec8f744e55570-0350012021/relati>

	Energy	Non-energy				
			Agriculture			
				Beverages	Food	Oils & Meals
	KIENERGY	KINONFUEL	KIAGRICULTU	KIBEVERAGE	KIFOOD	KIFATS_OILS
1960	11.15	96.68	113.16	129.37	108.81	115.38
1961	10.68	94.70	111.55	115.87	112.81	127.60
1962	10.28	91.72	108.06	110.90	109.67	113.67
1963	10.32	98.59	117.83	115.19	125.11	120.40
1964	9.83	101.43	116.23	123.30	122.54	122.20
1965	9.71	101.97	112.50	112.57	118.94	134.03
1966	9.09	102.21	112.65	112.36	118.66	127.88
1967	8.93	97.52	111.24	112.25	118.18	116.54
1968	8.95	96.32	108.05	117.98	111.86	105.61
1969	8.31	96.36	106.16	120.84	110.21	101.73
1970	9.10	95.03	106.16	119.28	112.40	117.14
1971	11.10	88.37	100.87	102.36	109.05	114.45
1972	10.98	84.65	98.00	102.01	107.21	104.87
1973	13.51	113.87	138.24	115.64	162.00	182.20
1974	36.67	125.90	149.29	117.88	182.28	172.11
1975	33.13	92.37	108.52	98.50	126.66	111.27
1976	36.32	95.05	114.06	173.33	112.51	111.91
1977	36.83	102.17	123.07	278.38	107.89	128.08
1978	33.24	88.68	105.16	182.16	102.46	113.26
1979	65.83	92.48	106.80	169.23	104.01	112.45
1980	71.73	91.99	105.43	134.70	108.84	97.90
1981	71.77	80.38	94.51	107.02	101.15	93.81
1982	69.47	70.41	83.12	109.13	83.34	79.79
1983	63.79	79.65	92.57	125.32	93.18	93.56
1984	62.71	82.73	99.06	146.67	97.64	110.66
1985	60.56	71.49	84.11	132.28	81.44	83.44
1986	31.83	59.72	69.82	128.82	62.03	55.92
1987	32.59	59.27	65.92	89.06	60.42	57.07
1988	26.45	69.82	70.93	81.34	72.31	71.48
1989	30.97	68.74	72.14	68.86	73.71	64.54
1990	37.42	62.00	66.59	60.97	66.14	54.24

1991	32.98	59.56	66.32	55.92	66.04	56.22
1992	31.43	57.40	64.40	47.56	64.56	57.01
1993	27.75	56.08	66.42	49.13	62.70	56.73
1994	26.67	65.53	76.48	80.94	69.43	66.15
1995	25.98	64.80	72.75	75.33	68.53	64.79
1996	32.18	64.86	75.71	66.29	75.37	69.47
1997	31.27	66.47	76.65	88.38	73.10	73.66
1998	23.20	59.67	69.76	81.75	69.73	70.73
1999	30.29	55.29	62.84	64.43	59.73	56.27
2000	49.27	56.63	61.42	54.17	58.93	53.15
2001	45.75	55.67	61.48	48.40	63.64	52.98
2002	44.88	58.83	66.90	59.50	68.44	61.47
2003	52.01	61.24	69.00	57.94	70.57	70.69
2004	62.87	66.57	71.17	56.03	74.79	73.65
2005	85.96	69.88	70.69	63.15	71.22	64.36
2006	94.99	84.02	76.04	66.08	75.33	64.21
2007	98.92	94.12	85.14	72.01	87.88	89.66
2008	125.52	102.81	99.92	81.40	108.90	112.52
2009	82.23	87.02	92.79	89.26	96.40	94.39
2010	100.00	100.00	100.00	100.00	100.00	100.00
2011	121.82	107.72	109.48	104.49	110.34	108.60
2012	121.71	99.40	103.73	84.08	112.69	113.77
2013	120.01	92.34	96.34	75.94	104.47	104.23
2014	111.28	89.14	94.15	94.04	97.97	98.27
2015	67.52	83.41	89.78	96.01	90.40	86.21
2016	59.60	84.40	93.00	97.09	95.31	93.87
2017	71.28	86.03	89.34	85.40	92.68	89.98
2018	87.69	83.54	84.94	77.61	88.76	83.42
2019	78.64	81.97	83.55	76.51	87.42	77.85
2020	53.87	86.02	89.03	82.18	95.22	91.81
2021	87.19	102.79	99.00	85.45	111.38	116.18
2022	138.20	112.65	111.19	96.27	130.14	131.53
2023	93.74	96.61	97.18	94.44	109.94	104.21
2024	88.21	97.62	99.74	153.23	100.65	92.85

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					Fertilizers **	Metals & Minerals
		Raw Materials				
Grains	Other Food		Timber	Other Raw Mat.		
KiGRAINS	KiOTHERFOOD	KiRAW_MATERIAL	KiTIMBER	KiOTHERRAW	KiFERTILIZERS	KiMETMIN
122.38	87.89	115.49	84.83	149.01	67.19	66.22
124.66	82.64	106.27	84.99	129.55	66.08	63.38
134.91	81.55	102.71	88.50	118.25	63.38	61.44
140.50	117.37	101.48	88.74	115.41	64.85	62.98
139.67	107.47	97.31	79.37	116.92	79.02	73.63
134.41	85.08	96.82	85.99	108.67	82.60	82.57
142.36	85.07	98.21	85.55	112.05	72.72	84.15
151.29	90.37	93.87	88.30	99.96	62.53	73.34
142.46	92.40	93.73	90.08	97.73	50.70	77.44
133.24	100.54	88.84	83.98	94.15	42.01	82.43
118.24	100.89	84.34	84.16	84.54	37.45	78.72
112.04	99.23	80.25	80.18	80.31	34.87	68.81
108.26	109.35	73.56	71.21	76.14	44.63	61.81
182.47	116.89	92.03	83.96	100.86	63.75	69.58
219.40	162.06	85.09	77.46	93.43	187.34	70.95
156.89	119.54	69.55	61.92	77.90	122.51	55.84
131.15	96.42	87.70	74.47	102.17	61.43	59.88
107.78	81.42	81.00	69.27	93.82	78.88	61.94
111.35	80.21	72.56	61.59	84.57	63.24	57.76
108.51	88.84	81.83	79.44	84.45	68.31	65.86
113.80	118.76	82.24	79.58	85.14	80.54	65.74
120.53	93.27	72.00	67.94	76.45	76.24	51.88
95.39	77.10	69.35	67.55	71.31	59.22	45.61
105.69	81.38	74.44	66.59	83.02	56.61	55.76
104.90	73.94	78.30	74.18	82.81	65.20	51.22
89.70	71.33	66.12	63.45	69.05	56.41	47.32
65.30	67.10	58.73	60.09	57.25	39.60	41.30
58.38	66.66	67.54	70.28	64.54	38.79	47.95
73.23	72.58	62.26	66.03	58.14	44.38	70.44
80.18	79.90	70.01	74.97	64.59	42.48	64.76
70.05	78.27	70.53	76.24	64.28	42.05	54.82

70.90	74.58	72.28	80.51	63.27	47.44	47.10
69.97	69.60	72.55	86.77	57.00	42.56	44.75
63.11	70.18	84.26	116.52	48.98	35.29	37.25
70.19	73.08	91.32	119.23	60.79	39.24	46.08
75.48	67.18	81.68	96.02	65.99	42.10	51.10
92.72	67.44	81.33	97.72	63.41	48.87	44.41
75.37	70.30	79.31	93.05	64.29	44.92	48.07
69.89	68.27	63.72	69.81	57.07	41.06	41.08
61.54	62.66	69.58	86.91	50.62	39.54	41.60
58.63	66.80	71.15	87.61	53.14	45.37	48.05
61.24	79.84	62.88	74.23	50.47	45.51	44.89
69.95	76.25	66.92	81.08	51.44	46.13	43.70
68.22	72.54	70.82	82.23	58.34	52.90	46.27
69.72	80.90	70.08	81.21	57.91	59.12	57.98
67.03	84.04	73.24	87.44	57.70	67.25	68.53
77.87	87.65	82.82	97.68	66.58	67.94	102.25
93.24	80.69	85.17	99.90	69.06	87.38	113.34
127.09	87.67	87.54	101.98	71.74	184.91	99.44
102.58	93.45	85.82	100.34	69.95	124.39	70.90
100.00	100.00	100.00	100.00	100.00	100.00	100.00
124.50	99.80	109.95	105.67	114.63	123.98	102.24
128.29	97.14	91.98	99.03	84.27	128.01	87.23
116.92	93.50	86.98	93.51	79.83	104.25	82.78
96.18	99.18	84.93	96.91	71.83	93.78	78.36
88.97	97.21	85.11	98.17	70.82	100.51	68.37
85.80	105.81	85.29	95.34	74.30	82.67	66.99
82.76	105.22	83.23	87.91	78.10	76.32	80.37
87.10	97.28	79.39	86.63	71.48	80.96	80.97
89.40	98.20	77.73	86.03	68.65	81.79	78.75
97.49	97.64	77.50	88.35	65.63	76.29	80.93
113.19	103.41	75.82	82.66	68.34	139.22	106.44
136.21	122.83	72.71	72.51	72.92	213.53	104.19
116.6	111.46	67.57	69.34	65.64	134.56	91.15
98.06	113.26	70.33	69.16	71.61	102.2	92.74

	Precious Metals	Deflator MUV Index
Minerals		
Base Metals (ex. iron ore)		
KiBASEMET	KiPRECIOUS	MUV
72.11	17.08	19.14
69.12	16.79	19.52
66.90	17.23	19.88
68.62	18.42	19.52
82.50	18.32	19.79
93.26	18.25	19.98
96.39	17.62	20.72
83.63	18.66	20.90
88.70	22.81	20.72
94.40	20.82	21.82
90.26	18.16	23.21
78.07	17.59	24.41
70.00	20.79	26.63
80.42	28.80	30.89
81.78	39.62	37.64
62.22	35.23	41.80
65.25	29.17	42.32
68.30	30.74	45.74
64.75	33.67	53.16
74.85	51.95	59.30
74.16	91.28	65.22
57.09	61.26	65.30
48.03	50.49	63.35
61.23	62.32	61.67
56.21	51.85	60.32
51.22	44.45	59.69
44.80	42.84	68.65
53.74	47.96	75.23
81.98	43.66	80.11
74.24	38.22	79.63
61.30	36.10	82.67

51.28	33.67	81.91
48.82	31.50	83.44
40.53	32.05	86.32
51.75	36.06	83.73
58.06	32.81	91.92
49.43	33.62	90.17
53.65	30.78	85.92
44.61	30.00	82.17
45.81	29.00	80.59
53.46	29.51	79.56
49.07	29.25	76.58
47.69	33.17	75.68
50.63	36.53	79.62
64.35	39.96	85.03
72.65	42.13	87.70
113.73	57.36	89.93
119.14	62.36	95.43
98.37	70.69	102.83
74.17	80.84	96.46
100.00	100.00	100.00
101.93	122.76	111.04
88.92	125.64	110.23
82.36	104.91	109.75
82.30	93.46	108.25
75.18	92.58	97.87
72.65	103.67	94.03
87.30	100.56	97.29
88.89	95.36	101.86
82.03	105.97	99.48
81.99	136.54	97.90
107.62	128.13	109.40
110.90	123.92	117.45
95.49	129.14	114.10
99.15	156.53	115.10